

CAPSTONE PROJECT 2
Final Report

**A Data-Driven Web-Based Service Quality Analysis for Enhancing
Customer Satisfaction in Malaysian E-Banking**

by

OON JING XUAN
22042063

Bachelor of Information Systems (Honours) (Data Analytics)

Supervisor: Dr Nor Hafizah Mohamed Halip

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School of Computing and Artificial Intelligence
Faculty of Engineering and Technology
Sunway University

Abstract

Malaysia's growing use of e-banking increases the need for banks to understand how digital service quality influences Customer Satisfaction. This project examines the relationships between seven e-banking service-quality dimensions—Reliability, Responsiveness, Efficiency, Privacy and Security, Ease of Use, System Availability, and Website Design—and Customer Satisfaction. It also develops an interactive Power BI dashboard and proposes data-driven recommendations to support service-quality improvement. The study applies a quantitative, cross-sectional design and analyses 426 valid questionnaire responses from e-banking users in Malaysia. Python is used for data cleaning and preparation, while IBM SPSS Statistics is used to conduct Cronbach's Alpha reliability analysis, descriptive statistics, Pearson correlation, and multiple linear regression. The findings show that respondents generally evaluate Malaysian e-banking services positively, with a mean Customer Satisfaction score of 4.08 out of 5. Nevertheless, 303 respondents, or 71.13%, report experiencing at least one e-banking issue. All seven service-quality dimensions have strong, positive, and statistically significant relationships with Customer Satisfaction. The regression model explains 69.7% of the variation in Customer Satisfaction, with Efficiency emerging as the strongest significant predictor, followed by System Availability, Responsiveness, and Ease of Use. Website Design records the lowest mean score, while Reliability has the highest number of issue selections, and transaction failure is the most frequently reported individual problem. The four-page Power BI dashboard presents the findings through key performance indicators, dimension comparisons, customer-segment filters, and issue-monitoring views, while establishing baseline measures for future comparisons. An Evidence-to-Action mapping process translates the combined findings into prioritised recommendations, measurable indicators, and responsible stakeholder teams. Overall, the project provides an integrated decision-support structure that connects customer feedback, statistical evidence, interactive visualisation, and practical service-improvement planning for Malaysian banks.

Keywords: E-banking, service quality, Customer Satisfaction, Power BI, multiple linear regression, Evidence-to-Action, Malaysia.

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
ANOVA	Analysis of Variance
BI	Business Intelligence
BSQ	Banking Service Quality
CS	Customer Satisfaction (construct code)
CSAT	Customer Satisfaction Score
DBSQ	Digital Banking Service Quality
E2A	Evidence-to-Action
ECT	Expectation–Confirmation Theory
EFF	Efficiency (construct code)
EOU	Ease of Use (construct code)
E-S-QUAL	Electronic Service Quality Scale
E-SERVQUAL	Electronic Service Quality (SERVQUAL adapted for electronic services)
ETL	Extract, Transform, and Load
FPX	Financial Process Exchange
IBM	International Business Machines
IC	Identity Card
IT	Information Technology
KMO	Kaiser–Meyer–Olkin Measure
KPI	Key Performance Indicator
KTA	Knowledge-to-Action
MPQe-BS	E-Banking Service Quality Measurement Instrument

Abbreviation	Meaning
OTP	One-Time Password
P1	Priority 1 (Immediate)
P2	Priority 2 (Next)
P3	Priority 3 (Monitor)
PDF	Portable Document Format
PS	Privacy and Security (construct code)
QR	Quick Response
R ²	Coefficient of Determination (R-squared)
REL	Reliability (construct code)
RES	Responsiveness (construct code)
RM	Malaysian Ringgit
SA	System Availability (construct code)
SD	Standard Deviation
SERVQUAL	Service Quality Model
SMS	Short Message Service
SPSS	Statistical Package for the Social Sciences
TAM	Technology Acceptance Model
UI	User Interface
UX	User Experience
VIF	Variance Inflation Factor
WBD	Website Design (construct code)
WBS	Work Breakdown Structure

Chapter 1. Introduction

1.0 Overview

Over the past few years, Malaysia's payment habits seem to have moved away from paper-based tools and toward digital options such as e-banking and e-wallets. Figure 1.1 gives a clear snapshot of this shift. It shows e-money usage rising strongly from 2019 onward, while cheque transactions keep falling across the same years (Malaysia, G. of., 2025). That pattern likely reflects a change in what people reach for in everyday situations, like paying bills online, transferring money through mobile banking, or using an app to pay at a store. Hence, more Malaysians appear to be relying on electronic channels for routine financial activities (Malaysia, G. of., 2025). As usage grows, service-quality issues now surface faster, but banks' measurement and response still lag behind real-time customer experience.

E-Money

Latest (Jul 2025)

545.4 million



Cheques

Latest (Jul 2025)

2.6 million

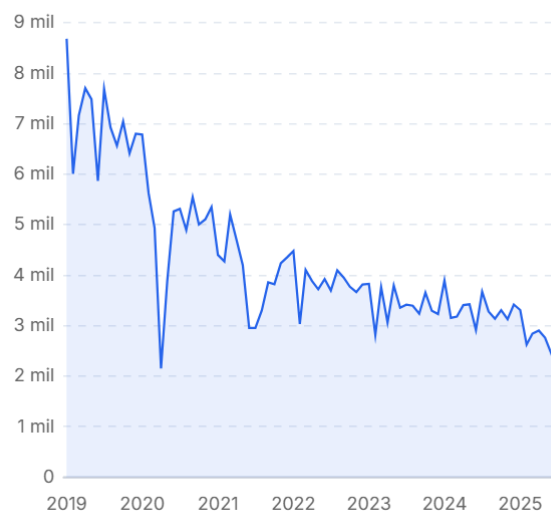


Figure 1.1: Trends in e-money and cheque transactions in Malaysia, 2019–2025 (Malaysia, G. of., 2025)

1.1 Background

Banks are moving routine services from branch counters to digital platforms, allowing customers to check balances, transfer funds and pay bills online within minutes (Bankuoru Egala et al., 2021). This change involves more than placing existing banking services within an application. Traditional processes must be redesigned into clear digital steps, while the system must remain stable, secure and accessible (Kumar, 2024). Consequently, customers increasingly evaluate banks through their digital experiences, including application speed, screen layout and the ease of locating and completing a task, rather than mainly through interactions with branch employees. Although fundamental banking functions such as deposits, payments and transfers remain unchanged, their continuous accessibility through mobile phones and computers has changed how customers interact with and perceive banking services.

Banking is also moving further towards a mobile-first environment because many customers now regard banking as a quick daily activity rather than a visit to a physical branch (Esmaeili et al., 2021). Users commonly access mobile applications to check account balances, transfer funds or pay bills, while online banking websites remain important for customers who use computers or prefer larger screens. At the same time, customer support has become increasingly digital through chatbots, help menus and automated guidance for common enquiries such as fee checks and password resets. However, customers may experience difficulty when dealing with more complex problems, particularly when digital support is slow, unclear or unable to provide suitable assistance. This highlights the importance of responsiveness, clear instructions and strong privacy and security safeguards within e-banking platforms.

As reliance on digital banking continues to increase, e-banking service quality has become an important factor influencing customer satisfaction. Users evaluate applications and websites based on whether transactions are completed accurately, services remain available, information can be accessed efficiently, the interface is easy to use and personal data are protected. These expectations are reflected in the seven service-quality dimensions examined in this project: Reliability, Responsiveness, Efficiency, Privacy and Security, Ease of Use, System Availability and Website Design. A stable, accessible and user-friendly experience may strengthen customer confidence and satisfaction, whereas slow performance, transaction failures, confusing navigation or ineffective support may weaken trust in the service (Bankuoru Egala et al., 2021). Therefore,

banks need to understand which dimensions are most strongly associated with customer satisfaction and present the findings in a form that supports clearer service-improvement decisions.

1.2 Problem Statement

Although Malaysia's financial sector has digitalised rapidly, the assessment of e-banking service quality often remains dependent on one-off surveys, separate feedback exercises and static reporting. Recent Malaysian research on e-banking satisfaction has similarly relied on single-instance online questionnaires, which provide useful evidence about customer perceptions at a particular point in time (Kah Boon Lim et al., 2023). However, the resulting findings are commonly presented through statistical tables or separate reports that may be difficult for non-technical stakeholders to interpret and apply directly.

As e-banking usage continues to increase, banks need a clearer way to consolidate customer feedback, identify the service-quality dimensions most strongly associated with customer satisfaction and examine whether weaker experiences are concentrated within particular customer segments. When service-quality findings are distributed across different reports, teams or systems, stakeholders may have limited visibility of the overall customer experience, making coordinated decision-making and improvement planning more difficult (Malc et al., 2023; Rowena, 2025). Statistical relationships alone may also be insufficient if they are not translated into clear priorities, responsible actions and measurable indicators.

Therefore, there is a need for an integrated and actionable decision-support structure that combines statistical analysis with interactive data visualisation. Such a structure should enable banking stakeholders to identify priority service-quality dimensions, compare customer segments, explore reported e-banking issues and connect the findings with practical improvement actions. It should also establish consistent baseline indicators that can support future comparisons when new customer-feedback data become available. By addressing these needs, the project aims to transform customer-perception data from a static research output into a more accessible and evidence-based resource for e-banking service-quality improvement (Nwanneka Ezechi et al., 2025).

1.3 Objectives

This project aims to evaluate how key e-banking service quality dimensions relate to customers' overall satisfaction using statistical testing. The main goal is to identify the strongest service quality drivers and translate the evidence into practical outputs that banks can use for decision support and service improvement. This project delivers three outputs that banks can use in practice. First, the analysis tests how each service quality dimension relates to satisfaction and reports the effect sizes, significance levels, and a simple ranking of the strongest drivers. Second, the findings come in a Power BI dashboard in which KPI panels, comparison views, and interactive filters allow people to explore patterns in more detail. Third, the project offers specific recommendations by linking actions to the highest-impact drivers. This approach enables different teams to implement the suggestions based upon their roles and priorities. Taken together, these outputs are a combination of evidence, clear visuals, and actionable guidance to support stronger service quality and higher customer satisfaction.

1. To examine the relationship between key e-banking service quality dimensions and customers' overall satisfaction using statistical analysis.
2. To develop an interactive Power BI dashboard that visualizes and tracks e-banking service quality and customer satisfaction results by dimension and segment.
3. To propose data-driven recommendations to improve e-banking service quality based on the empirical findings and dashboard insights.

1.4 Significance of Study

Many e-banking studies identify the service-quality factors associated with customer satisfaction, but the discussion often ends after reporting statistical relationships. As a result, banking stakeholders may understand which dimensions are related to satisfaction but still lack a clear way to interpret the findings, compare customer groups and translate the evidence into practical improvement priorities. This project addresses that gap by combining statistical analysis, interactive visualisation and data-driven recommendations.

From an academic perspective, this study provides evidence on the relationships between seven e-banking service-quality dimensions and customer satisfaction within the Malaysian context. The findings contribute to existing e-banking research by showing which dimensions are

significantly associated with satisfaction and which dimensions make unique contributions when considered together. Since customer expectations and digital-banking experiences may differ across countries, the Malaysian findings can also provide useful local evidence for future studies.

From a practical perspective, the project presents customer-feedback findings through an interactive Power BI dashboard. The dashboard allows banking stakeholders to review key service-quality and satisfaction indicators, compare dimensions, examine customer segments and explore reported e-banking issues more efficiently than through static statistical tables alone. This makes the results more accessible to non-technical users and helps them identify areas such as transaction reliability, system availability, responsiveness, ease of use and interface design that may require further attention.

The project also provides decision-support value by connecting important findings to proposed improvement actions, responsible stakeholder teams and measurable indicators. Instead of presenting general recommendations, the Evidence-to-Action approach shows why an area should be prioritised, which team should address it and which indicator can be reviewed in future assessments. Therefore, the project contributes not only by identifying service-quality relationships but also by providing a structured basis for prioritising and planning e-banking service improvements.

1.5 Scope of Project

This project examines e-banking service quality in Malaysia using a data-driven approach, focusing on how customers' experiences with digital banking relate to their overall satisfaction. Malaysia is a suitable setting because mobile banking and electronic payments are widely used for routine activities such as paying bills, transferring funds and checking account balances. High internet and smartphone usage also means that service problems can affect customers directly when an application slows down, becomes unavailable or fails to complete a transaction (Kemp, 2025; BERNAMA, 2025; InvestMalaysia, 2024). The analysis focuses on seven service-quality dimensions: Reliability, Responsiveness, Efficiency, Ease of Use, Privacy and Security, System Availability, and Website Design. These dimensions were drawn from established e-banking frameworks, including adapted E-SERVQUAL, E-S-QUAL and banking-specific measures, and

have been linked to customer-satisfaction outcomes in previous studies (Khatoon et al., 2020; Raza et al., 2020).

The project also develops an interactive Power BI dashboard to present the findings in a form that decision-makers can explore more easily. Power BI allows users to review key performance indicators, apply filters, compare service-quality dimensions and examine patterns across customer segments without requiring programming knowledge (Ahadi et al., 2024; Sethi, 2023; Zamil et al., 2024). The dashboard follows a decision-first flow, guiding users from an overall view of customer satisfaction to the dimensions, segments and reported issues that may require further attention. It visualises the indicators contained in the completed survey dataset and establishes baseline measures that may support future comparisons when new data become available.

The study covers individual retail customers who use e-banking services provided by banks operating in Malaysia. Participants include Malaysian and non-Malaysian residents who hold an account with a Malaysian bank and use its mobile application or online banking website for activities such as checking balances, transferring funds or paying bills. A total of 426 valid responses were analysed using a quantitative, cross-sectional approach. The structured questionnaire measured customers' perceptions of the seven service-quality dimensions, Customer Satisfaction and reported e-banking issues. The cleaned and summarised findings were subsequently presented through charts, KPI cards and interactive dashboard filters.

The project is limited to customer-perception and self-reported issue data collected at one point in time. It does not examine internal banking records, operational system performance, customer churn, profitability or backend system architecture. Therefore, the findings identify statistical relationships, relative service-quality weaknesses and reported customer concerns but do not establish causality or confirm the technical root causes of individual issues. Within these boundaries, the project provides a structured basis for comparing customer experiences, identifying improvement priorities and supporting evidence-based e-banking service-quality decisions.

1.6 Report Organization

This report is structured into five main chapters. Chapter 1, Introduction, presents the Malaysian e-banking context, problem statement, research objectives, significance of the study, project scope and overall report organisation. Chapter 2, Literature Review, reviews previous studies on e-banking, Malaysian e-banking, the seven service-quality dimensions—Reliability, Responsiveness, Efficiency, Privacy and Security, Ease of Use, System Availability and Website Design—Customer Satisfaction, statistical analysis methods, dashboard development and Evidence-to-Action mapping. It also identifies the research gaps that guide the direction of this project.

Chapter 3, Methodology, explains the three-phase approach adopted in the project. Phase 1 covers survey data collection, preprocessing, reliability analysis, descriptive statistics, Pearson correlation and multiple linear regression. Phase 2 explains the development of the interactive Power BI dashboard, including KPI design, ETL procedures, data modelling, dashboard use cases and interaction design. Phase 3 describes how the statistical findings and dashboard insights are translated into prioritised, measurable and team-owned recommendations through an Evidence-to-Action mapping process.

Chapter 4, Results and Discussion, presents and interprets the findings according to the three project phases and research objectives. It reports the respondent profile, reliability results, descriptive statistics, correlation and regression findings, dashboard outputs, issue patterns, customer-segment comparisons and prioritised recommendations. The findings are also compared with previous studies, and their practical implications are discussed.

Chapter 5, Conclusion and Future Work, summarises the main findings, evaluates the achievement of the three research objectives and explains the academic and practical contributions of the project. It also discusses the study's limitations and proposes directions for future research, dashboard development and recommendation evaluation. After the five main chapters, the report includes the Work plan, References and supporting appendices.

1.7 Summary

Malaysia's growing reliance on e-banking has increased the importance of understanding how digital service quality relates to Customer Satisfaction. However, customer-feedback findings are often presented through separate statistical reports, making it difficult for banking stakeholders to

identify priority service-quality dimensions, compare customer groups and translate the evidence into practical actions. Therefore, this project has three objectives: to examine the relationships between seven e-banking service-quality dimensions and Customer Satisfaction, to develop an interactive Power BI dashboard for visualising and comparing the findings, and to propose data-driven recommendations based on the statistical and dashboard results. The project follows a three-phase approach in which survey data are first analysed using statistical methods, the findings are then presented through an interactive dashboard, and the combined evidence is finally converted into prioritised recommendations linked to responsible teams and measurable indicators. Through this approach, the project is expected to contribute Malaysian evidence on e-banking service quality and provide a practical decision-support structure for identifying, prioritising and planning service-quality improvements.

Chapter 2. Literature Review

2.0 Overview

This chapter reviews findings from earlier investigations into e-banking service quality and customer satisfaction, taking Malaysia as the primary case. It initially describes e-banking as a digital service channel and then defines the expectations of customers, including convenient access, system reliability, and ease of use on mobile applications. Next, this chapter discusses how e-banking grew alongside broader nationwide payment initiatives and the increase in smartphone ownership. This is done to illustrate how technological change and institutional adaptations facilitated the growth of online banking in the day-to-day life of citizens. At the same time, the chapter examines factors particular to the Malaysian context that affect its environment, including the coexistence of traditional and Islamic banking and the advent of new digital banks. Finally, the chapter discusses how service quality has been defined and measured in earlier research, and how these measures have been linked to customer satisfaction through prior studies. It also looks at how the literature describes dashboards and other data-driven tools as ways to monitor service performance on an ongoing basis. Finally, the chapter points out the main gaps that remain in this work and explains how these gaps guide the direction of the present project.

This review was designed to cover both classic and recent work on e-banking service quality. The sources were found through keyword searches on Google Scholar and other academic databases often used in business and information systems studies. Keywords were combined in different ways, including “*e-banking service quality*,” “*E-SERVQUAL*,” “*E-S-QUAL*,” “*digital banking*,” “*customer satisfaction*,” “*Malaysia*,” “*Power BI dashboard*,” “*business intelligence*,” and “*evidence-to-action*.” The review mainly focuses on peer-reviewed journal articles, especially studies from 2020 to 2025 that measure service-quality dimensions and examine how they relate to customer satisfaction. Industry reports and news were also included when they help explain what happens in real settings in Malaysia, such as how banks collect feedback and how service-quality signals move across different teams.

2.1 Concept, Functions, and Customer Experience of E-Banking

Banking has gradually moved from counters and paper forms to screens and self-service menus. In this project, e-banking refers to completing routine banking tasks through a bank's website or mobile app rather than visiting a physical branch. Many scholars describe e-banking services in three layers. The first layer is informational and gives secure access to the basics like account balances and statements (Gaviyau & Godi, 2025). The second layer is transactional and enables users to take actions like fund transfers, paying bills, or phone top-ups. The third layer is interactive and helps users if problems occur. The support can be provided through live chat and an AI chatbot that can answer common questions and guide the user through the process step by step during the time of using digital tools (Manser Payne et al., 2021).

When it comes to user experience, e-banking users assess it by how well it works, but they also want it to be as seamless and intuitive as possible. Users expect quick transfer and balance updates as they conduct everyday business and they are also quick to respond to customer enquiries (Laws, 2023; Ranjan, 2024). Helpful, relevant messages about when a payment is due or how much money has been spent can make the e-banking experience smoother, but users can become annoyed by intrusive or unnecessary notifications or messages that do not add valuable value. Personalisation that is helpful and not intrusive can help in retaining users in the long term (Ayinaddis et al., 2023). Moreover, trust is shaped by how directly a bank explains how it handles data. Users will expect a clear option of consensual actions and simple explanations when privacy rules are tightening. The whole digital experience will be about quick functioning, relevant assistance, and the clear way of handling individual details.

E-banking is rarely a smooth experience. Long loading times and timeouts, particularly during peak hours, can leave users feeling uncertain and mistrustful of the platform (Rosdiana Sijabat, 2024). The existence of usability issues can also add a layer of dissatisfaction. Waluyo et al. (2025) indicate unclear layouts, buttons that are overly similar, and error messages without explanations, leading users into trial and error. The experience can feel worse when the desktop site and the mobile app behave differently, because users usually expect the same steps and visual cues across devices. Over time, these delays and design issues may weaken trust and lower satisfaction, even if the core functions are technically available. These recurring problems suggest that technical teams and interface designers need to coordinate closely so that platforms remain stable, predictable, and straightforward to use.

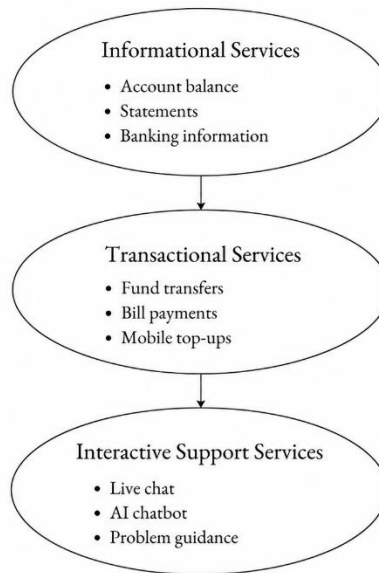


Figure 2.1: Three Layers of E-Banking Services

Figure 2.1 illustrates the three main layers of e-banking services. Informational services provide customers with access to balances, statements and banking information, while transactional services allow activities such as fund transfers, bill payments and mobile top-ups. Interactive support services assist customers through live chat, AI chatbots and problem guidance.

2.2 Development and Current Landscape of E-Banking in Malaysia

E-banking in Malaysia developed significantly after Bank Negara Malaysia authorised internet banking around 2000 and major financial institutions introduced online banking platforms. These platforms allowed customers to perform routine transactions, including checking account balances, transferring funds and paying bills, without visiting a physical branch. The subsequent growth of smartphone ownership further shifted e-banking towards mobile applications. National payment infrastructure, including FPX and DuitNow, also supported faster and more convenient electronic transactions. DuitNow, introduced in 2018, enabled customers to transfer funds using identifiers such as mobile-phone numbers and national identification numbers, making digital payments more accessible for everyday use (Madihah Jamian, 2025).

Recent patterns of internet and mobile usage demonstrate the increasing importance of digital banking in Malaysia. By 2025, approximately 97.7% of the Malaysian population used the

internet, while the number of mobile connections exceeded the total population (Kemp, 2025). BERNAMA (2025) reported that approximately 62% of online banking activity in 2024 was conducted through mobile applications. Electronic-payment transactions also increased by 19% in 2024, reaching approximately 409 transactions per person (InvestMalaysia, 2024). Banking applications such as MAE, CIMB OCTO, MyPB, HLB Connect and Be U now provide services beyond basic account management, including QR payments, investment access, bill settlement and other financial activities. These developments demonstrate that digital banking has become an important part of the daily financial activities of Malaysian customers.

Malaysia's dual banking system also influences the development of e-banking services. Conventional and Islamic banks generally provide similar digital functions, although Islamic banks may offer additional Shariah-compliant services such as zakat and waqf payments (Meerangani et al., 2022; Mohsin et al., 2022). At the same time, customers do not have the same level of digital confidence or access. Older users, rural customers and individuals with lower digital familiarity may experience greater difficulty when completing online transactions or adapting to security procedures such as device binding and cooling-off periods. Although these controls strengthen security, they may also introduce additional steps that affect convenience and ease of use (Khamaj & Ali, 2024). Banks must therefore balance security requirements with accessibility and usability.

The emergence of licensed digital banks, including GXBank, AEON Bank, Boost Bank, KAF Digital Bank and Ryt Bank, has further increased the importance of mobile-based financial services in Malaysia (Abdullah, 2025). These institutions operate primarily through digital platforms and use technologies such as electronic Know Your Customer processes and cloud-based systems to deliver banking services. Their development increases competition within the banking sector and places greater emphasis on the quality of digital customer experiences.

As customers increasingly depend on mobile applications and online banking websites, weaknesses in digital service delivery can directly affect their ability to complete important financial activities. Transaction failures, delayed confirmations, system interruptions, confusing navigation and security concerns may reduce customers' confidence in a bank's digital services. Therefore, the growth of e-banking in Malaysia creates a need to examine perceived service quality systematically and identify the dimensions that are most strongly related to customer satisfaction.

Banks in Malaysia



Figure 2.2: Selected Conventional, Islamic, Foreign and Digital Banks Operating in Malaysia

Figure 2.2 presents examples of major banks operating in Malaysia, including local conventional banks, Islamic banks, foreign banks and newly licensed digital banks. The variety of institutions reflects Malaysia's diverse banking landscape and the increasing competition in providing mobile and online banking services.

2.3 Conceptualisation and Measurement of E-Banking Service Quality

Service quality was originally explained through SERVQUAL, which evaluates areas such as reliability, assurance, tangibles, empathy and responsiveness (Parasuraman et al., 1988; Hadid et al., 2020). The traditional SERVQUAL approach commonly examines differences between what customers expect and what they perceive that they receive. However, physical service-quality measures do not fully represent e-banking because customers interact mainly with applications, websites, automated processes and digital-support channels rather than branch employees.

For this reason, researchers have adapted SERVQUAL into digital service-quality frameworks such as E-SERVQUAL and E-S-QUAL. These frameworks place greater emphasis on digital-service characteristics, including efficiency, system availability, fulfilment, privacy, security and interface usability (Raza et al., 2020; Lama & Mijala Kayestha, 2025). Banking-specific measures have also incorporated factors such as reliability, responsiveness, access and service variety to reflect the characteristics of financial services delivered through digital channels (Bankuoru Egala et al., 2021).

The present project adopts relevant dimensions from E-SERVQUAL, E-S-QUAL and banking-specific service-quality studies. However, it uses a performance-only perception approach rather than a traditional SERVQUAL gap analysis. Respondents rate their perceptions of their bank's e-banking performance, but they do not provide separate expectation and performance scores. Therefore, this project does not calculate an expectation–performance service-quality gap. Instead, perceived e-banking service quality refers to customers' evaluations of how reliably, efficiently, securely and conveniently their bank's e-banking services perform.

Table 2.1: Summary of e-banking service quality dimensions used in prior studies

Author (Year)	Title	Model Used	Service Quality Elements
Hadid et al. (2020)	The Effect of Digital Banking Service Quality on Customer Satisfaction: A Case Study on the Malaysian Banks	Digital Banking Service Quality (DBSQ)	reliability; tangibility; responsiveness; assurance; empathy
Khatoon et al. (2020)	The Mediating Effect of Customer Satisfaction on the Relationship Between Electronic Banking Service Quality and Customer Purchase Intention: Evidence From the Qatar Banking Sector	Modified E-SERVQUAL	reliability; efficiency; responsiveness and communication; security and privacy
Raza et al. (2020)	Internet banking service quality, e-customer satisfaction and loyalty: the modified e-SERVQUAL model	Modified E-SERVQUAL	responsiveness; reliability; user friendliness; efficiency; personal need; site organisation
Bankuoru Egala et al. (2021)	To leave or retain? An interplay between quality digital banking services and customer satisfaction	E-S-QUAL + BSQ	ease of use; efficiency; privacy/security; reliability; interoperability; responsiveness; service charges; service portfolios
Ighomereho et al. (2022)	From Service Quality to E-Service Quality: Measurement, Dimensions and Model	Modified E-SERVQUAL	website appearance; ease of use; reliability; security; personalization; fulfillment; responsiveness

Author (Year)	Title	Model Used	Service Quality Elements
Serrah & Binti Haja Maideen (2022)	E-Banking Services and Customer Satisfaction: Focus on a Private Bank in Malaysia	Modified E-SERVQUAL	Security & privacy; reliability; system availability; website design
Ayinaddis et al. (2023)	Examining the effect of electronic banking service quality on customer satisfaction and loyalty: an implication for technological innovation	Modified E-SERVQUAL	responsiveness; reliability; security and privacy; speed; convenience; system availability; easiness; service charge
Kah Boon Lim et al. (2023)	Customer Satisfaction on E-Banking Service Quality in Malaysia	Modified E-SERVQUAL	efficiency; website design; security; reliability; responsiveness
Waluyo et al. (2025)	Driving Factors Influencing Customer Satisfaction in Digital Banking Service	Extended TAM	ease of use; usefulness; security; reliability
Lama & Mijala Kayestha (2025)	From Swipe to Stay: How Digital Banking Service Quality Drives Customer Retention through Customer Satisfaction	Modified E-S-QUAL	ease of use; efficiency; interoperability; privacy/security; responsiveness
Nguyen (2025)	E-Banking Service Quality and Customer Satisfaction with Moderator Factor	Modified E-SERVQUAL	transaction speed; efficiency; reliability; responsiveness; confidentiality

Author (Year)	Title	Model Used	Service Quality Elements
Malic et al. (2023)	The Development of E-Banking Services Quality Measurement Instrument: MPQe-BS	MPQe-BS	Efficiency; system availability; Contact; Design/aesthetics/guide; privacy and Security; Fulfillment
Ul Haq & Awan (2020)	Impact of e-banking service quality on e-loyalty in pandemic times through interplay of e-satisfaction	Modified E-SERVQUAL	Reliability; Privacy & Security; Website Design; Customer Service & Support

Based on the dimensions identified in Table 2.1, this project examines Reliability, Responsiveness, Efficiency, Privacy and Security, Ease of Use, System Availability and Website Design. These dimensions represent the functional, security-related and user-interface characteristics that customers encounter when using e-banking. Any dashboard indicator calculated by subtracting a dimension mean from the maximum score of five should be described as the distance from the maximum possible score, rather than as a SERVQUAL expectation–performance gap.

2.4 E-Banking Service Quality Dimensions and Their Relationships with Customer Satisfaction

Perceived e-banking service quality is examined through separate dimensions because customers evaluate several aspects of a digital-banking experience at the same time. A platform may, for example, process transactions accurately but remain difficult to navigate, or it may provide an attractive interface while responding slowly when a customer experiences a problem. Separating perceived service quality into dimensions therefore supports clearer questionnaire measurement and helps identify which areas are associated with customer satisfaction (Raza et al., 2020; Ighomereho et al., 2022).

The following subsections synthesise previous findings concerning the seven service-quality dimensions adopted in this project. Each subsection defines the dimension, identifies the general pattern in previous findings, considers conflicting evidence and explains its relevance to the relationship tested in the present study.

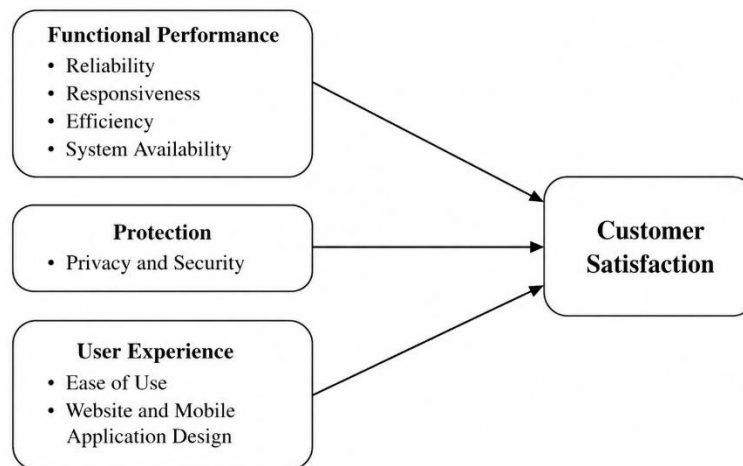


Figure 2.3: Literature-Based Relationships between E-Banking Service-Quality Dimensions and Customer Satisfaction

Figure 2.3 groups the seven e-banking service-quality dimensions into functional performance, protection and user experience. The arrows indicate that these three areas are expected, based on the reviewed literature, to be positively related to customer satisfaction. Functional performance includes reliability, responsiveness, efficiency and system availability,

while protection covers privacy and security, and user experience includes ease of use and website or mobile application design.

2.4.1 Reliability and Customer Satisfaction

Reliability refers to the ability of an e-banking platform to provide promised services accurately, consistently and within the expected time. It includes correct transaction processing, accurate account information, dependable records and clear confirmation that an activity has been completed successfully. When customers believe that their transfers, payments and other transactions are processed correctly, they are more likely to view the platform as dependable and trustworthy (Hadid et al., 2020; Nguyen, 2025).

Previous research generally indicates that reliability has a positive relationship with customer satisfaction. Studies conducted in Malaysia and other banking environments have found that accurate and dependable digital services support more favourable customer evaluations (Khatoon et al., 2020; Raza et al., 2020; Hadid et al., 2020; Kah Boon Lim et al., 2023). Reliability has also been associated indirectly with continued usage and loyalty through its contribution to customer satisfaction (Ul Haq & Awan, 2020; Bankuoru Egala et al., 2021). These findings suggest that customers consider dependable transaction processing to be a fundamental requirement of e-banking.

However, the strength of this relationship varies across studies. Reliability may overlap with related dimensions such as Efficiency, System Availability and Website Design because accurate performance also depends on whether the platform remains accessible and communicates transaction progress clearly. Waluyo et al. (2025) further suggest that clear interface feedback may reduce customers' concerns during minor service delays. A customer may tolerate a short delay when the platform clearly displays whether the transaction is pending, successful or unsuccessful.

Overall, previous research supports the importance of reliability, although its unique contribution may depend on how it interacts with other service-quality dimensions. Because financial transactions require accuracy and consistency, Reliability is expected to have a positive relationship with Customer Satisfaction in the present project.

2.4.2 Responsiveness and Customer Satisfaction

Responsiveness refers to the speed, clarity and usefulness of support provided when customers request assistance or encounter a service problem. It includes prompt responses to enquiries, clear instructions, appropriate follow-up and effective resolution of failed or delayed transactions. Responsiveness therefore concerns not only how quickly a response is provided but also whether the response helps the customer understand and resolve the problem (Hadid et al., 2020; Ighomereho et al., 2022; Kah Boon Lim et al., 2023).

Previous findings regarding responsiveness are mixed. Several studies report a positive relationship between responsiveness and customer satisfaction. Customers tend to evaluate e-banking more favourably when banks reply promptly, provide accurate information and respond appropriately when transactions fail or other difficulties occur (Khatoon et al., 2020; Raza et al., 2020; Ayinaddis et al., 2023; Kah Boon Lim et al., 2023). Nguyen (2025) similarly suggests that responsiveness may contribute to satisfaction indirectly by strengthening customers' overall perceptions of service quality.

Other studies have found that responsiveness is insignificant or negatively related to satisfaction or perceived e-service quality (Hadid et al., 2020; Bankuoru Egala et al., 2021; Ighomereho et al., 2022; Lama & Mijala Kayestha, 2025). These differences may reflect variations in the support channels available, customers' expectations and the effectiveness of digital service recovery. An automated response may be fast but still be judged negatively when it does not answer the customer's question or resolve the underlying issue. Some customers may also rely on telephone or branch support for complicated problems, reducing the influence of support provided through the digital platform itself.

Therefore, previous evidence does not support a universal conclusion that responsiveness always increases or decreases satisfaction. Its effect appears to depend on both response speed and response quality. Nevertheless, because timely and useful assistance can reduce uncertainty when e-banking problems occur, Responsiveness is expected to have a positive relationship with Customer Satisfaction in the present project.

2.4.3 Efficiency and Customer Satisfaction

Efficiency refers to the extent to which customers can complete e-banking activities quickly and with minimal effort. It includes platform speed, logical navigation, clear information, simple transaction steps and the ability to locate required functions without unnecessary difficulty. Efficient e-banking services reduce the time and effort required to complete activities such as fund transfers, bill payments and account enquiries (Malc et al., 2023; Nguyen, 2025).

Most previous studies report a positive relationship between efficiency and customer satisfaction. Efficient websites and applications may improve satisfaction by reducing transaction time, unnecessary steps and interruptions (Khatoon et al., 2020; Raza et al., 2020). Several studies also identify Efficiency as one of the strongest service-quality dimensions affecting satisfaction, perceived quality or retention intentions (Bankuoru Egala et al., 2021; Malc et al., 2023; Lama & Mijala Kayestha, 2025; Nguyen, 2025). These findings indicate that customers value digital channels partly because they expect them to provide a faster and more convenient alternative to branch banking.

However, Kah Boon Lim et al. (2023) found that Efficiency did not have a significant relationship with customer satisfaction in their Malaysian sample. Differences between studies may result from respondent demographics, the maturity of the e-banking environment or the extent to which efficient performance has become a basic customer expectation. When most banking applications already provide similar transaction speeds, customers may pay greater attention to other areas such as reliability, security or support.

Despite this mixed evidence, efficiency remains closely connected to the main purpose of e-banking: enabling customers to complete financial tasks conveniently and quickly. Efficiency is therefore expected to have a positive relationship with Customer Satisfaction in the present project.

2.4.4 Privacy & Security and Customer Satisfaction

Privacy and Security refer to customers' perceptions that their personal information, financial details and online transactions are protected from fraud, theft, unauthorised access and misuse. This dimension includes confidentiality, secure authentication, protection of transaction information and confidence that the bank handles customer data responsibly (Ighomereho et al., 2022; Serrah & Binti Haja Maideen, 2022; Kah Boon Lim et al., 2023).

Most previous studies report a positive relationship between Privacy and Security and customer satisfaction. Customers who feel confident that their information and transactions are protected tend to evaluate e-banking services more favourably (Khatoon et al., 2020; Bankuoru Egala et al., 2021; Ul Haq & Awan, 2020; Serrah & Binti Haja Maideen, 2022). Malaysian evidence similarly suggests that effective protection and responsible data handling support customer satisfaction and continued e-banking usage (Kah Boon Lim et al., 2023). Privacy and Security may be especially important in financial services because the consequences of fraud or unauthorised access can be serious.

Nevertheless, some studies report that Privacy and Security do not have a significant direct relationship with satisfaction. Ayinaddis et al. (2023) found different results across bivariate and multivariate analyses, while Waluyo et al. (2025) found that user experience moderated the relationship between perceived security and satisfaction. These differences suggest that security may influence customers indirectly or may be treated as a basic requirement rather than a feature that independently increases satisfaction. Security measures may also reduce convenience when they introduce complicated authentication procedures or additional transaction steps.

Overall, customers are unlikely to be satisfied with an e-banking service that they consider unsafe. However, the strength of the direct relationship may depend on how security measures are implemented and communicated. Privacy and Security are therefore expected to have a positive relationship with Customer Satisfaction in the present project.

2.4.5 Ease of Use and Customer Satisfaction

Ease of Use refers to how easily customers can understand, navigate and operate an e-banking platform. It includes clear instructions, intuitive menus, understandable labels and transaction processes that can be completed without extensive guidance. An easy-to-use platform reduces customers' cognitive effort and helps them complete banking activities with fewer errors (Ighomereho et al., 2022).

Previous studies generally suggest that Ease of Use supports customer satisfaction, adoption and continued usage. User-friendly interfaces can help customers learn digital processes more quickly, complete activities with less effort and feel more confident when using e-banking services (Izni et al., 2022; Bankuoru Egala et al., 2021; Lama & Mijala Kayestha, 2025). These

findings are consistent with the Technology Acceptance Model, which identifies perceived ease of use as an important explanation of technology acceptance.

However, some studies found that Ease of Use did not have a significant direct relationship with customer satisfaction (Ayinaddis et al., 2023; Waluyo et al., 2025). One possible explanation is that experienced users may already consider basic usability to be a standard feature of digital banking. In mature e-banking environments, ease of use may influence satisfaction indirectly through efficiency, user experience or perceived usefulness rather than as an independent factor.

The importance of Ease of Use may also differ between customer groups. Older users and customers with lower digital familiarity may place greater importance on simple navigation and clear guidance than frequent or experienced users. Because understandable and manageable transaction processes can reduce customer effort and confusion, Ease of Use is expected to have a positive relationship with Customer Satisfaction in the present project.

2.4.6 System Availability and Customer Satisfaction

System Availability refers to whether customers can access and use a bank's website or mobile application when required, without frequent interruptions, crashes or unresponsive functions. It reflects the stability and accessibility of the digital platform during routine and time-sensitive financial activities (Serrah & Binti Haja Maideen, 2022; Malc et al., 2023).

Previous evidence generally indicates a positive relationship between System Availability and customer satisfaction. Malaysian research has shown that continuous access to digital-banking services supports more favourable customer evaluations, particularly because customers expect banking applications and websites to be available beyond normal branch operating hours (Serrah & Binti Haja Maideen, 2022). Malc et al. (2023) similarly identify availability as part of the wider relationship between e-banking service quality and customer satisfaction. When services remain accessible, customers can complete urgent transfers, payments and other transactions with less uncertainty.

Some findings are less consistent. Ayinaddis et al. (2023) reported a significant association between availability and satisfaction in part of their analysis, although the reported regression findings were not fully consistent across the study. Such variation may result from differences in the measurement of availability, the frequency of service interruptions or customers' tolerance of occasional downtime.

Nevertheless, digital banking is valuable partly because customers expect to access it whenever necessary. Interruptions during important transactions may therefore have an immediate negative effect on the customer experience. System Availability is consequently expected to have a positive relationship with Customer Satisfaction in the present project.

2.4.7 Website Design and Customer Satisfaction

Website Design refers to the appearance, organisation and presentation of the e-banking interface. Although the formal construct is named Website Design, it applies to both online banking websites and mobile-banking application interfaces in this project. It includes layout, menus, fonts, colours, icons, visual hierarchy and the organisation of information required to navigate and complete transactions (Ighomereho et al., 2022; Kah Boon Lim et al., 2023).

Most previous studies report a positive relationship between Website Design and customer satisfaction. Clear layouts, consistent visual elements and easily accessible information can help customers understand the platform and complete tasks more confidently (Serrah & Binti Haja Maideen, 2022; Kah Boon Lim et al., 2023). Website Design has also been linked to e-banking satisfaction and loyalty because the interface represents the main point of interaction between the bank and its digital customers (Ul Haq & Awan, 2020; Malc et al., 2023).

However, Ighomereho et al. (2022) found that website appearance had a negative and statistically insignificant relationship with overall perceived e-service quality. This finding suggests that attractive visual elements may have limited value when they do not improve functionality. Customers may prefer a simple interface that allows them to complete transactions efficiently rather than a visually complex design that introduces distraction or confusion.

Differences across studies may therefore result from how Website Design is defined. Some studies emphasise visual attractiveness, while others include navigation, information organisation and functional clarity. In the present project, Website Design includes both appearance and interface organisation. Because clear and professional design can support confidence, navigation and task completion, Website Design is expected to have a positive relationship with Customer Satisfaction.

2.5 Concept, Theoretical Foundations, and Importance of Customer Satisfaction in E-Banking

Customer satisfaction in e-banking refers to how users judge their overall experience after using a bank's website or mobile application. It shows whether customers feel that the digital service suits their needs well enough for them to keep using it over time (Bankuoru Egala et al., 2021). Satisfaction matters because it often shapes the next decision. When people are satisfied, they are more likely to continue using the same bank for everyday tasks, including transferring funds and paying bills (Lama & Mijala Kayestha, 2025). On the other hand, disenchantment with service can result in direct expenses to banks. Some customers go so far as to uninstall the application, others to share negative reviews on social media or online review sites. A significant percentage may never lodge a complaint at all but will move routine transactions to competing banks without even noticing. Because the switching costs in digital banking are low by most standards, customers can download and install a different application within the same short time horizon, making customer retention more vulnerable. Negative reviews will also stay on online listing sites for weeks or even months and mold prospective users' views of the service before they attempt to use it. In addition, customer satisfaction cannot be considered merely as a distinct assessment, but also as a determinant of tenure and digital reputation for a particular bank.

The theory of satisfaction in e-banking can be explained through the usual two common theories. The first is the expectation–confirmation Theory (ECT). This theory states that whether someone is satisfied depends on the match between their real experience and the one expected in the first place. This matters for digital banking as expectations can fluctuate over time while users build more experience with an app and compare an app's offerings across banks (Ayinaddis et al., 2023). The second common theory is the Technology Acceptance Model (TAM). The TAM proposes continued use based on perceived usefulness and ease of use. In practice, this can occur when an app offers clear layout, simple steps, and saves time through quick payments, saved favourites, or repeat transfer templates (Rahi et al., 2020). Further versions of TAM have also incorporated the perception of risk. This means that if someone is worried about fraud or data leaking or any type of security, this can reduce satisfaction despite the app's overall good performance (Esmaeili et al., 2021). Overall, the two theories suggest that satisfaction is not only influenced by the system performing well but also through the fluctuation of expectations, usability, and security trust.

Satisfaction with e-banking is based on varying factors than with branch banking. In a branch, customer perception often hinges on staff attitude, how clearly staff explain information, and the overall atmosphere. The core of a system in a smartphone app is certainly the experience. Users will be more satisfied when they are able to handle common tasks in the app like checking balances, moving payments, or paying bills conveniently, without delays, or points of failure. In times when there are problems, good instructions on screen and support are easily available and become increasingly important. It can also foster feelings of confidence, since the issue is immediately resolved, there is no additional time, effort, or stress involved (Ighomereho et al., 2022; Runsewe et al., 2024). That means that digital satisfaction does not depend as much on interpersonal connections and relies more on a consistent framework that is convenient to use. Ranjan (2024) introduces an applicable perspective here. Clients who blend digital and branch channels can experience boosted satisfaction on the whole, due to their capability to transact everyday tasks exclusively via the app, and to depend on representatives in the case of complicated or sensitive matters. Banks that manage both channels in a consistent way often face fewer complaints and may keep customers for longer periods. This pattern supports the need for dashboard-based monitoring discussed in Section 2.7, so that banks can track satisfaction-related indicators across channels and respond before small problems become larger trust issues.

2.6 Data-Driven Approaches for Examining E-Banking Service Quality and Customer Satisfaction

Previous e-banking studies apply a range of statistical techniques depending on their research objectives, dataset structure, and modelling needs. This section reviews commonly used methods in literature. For this project, Objective 1 focuses on Pearson correlation analysis and multiple linear regression because the project uses a cross-sectional survey design and these techniques are sufficient to test how service quality dimensions relate to overall satisfaction.

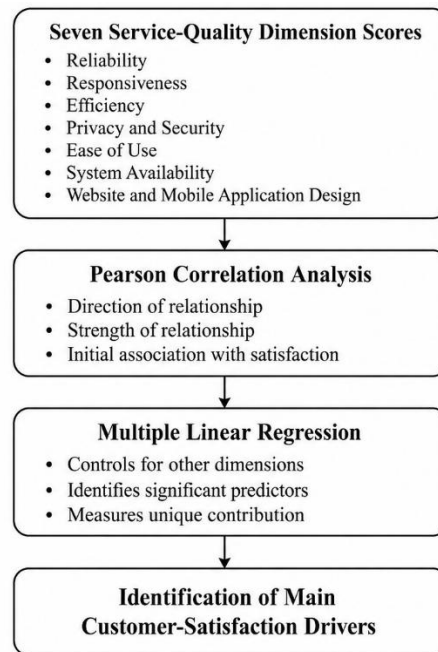


Figure 2.4: Statistical Analysis Workflow for Identifying E-Banking Customer-Satisfaction Drivers

Figure 2.4 illustrates the statistical analysis process used to examine the relationships between the seven e-banking service-quality dimensions and customer satisfaction. Pearson correlation analysis first assesses the direction and strength of each relationship, while multiple linear regression determines the significant predictors and unique contribution of each dimension. The process ultimately identifies the main service-quality drivers of customer satisfaction.

2.6.1 Pearson Correlation Analysis of Service Quality and Customer Satisfaction

Correlation analysis is a statistical method used to examine the direction and strength of the relationship between two variables. In e-banking service quality research, correlation analysis is

commonly applied to identify whether service quality dimensions are associated with customer outcomes such as satisfaction, loyalty, retention, and perceived security. The correlation coefficient shows whether the relationship is positive or negative, and whether the association is weak, moderate, or strong. Correlation analysis is useful because it provides an initial understanding of how variables move together before more complex analysis is conducted. For example, Dahiya and Singh (2024) used correlation analysis to examine whether loyalty, retention, and satisfaction move in the same direction. This helped provide a clearer foundation for further statistical analysis. However, correlation analysis has limitations. It only measures association and does not prove cause-and-effect relationships. In addition, unusual or extreme data values may affect the strength of the correlation. In this project, correlation analysis is used to examine the initial relationships between the seven e-banking service quality dimensions and customer satisfaction. It also helps identify which dimensions have stronger or weaker associations with satisfaction before conducting multiple linear regression.

2.6.2 Multiple Linear Regression for Identifying Customer Satisfaction Drivers

Compared with correlation analysis, multiple linear regression provides a deeper understanding because it estimates the unique contribution of each service quality dimension while controlling for the effects of the other dimensions. This allows the researcher to identify which dimensions remain statistically significant when all predictors are considered together. The model also provides overall evaluation indicators such as R^2 , adjusted R^2 , and the F-test. For example, Ighomereho et al. (2022) found that several e-service quality dimensions were strongly related to overall e-service quality. Their regression model was statistically significant, with $R^2 = 0.958$, suggesting that the selected dimensions explained a large proportion of variation in the outcome variable. Their results also showed that some dimensions remained significant predictors, while others became non-significant after all variables were entered into the same model. This shows the usefulness of multiple linear regression because a variable may appear important in correlation analysis but may not remain significant once overlap with other predictors is controlled. In this project, multiple linear regression is used to identify which e-banking service quality dimensions have the strongest unique relationship with customer satisfaction.

2.7 Business Intelligence Dashboards for Monitoring E-Banking Service Quality

Dashboards are now a common support tool in environments where performance signals shift quickly, but published work tends to pay more attention to financial or operational indicators than to customer-perceived e-banking service quality. Across different sectors, dashboards are usually described as compact visual displays that pull key performance indicators (KPIs) into one place so managers can spot patterns and react more quickly (Ahadi et al., 2024; Sethi, 2023). In banking and call-centre settings, for example, dashboards often track waiting times, ticket volumes, satisfaction scores, and system errors instead of relying only on static monthly reports. Because many of these tools are web-based, different teams and branches can refer to the same, frequently updated information. Reviews in the finance and BI literature suggest that when dashboards are built on integrated data infrastructures, they can support monitoring, forecasting, risk management, and regulatory reporting (Zamil et al., 2024). However, their usefulness still depends on whether the chosen KPIs reflect real business goals, whether data governance practices keep the figures reliable, and whether users actually understand and trust the visualisations they see (Ahadi et al., 2024; Sethi, 2023; Zamil et al., 2024).

Banking case studies illustrate typical dashboard applications. Tantipisut and Yenradee (2020) design a performance dashboard for a Thai bank that links KPIs to Balanced Scorecard perspectives and allows users to drill down from corporate-level results to individual staff performance. Rachansa and Meditya (2024) describe a Power BI dashboard developed for a Japanese bank after a merger, which brings together information on funding, loans, foreign exchange, trade finance, and fee income for portfolio review. Overall, these studies suggest that dashboards are useful because they bring many indicators into one place and make it easier to run regular review meetings. But the same studies also show a clear pattern in what most dashboards emphasise. Many designs mainly track operational and financial outcomes, such as efficiency, costs, or performance targets. In contrast, service-quality constructs are less visible, especially measures that reflect how customers perceive e-banking service quality. As a result, customer-based views of service quality often receive less direct focus in dashboard design.

The design of the dashboard and its layout is demonstrated to affect users' reading of data and their action-taking based on the insight gained. As stated by Ahadi et al. (2024), clear layout and simple visuals in a dashboard allow users to quickly scan for key indicators, whereas a

cluttered screen can be a distraction and hinder the user from noticing early Warning signs. Hjelle et al. (2024) add that design is not necessarily about how a dashboard looks. Users must also be able to trust the data. Freshness, completeness, and accuracy are some of the fundamental requirements for a good decision-making purpose. This can be even more concerning when considering that problems can begin even before a dashboard is launched. Some examples include choosing the wrong KPIs, weak integration of relevant systems, limited training of users, and poor governance. Moreover, in the longer term, this can result in poorer data quality and trust in the dashboard (Ahadi et al., 2024; Hjelle et al., 2024; Sethi, 2023). Indeed, an effective dashboard needs both clear design and strong data foundations to support user confidence.

Business Intelligence (BI) extends beyond the dashboard screen. The process is also dependent on the design of the data structure and the backend of the dashboard that, in turn, supports it. Gonçalves et al. (2023) describe a dashboard that adopts the Vercellis approach to Power BI design, which uses ETL processes to retrieve, sanitize and load data into a star-schema data warehouse. There, KPI tracking becomes more interactive, uniform, and easily discoverable. A similar backend configuration can also be applied to e-banking dashboards, particularly when the desired goals are stable filtering, reliable drill-down, and consistent totals across views. When a clean data pipeline is combined with clear visuals, banks can create dashboards that feel well designed and carry the same type of insights that users can trust to make decisions. In that setting, the dashboard can display customer indicators such as CSAT scores and complaint counts together with operational measures such as response time, and it can break the results down by channel or customer segment (Gonçalves et al., 2023; Zamil et al., 2024). Overall, the work reviewed tends to agree that dashboards make analysis faster and make key indicators easier to access. However, the papers also keep returning to the same obstacles, including messy data integration across systems, uneven adoption by users, and weaker governance or analytical skills that reduce the dashboard's long-term value (Gonçalves et al., 2023; Zamil et al., 2024). These points support the case for using dashboards to monitor e-banking service quality, especially when the goal is continuous oversight rather than one-off reporting.

2.8 Decision-First Dashboard Storytelling for Managerial Interpretation and Action

A decision-first dashboard treats the screen as help for making decisions rather than a place to display many charts. The design starts by being clear about the outcome the user needs to reach, such as which branch to prioritise, what pattern to investigate next, or which process to adjust. After that, the dashboard is built around a small set of key measures, simple comparisons, and short explanations that are enough to support that choice with reasonable confidence. This view fits decision-first thinking, where analytics are judged by how they guide judgement and action, not by the volume of data or number of visuals shown (Cassimally, 2023; Wharton, 2025). In practice, even accurate charts can still lead to weak decisions if users have to guess the relevance and trade-offs on their own, so storytelling provides the structure that links results into a clear line of reasoning (Wharton, 2025).

Decision-first dashboard work therefore keeps coming back to coherence and sequence. Instead of starting from whatever data fields are available, or which chart type looks attractive, the design follows an ordered set of analytic steps that matches how decision-makers usually think through a problem. Lavalley et al. (2025) suggest capturing requirements in terms of decision needs and then turning these into a visual task flow on the dashboard. The aim is not to push users toward one “correct” answer but to give them enough context to read signals in a sensible order. When users understand why each indicator is shown, how it links to others, and what it suggests for the choice they face, the dashboard supports reasoning instead of guesswork (Cassimally, 2023; Wharton, 2025).

The need for decision-first storytelling becomes sharper in mixed-literacy settings, which are very common in business intelligence work. Many organisational dashboards are used by people who understand the business context but do not feel confident with statistics or advanced visual analysis. Fernandez Nieto et al. (2022) show that simply putting several indicators on a screen does not ensure understanding or follow-through. Users may be unsure what an index really means, how to read multiple measures together, or whether a change is large enough to matter. Their results suggest that small narrative cues can make dashboards easier to understand. The simplest elements of design can make a dashboard more user-friendly and help people better comprehend the information presented on it. Mini notes can provide information on what a diagram is illustrating, while straightforward titles can let people see exactly what each view is

intended to express. A fixed sequence of information can help guide people throughout the page, so the overall layout does not appear unstructured and is less confusing. Likewise, in some cases, a straightforward table or a concise written overview can be more reliable than more difficult charts if the accurate interpretation of the information is paramount over the visual treatment (Fernandez Nieto et al., 2022). This indicates that good dashboard design should not be about including more visuals. Instead, it is about making the conveyed message as straightforward and easily readable as possible.

Gunklach et al. (2023) provide an explanation of dashboards using the “theory of effective use,” which posits that a dashboard generates value only if it helps users make better decisions and take actions. They point out that many dashboards do not work for simple reasons. Some of the screens are too congested. Other dashboards lack clarity in regards to the dashboard’s goals. Others fail to provide a rationale for how the data under consideration was processed. These issues can cause confusion in users, induce them to form the wrong conclusion, and erode the quality of their decision-making. To minimize this issue, Gunklach et al. (2023) point out the use of storytelling as an effective design approach. If a dashboard follows an understandable story, users can grasp the situation, the primary message, and related insights to specific actions. Storytelling can simplify a dashboard and enhance usability when it reduces the complexity of navigation and clarifies the importance of each metric. It can also reduce uncertainty by guiding users through the insights in a logical order.

Decision-first storytelling also needs legitimacy. The narrative should reflect stakeholder priorities, not just an analyst’s personal view. Pignatelli et al. (2023) show that involving stakeholders in choosing KPIs and in shaping the story structure can turn a dashboard into a shared space for sense-making and action. When indicators are tied to agreed priorities, users can debate what should change and why, rather than stop at description. In this way, the dashboard makes the decision object visible and links evidence to an action logic that stakeholders recognise (Pignatelli et al., 2023; Wharton, 2025).

Hence, these studies point to a common pattern for decision-first dashboard storytelling. It begins with a clear decision target and a simple definition of success, then turns that target into a short sequence of analytical tasks that follows how users usually think through the issue (Cassimally, 2023; Wharton, 2025; Lavalley et al., 2025). The dashboard is also treated as revisable

and is judged mainly by the quality of decisions and actions it supports, which shifts its role from being a data display to working as a decision aid (Cassimally, 2023; Wharton, 2025).

2.9 Evidence-to-Action Mapping for Translating Findings into Improvement Actions

An evidence-to-action (E2A) mapping approach fits this project because it keeps a clear link between what the data shows and what should actually change in practice. E2A encourages a context-aware view of evidence, where measurement choices, how results are read, and whose experience is taken seriously can all depend on local conditions and stakeholder roles (Das et al., 2023). It also treats evidence use as a learning cycle rather than a single report. Das et al. (2023) describe E2A as a loop in which teams generate evidence, act on it, and then look back at implementation results. In this project, E2A logic means at least four elements must be stated clearly: the empirical signal that triggers a decision, the action to follow, the group responsible, and the indicators used to see whether change has actually happened. Figure 2.5 summarises the Evidence-to-Action (E2A) framework used to translate findings into actionable recommendations.

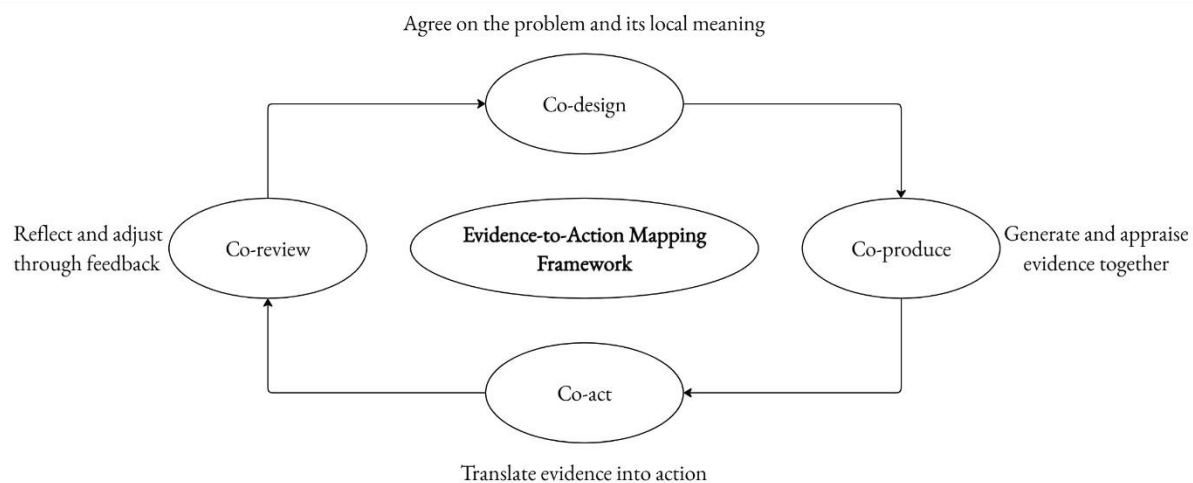


Figure 2.5: Overview of the Evidence-to-Action (E2A) Framework Stages and Core Purposes

In Malaysian e-banking service quality work, an evidence-to-action (E2A) mapping approach matters because the project only has value if banks go beyond describing gaps and actually change how services run. In practice, this means using findings to refine procedures, adjust response routines, or redesign specific parts of digital interfaces instead of leaving results in a

report. This line of thinking fits with knowledge translation studies that describe a “know–do” gap, where findings stay on paper instead of shaping daily practice (Ejaz et al., 2025; Chamathi Kossinna et al., 2025). The Knowledge-to-Action (KTA) model separates knowledge creation from an action cycle that adapts and applies that knowledge to local barriers and supports (Ejaz et al., 2025). Reviews of applied KTA show that many teams start by naming a specific gap and then tailoring interventions to local conditions rather than assuming one solution will work everywhere (Chamathi Kossinna et al., 2025). For this project, that logic supports giving different recommendations by bank type, delivery channel, or user group when the evidence shows meaningful differences.

Implementation research also highlights that evidence on its own seldom leads to change when responsibility is vague. Actual uptake depends on who is involved, whether they agree with the need for change, and whether the proposed steps feel feasible within existing constraints (Nørgaard et al., 2025). Because of this, E2A mapping should identify concrete owner groups, such as digital product teams, customer service leads, UX units, and compliance or security stakeholders, together with clear responsibilities. Dashboard-based audit-and-feedback, as discussed in Section 2.7, can support this process by letting teams check on a routine basis whether agreed actions were carried out and whether key indicators are moving in the expected direction (Chamathi Kossinna et al., 2025; Nørgaard et al., 2025). In this project, Power BI KPI views are intended to play that role by linking each recommendation to measurable indicators and making it easier to review progress over time.

E2A mapping can also help make the reasoning behind recommendations more transparent, especially when actions need to be ranked or phased. Evidence-to-decision frameworks, including GRADE-style logic, encourage teams to consider likely benefits and harms, the certainty of evidence, cost, feasibility, acceptability, and equity before settling on priorities (Neumann et al., 2023). Using this type of thinking in a simplified way supports explaining why some actions are treated as high priority and others are framed as conditional or context dependent. This, in turn, helps connect the empirical findings from e-banking service quality data to concrete, defensible choices about what should change first in practice.

2.10 Research Gaps and Limitations of Previous Studies

Several of the reviewed studies point to fairly similar limits in how their samples were drawn and who actually answered their surveys. Ul Haq and Awan (2020), for instance, stress that their results come from a single country, so the patterns they find may not travel well to other settings. Ubam et al. (2021) note that their use of convenience sampling and a geographically narrow area weakens generalisability, and they also admit that an English-only questionnaire can leave out older respondents who are less comfortable using English. Several studies point out sampling and design limits that affect how far findings can be generalised. Kah Boon Lim et al. (2023) note that small samples and a narrow age range reduce the strength of conclusions, and repeated gaps in location, age, and language can leave some user groups out. This makes it harder to claim that the results represent the full population of e-banking users. Waluyo et al. (2025) also argue that interface and user experience features may not perform the same way across different groups, so design findings should be interpreted together with user background and context. Dewi and Zulkifli Zulkifli (2024) argue that future studies should use stronger causal designs to test relationships more directly, rather than relying mainly on correlational findings. Another limit identified by Ubam et al. (2021), they also urge wider coverage in Malaysian states as well as multilingual survey instruments. With exclusively English questionnaires, studies could exclude respondents with lower English fluency, thus decreasing the representativeness of the sample and undermining the legitimacy of conclusions. Therefore, as suggested by these works, e-banking studies could be strengthened with more solid approaches and more receptive ways of collecting data.

This section identifies gaps in both research and banking practice. Many banks still collect customer feedback slowly, and the records are often stored across separate systems. One-time surveys, such as the *Malaysian Retail Banking Industry Customer Satisfaction Survey 2023*, provide a snapshot of user experience at one point in time, even though e-banking changes quickly. As a result, issues can grow before teams review the results and start corrective action. The situation becomes more serious when feedback, complaints, and service data sit in silos across different units, because each unit only sees part of the problem. In Malaysia, financial data has been described as “fragmented and inconsistent” (Rowena, 2025), and the *Maybank Malaysia: Redefining Customer-Centric Banking* (2025) report also notes that customer input is split across units. This separation makes it harder to combine complaints, usage signals, and service logs into

one unified view, which can delay early pattern detection and slow response (Malc et al., 2023). Overall, slow feedback cycles and fragmented systems can reduce a bank's ability to act quickly, which can weaken customer satisfaction and operational resilience.

Some banks are beginning to accelerate efforts with the way they receive feedback. For instance, RHB bank was heading toward real-time "Voice of Customer" systems in 2025 (Fintech News Malaysia, 2025). However, many banks still rely on slow routines, with feedback disseminated across various teams and channels. As apps continue to evolve, and customer expectations increase substantially, this fragmented structure makes it challenging to manage service quality and enhancements. In this case, early warning signals are often overlooked, while plans for improvement lose the capacity to help ensure a bank's competitive position in the digital realm (Raza et al., 2020). For this reason, case studies often highlight dashboards as a tactical tool for aggregating key signals into a single location for quicker review and follow-up.

The literature reveals weaknesses in the way banks collect and utilize customer feedback, wherein information is delayed and scattered across units of service. To decrease these shortcomings, the present work applies a formal methodology that enables survey indications to be refreshed regularly and compared across categories of users. The analysis does not stop at describing patterns. It also uses correlation and regression tests to examine which service quality dimensions are most strongly associated with satisfaction. As a result, improvement priorities can be set based on evidence, not assumptions, and the recommendations can be linked to measurable relationships rather than general impressions. To address interpretability limits for managers, findings are presented through a decision-first Power BI dashboard with clear KPI definitions and drill-down views that help stakeholders move from "what is weak" to "where it is weak." Finally, because insights often fail to translate into operational follow-through, the project converts key findings into an evidence-to-action map that links each signal to a specific improvement action, an owner, and an indicator for monitoring progress.

2.11 Summary

This chapter explored e-banking in Malaysia as a self-service digital channel, in a setting where national payment systems and everyday app-based payments have gradually pushed banking toward a mobile-first pattern. It then brought together studies that describe e-banking service quality through seven main dimensions (reliability, responsiveness, efficiency, privacy or security, ease of use, system availability, and website design) and noted that many of these studies find links to customer satisfaction, although the strength of these links can differ across contexts and methods. Customer satisfaction was discussed as an outcome shaped by changing expectations and adoption factors, often framed using Expectation–Confirmation Theory and the Technology Acceptance Model. The chapter also reviewed work on dashboards and business intelligence, treating dashboards as tools for continuous monitoring by decision-first dashboard storytelling approach, and outlined an evidence-to-action mapping approach that connects empirical signals to specific responsibilities, actions, and indicators. All these parts work toward the same goal. The project aims to identify the main service quality drivers in Malaysian e-banking and translate the findings into practical steps for monitoring and improvement.

Chapter 3. Methodology

3.0 Overview

This study examines the relationship between e-banking service quality and Customer Satisfaction in Malaysia. Respondents completed a structured questionnaire that measured seven service-quality dimensions and their overall satisfaction. After data collection, the raw responses were processed in Python to standardise variable names, handle missing values, and identify common data errors. The cleaned dataset was then imported into IBM SPSS Statistics for Phase 1 analysis. Phase 1 began with Cronbach's Alpha reliability analysis to assess the internal consistency of each construct. Descriptive statistics were then used to summarise the respondent profile and the main study variables. Pearson correlation analysis examined the strength and direction of the relationship between each service-quality dimension and Customer Satisfaction, while multiple linear regression identified the dimensions that made significant unique contributions when all predictors were considered together. The Phase 1 results were then prepared for use in Power BI during Phase 2. The dashboard visualises and tracks the current service-quality and Customer Satisfaction indicators through KPI cards, dimension comparisons, segment filters, and issue-monitoring views. Tracking in this project refers to reviewing the current indicators and using them as baseline measures. The dashboard can support future comparisons when new data are added, but the present version is not connected to live banking systems. Phase 3 moved from analysis to action. An Evidence-to-Action mapping process was used to convert the statistical findings and dashboard insights into prioritised recommendations. Each recommendation was connected to a service-quality issue, a responsible team, and a measurable indicator. Figure 3.1 summarises the full three-phase workflow and shows how each phase supports the research objectives.

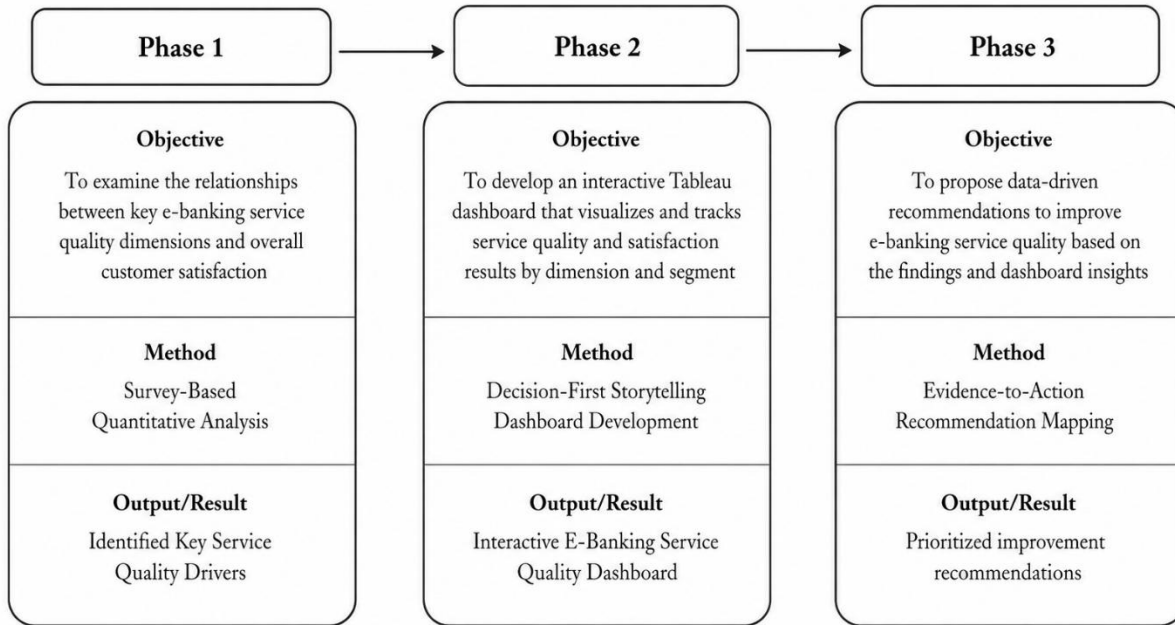


Figure 3.1: Overall Project Design

Table 3.1 summarises the main tools and technologies used throughout the project, from data collection and preprocessing to statistical analysis, dashboard development, and recommendation formulation. It also shows the purpose and output produced by each tool.

Table 3.1: Tools and Technologies Used

Project stage	Tool	Purpose	Output
Data Collection	Google Forms	Distribute the questionnaire and collect responses	Raw survey responses
Data Export	Microsoft Excel	Export responses from Google Forms for preprocessing	Raw .xlsx dataset
Data Preprocessing	Python with pandas	Rename variables, apply inclusion filters, clean values, validate missing data and transform issue selections	Cleaned respondent dataset and long-format issue dataset

Project stage	Tool	Purpose	Output
Statistical Analysis	IBM SPSS Statistics Version 30	Conduct Cronbach's Alpha, construct composite scores, descriptive analysis, Pearson correlation and multiple linear regression	Statistical tables and composite-score dataset
Dashboard Data Preparation	Python and Power Query	Prepare data types, reshape issue records and load dashboard-ready tables	FactResponses and FactIssues_Long tables
Dashboard Development	Microsoft Power BI Desktop	Create the data model, DAX measures, KPI cards, filters, charts and dashboard pages	Interactive Power BI dashboard
Recommendation Development	Power BI findings and Evidence-to-Action mapping	Convert statistical and dashboard findings into prioritised actions	Recommendation and action-priority table

3.1 Phase 1: Key Service Quality Drivers

Phase 1 aimed to identify the e-banking service-quality dimensions most strongly associated with Customer Satisfaction. Data were collected through a structured questionnaire completed by existing e-banking users. After the responses were cleaned and prepared, Cronbach's Alpha was used to assess the internal consistency of each construct. The analysis then proceeded with descriptive statistics, Pearson correlation, and multiple linear regression to examine the relationships between the seven service-quality dimensions and Customer Satisfaction.

3.1.1 Participants

The target population is e-banking users in Malaysia, meaning people who use e-banking services from banks that operate in the Malaysian market. It includes Malaysian citizens and also foreigners who live, work, or study in Malaysia, as long as they actively use these services. In practical terms, the survey focuses on individuals who use mobile banking apps and/or internet banking websites from Malaysian banks for routine tasks, such as checking balances, paying bills, transferring money, or topping up e-wallets. At the same time, the term "e-banking user" is quite broad. The survey therefore needs to capture different usage patterns, because someone who logs in once a month may judge service quality differently from someone who uses the app almost every day.

For sampling, non-probability convenience sampling is used, since the survey is distributed through channels that are easy to reach. The questionnaire link is shared through messaging apps, such as WhatsApp, and also through social media, for example Facebook and Xiaohongshu, where people can open the form on their phone and respond quickly. However, convenience sampling can easily attract respondents who are similar to each other, like friends from the same university or workplace, so this project applies three inclusion criteria to keep the responses aligned with the research focus and to reduce irrelevant submissions. First, respondents must provide consent to participate. Second, respondents must be 18 years old or above. Third, they must use e-banking at least once a month.

According to Nguyen (2025), data collection requires a sample size of at least five observations per variable. Considering the 24 Likert-scale items used to measure service quality and customer satisfaction, the minimum required sample size is around 120 respondents. This

project therefore aims for approximately 300 valid responses to provide more stable estimates and allow for possible data screening and subgroup comparisons. This figure is chosen because it should be large enough to support multiple linear regression with several predictors and to provide reasonably stable estimates of means and correlations.

The final sample of 426 valid respondents substantially exceeded the minimum requirement and provided sufficient observations relative to the number of predictors. A total of 491 raw responses were collected. After applying the eligibility criteria, 65 responses were excluded: five respondents did not provide consent, three respondents were below 18 years old, and 57 respondents used e-banking less than once a month. Therefore, 426 valid responses were retained for the final analysis.

3.1.2 Instrument

The main data collection instrument is a structured survey prepared using Google Forms. The questionnaire is designed to understand how users in Malaysia experience e-banking and how satisfied they feel overall. Respondents rate the service quality of e-banking based on their actual experience with the service when they are: logging in, checking balances, transferring money, paying bills, or using in-app support to solve a problem. The survey is based on recent usage, as this is easier for respondents to remember, but ratings can still vary between individuals due to individual judgement. For example, a single failed step in the process might lower a score and a smooth transaction may enhance it. In order to minimise this limitation, this survey uses the same format for each of the questions asked, thus providing a more equal basis on which responses can be compared and analysed clear-cut across all types of users. This survey is designed in four main sections, which give a structured framework for the systematic evaluation of service quality and satisfaction.

- Section A: Demographic and Usage Profile
- Section B: E-Banking Service Quality
- Section C: Customer Satisfaction
- Section D: Issue Monitoring

The questionnaire follows a straightforward layout. It begins with respondent information and proceeds to questions on e-banking service quality, overall satisfaction, and issues encountered by the user. Each question is posed in straightforward terminology, drawing from the actual use of the service. The five-point scale is used for all items, making it readily answerable and leaving a space for a reply that can be sent back to the analyst for further investigation. This layout guarantees straightforward comparisons and the ability to notice trends between users. The scale itself does, however, reduce complicated emotions to simply a score, so it cannot completely account for the decision of a respondent to rate in a certain way.

Section A: Demographic and Usage Profile

Section A compiles a basic background on respondents and their usage of e-banking in Malaysia. The reason is because perceptions of service quality can vary between user groups. More frequent users will be more likely to perceive delays, errors, or system change than occasional users who only use a service a few times. The same section is also used to aid screening, such that responses that do not fit within the scope of this study can be excluded. This way, the analysis can report on service quality ratings, satisfaction levels, and user-reported issues more clearly, without conflating very disparate user profiles into a single dataset.

Question 1: Age Group

Respondents pick one age band: Below 18 years, 18–20 years, 21–30 years, 31–40 years, 41–50 years, and Above 50 years. These groups make it easier to compare how different ages experience e-banking and what they expect from it. During cleaning, any response that selects “Below 18 years” is removed so the final dataset includes adults only.

Question 2: Nationality

Respondents choose Malaysian or Non-Malaysian. This item separates local users from foreign users who still rely on Malaysian banks. Later analysis can then check whether these two groups rate service quality and satisfaction differently.

Question 3: Ethnicity

Respondents select Malay, Chinese, Indian, or Other (please specify). This question describes the sample’s ethnic mix in simple terms. It also supports interpretation in Malaysia’s multicultural setting when patterns differ across groups.

Question 4: Gender

Respondents choose Male or Female. This information allows comparison of service quality ratings and satisfaction between the two groups. Any differences are treated as patterns to explore, not as fixed conclusions.

Question 5: Employment Status

Respondents select Employed, Student, Retired, or Unemployed. People in different situations often use e-banking for different reasons, such as salary payments, study expenses, or day-to-day transfers. This item supports subgroup comparisons in the later analysis.

Question 6: Monthly Income (RM)

Respondents choose one range: Below 2,000; 2,001–4,000; 4,001–6,000; 6,001–8,000; or Above 8,000. Income can shape how often a person uses banking services and what features they rely on. This question helps segment the sample into clearer user groups for comparison.

Question 7: Number of Bank Accounts

Respondents select One, Two, or More than two. Users with multiple accounts may switch between apps or compare banks more often. This question helps to determine whether that difference shows up in satisfaction and service quality ratings.

Question 8: Experience Using E-Banking

Respondents choose Less than 1 year, 1–3 years, 4–6 years, or More than 6 years. This item reflects how familiar the respondent may be with e-banking. It also allows comparison between newer users and long-time users.

Question 9: Frequency of E-Banking Use

Respondents select Daily, Weekly, Several times a month, About once a month, or Less than once a month. This question captures usage intensity, which often affects expectations. It also supports the inclusion rule, so any response that selects “Less than once a month” is removed during cleaning.

Question 10: Main Device Used for E-Banking

Respondents choose Smartphone / Mobile App, Computer / Laptop, Tablet, or Both Mobile and Web. The platform matters because the experience can differ across devices, such as layout, loading speed, and navigation flow. This item supports clearer comparisons during analysis.

Question 11: Region / Residence

Respondents indicate Urban or Rural. Location can relate to factors like internet stability or network coverage, which may shape the e-banking experience. This item helps check whether satisfaction or reported issues vary across these two groups.

Question 12: Main Bank Used for E-Banking

Respondents state the bank they use most often, with an option for “Other bank not listed above (please specify)”. This question supports bank-level grouping in both analysis and the Power BI dashboard. It also allows comparison of service quality and satisfaction across different providers.

Section B: E-Banking Service Quality

Section B assesses user experience with e-banking in seven dimensions on a uniform five-point scale that captures the perspective of everyday use. The items address pragmatic concerns: Does the app work smoothly? Can transactions and key actions be completed without fault? Is the system responsive when assistance is needed? The same scale is applied to all items, making responses comparable and allowing for comparisons of outcomes between different user groups. The scores also serve as a foundation for statistical tests and dashboard visualisation. The findings are not a measure of technology-corrected system performance, rather user perceptions.

Table 3.2: Definitions of E-Banking Service Quality Dimensions

Dimensions	Definition
Reliability	Accuracy and consistency of e-banking services and transactions.
Responsiveness	How promptly and clearly the bank responds when users need assistance or when services are requested.
Efficiency	How easy and quick it is to complete transactions and find information, and whether the platform runs smoothly.
Privacy and Security	The extent to which users feel safe providing personal and financial information and trust that their data are protected.

Ease of Use	How easy the interface is to understand, navigate, and use without needing guidance.
System Availability	Whether users can access the e-banking system whenever they need it and whether pages and functions load quickly and rarely crash.
Website Design	The attractiveness and professionalism of the interface, including layout, menus, colors, icons, and fonts.

Table 3.2 outlines the service quality dimensions used in this paper and details how each dimension is defined using the questionnaire and subsequent analysis. Each dimension in the survey is measured by three differently worded statements. The respondent is asked to score the following items on a 5-point Likert scale from 1 “Strongly Disagree” to 5 “Strongly Agree”. Measuring all items on the same scale simplifies scoring and makes it easy to discern across dimensions. This, combined with the fact that a higher score will generally indicate a more favorable opinion of the related service quality dimension, allows a strong interpretation.

The seven dimensions were selected due to the repeated appearance of these dimensions as important factors associated with perceived quality and customer satisfaction across many studies focusing on e-banking and e-service, including studies that focus on Malaysia and the surrounding region (Hadid et al., 2020; Izni et al., 2022; Kah Boon Lim et al., 2023; Nguyen, 2025; Serrah & Binti Haja Maideen, 2022). These dimensions, when considered together, mirror the system side of e-banking, such as reliability, efficiency, privacy and security, and system availability and the client-facing side, such as ease of use and website or app design. This combination is of particular importance since in many cases the user will judge the service as a whole by experiencing it in real use, thus all these aspects will be usually judged together rather than individually. In real use, they experience them at the same time, so a service only feels good when the system works properly and the interface is also easy to navigate. A transfer that is secure but slow can still feel frustrating, while a smooth interface is not enough if the system goes down or users worry about scams. In Malaysia, where round-the-clock access is often expected and security concerns are hard to ignore, these dimensions seem broad enough to reflect what users pay attention to in real usage. At the same time, using them keeps the measurement aligned with

prior empirical work, while still covering the main service quality areas that are likely to shape customer satisfaction in this setting. Table 3.3 lists the adapted measurement items used in the questionnaire. The item codes are used later for composite score construction.

Table 3.3: Measurement Items for E-Banking Service Quality Dimensions

Constructs	Items	Measures	References
Reliability	REL1	My bank's digital banking services are delivered accurately and without errors.	Ul Haq & Awan (2020), Bankuoru Egala et al. (2021), Raza et al. (2020)
	REL2	My bank consistently provides the digital services it promises.	
	REL3	Transactions are completed correctly the first time.	
Responsiveness	RES1	My bank responds promptly when I need assistance with digital services.	Bankuoru Egala et al. (2021), Raza et al. (2020)
	RES2	Digital services are processed quickly when requested.	
	RES3	My bank clearly informs me when online services will be completed.	
Efficiency	EFF1	It is easy and quick to complete transactions using my bank's online platform.	Bankuoru Egala et al. (2021), Raza et al. (2020), Malc et al. (2023)
	EFF2	I can easily find the information or service I need on the platform.	
	EFF3	The digital platform loads and operates smoothly without delay.	
Privacy and Security	PS1	I feel safe providing my personal or financial information online.	Ul Haq & Awan (2020), Waluyo et al. (2025),
	PS2	My bank protects my transaction details from unauthorized access.	Bankuoru Egala et al. (2021),

	PS3	I trust my bank's digital services to handle data securely.	Malc et al. (2023)
Ease of Use	EOU1	The e-banking interface is easy to understand and use.	Waluyo et al. (2025), Bankuoru Egala et al. (2021)
	EOU2	I can navigate the e-banking platform confidently without guidance.	
	EOU3	Performing any banking task on the website or app requires little effort.	
System Availability	SA1	I can access the e-banking system whenever I need it.	Malc et al. (2023)
	SA2	The pages and functions load quickly and rarely crash.	
	SA3	The e-banking platform works smoothly across all my devices.	
Website Design	WBD1	The e-banking website has an attractive and professional design.	Ul Haq & Awan (2020), Malc et al. (2023)
	WBD2	The layout and menus are well organized and easy to follow.	
	WBD3	Visual elements such as colors, icons, and fonts make the interface appealing.	

Section C: Customer Satisfaction

Section C evaluates user's general satisfaction with the bank's e-banking services via a simple scoring system. This section contains three statements on a Likert scale. These items inquire whether the users are satisfied with the overall quality of the service, whether their experience aligned to their expectations, and whether the customers would recommend the service to other people. Such construction offers a brief yet reliable satisfaction measure as it captures both individual assessment and the will to recommend the service. Each item is designed to give a simple satisfaction score that can be compared across different respondent groups. All three items use the same five-point scale from 1 ("Strongly Disagree") to 5 ("Strongly Agree"), so the scoring

matches the earlier sections. After respondents' answer, the three ratings are combined into one satisfaction score for each person. This score becomes the main outcome variable in the project, because it is used to test how the seven service-quality dimensions relate to customer satisfaction. Moreover, the same score is also useful for the Power BI dashboard because it functions as a clear headline indicator that stakeholders can interpret quickly, even though it still reflects perceived experience rather than direct system performance metrics. Table 3.4 summarises measurement items for e-banking customer satisfaction.

Table 3.4: Measurement Items for E-Banking Customer Satisfaction

Constructs	Items	Measures	References
Customer Satisfaction	CS1	I am satisfied with the overall quality of my bank's e-banking services.	Ul Haq & Awan (2020), Waluyo et al. (2025), Raza et al. (2020) Malc et al. (2023)
	CS2	My experience using the e-banking system meets my expectations.	
	CS3	I would recommend my bank's e-banking services to others.	

Section D: Issue Monitoring

Section D focuses on specific incidents. It captures the real problems users faced when using e-banking. In comparison, the questions in Sections B and C primarily quantify general impression and satisfaction levels. As with a high satisfaction score, this can occur even if the user encounters substantial issues during the actual task. For instance, the service may feel "good overall," but a transfer can fail, the app can crash, or the payment flow can become confusing at a key step. To get a picture of this other side of the experience, Section D asks questions about common issues such as failed transfers, app crashes, slow customer support replies, and unclear steps during payments or account verification. Reporting these experience problems makes it much easier to map common pain points to specific dimensions of service quality and present them in clear view in the Power BI dashboard for action. But the findings are based on what respondents recall, so the findings are a reflection of user recall and perceived importance instead of a complete log of technical errors.

Question 1: Issue Experience

Respondents first answer whether they have ever faced issues when using their bank's e-banking service (Yes / No). This question splits users into two clear groups: those with issue experience and those without it. Only respondents who select "Yes" continue to the follow-up questions. This structure keeps the survey relevant because users who have never faced a problem do not need to answer detailed issue questions. Moreover, it supports later analysis by allowing comparisons between the two groups, especially when checking whether issue experience is linked to lower service quality ratings or lower overall satisfaction.

Question 2: Timing of the Issue

Respondents indicate when the issue happened within the last three months by choosing one option: "Within the last 7 days," "1–4 weeks ago," or "1–3 months ago." These time windows help place each report into a recent and consistent period, which reduces confusion and makes recall easier. The time marker also helps interpretation, because a very recent failed transaction may affect satisfaction differently from a problem that happened months ago. In the Power BI dashboard, this field can show whether issue reports are mostly current, fairly recent, or less recent, and whether similar issues keep appearing across time.

Question 3: Platform Used During the Issue

Respondents select the platform they were using when the issue occurred: Mobile App, Web / Online Banking, or Both. This item matters because the user experience often differs across channels. A problem like an app crash is obviously tied to the mobile app, while a login error might appear on web banking, or across both platforms. By collecting platform details, the analysis can check whether issues cluster more on mobile, on the website, or across both, and the dashboard can highlight where improvements may be needed first.

Question 4: Bank Involved in the Issue

Respondents are asked which bank they encountered the problem with, using a list of common Malaysian banks and an "Other (please specify)" option. Linking each issue to a specific provider enables the analysis to group incidents by bank and to compare the frequency and types of issues across different institutions. This is important for benchmarking and for providing bank-level insights rather than only a general picture of e-banking issues.

Question 5: Type of Issue Experienced

Respondents select all issue types they experienced from a list that is grouped according to the service quality dimensions used in Section B. Table 3.5 summarises e-banking issue types by service quality dimension.

Table 3.5: E-Banking Issue Types by Service Quality Dimension

Dimensions	Issue
Reliability	• Transaction failure • Delayed confirmation • Service outage • Transfer credited very late • Scheduled payment not processed as planned
Responsiveness	• Slow customer support • Unhelpful chatbot / automated replies
Efficiency	• Slow performance • OTP takes too long / expires
Privacy and Security	• Suspicious activity / unusual login • Concern about data security • Not sure if messages/links are genuine (scam concern)
Ease of Use	• Confusing interface • Hard-to-find functions • Difficulty logging in • Too many steps for simple tasks
System Availability	• App crashes • Site/app unresponsive
Website Design	• Poor layout • Unattractive/confusing colour scheme

By organizing the response options in this way, the question connects concrete user problems to the same dimensions that are measured in Section B. This alignment makes it possible to compare perceived quality (how users rate each dimension) with experienced problems (what

actually went wrong). The data from this question can later be used to count how many times each type of issue occurs and to build visualizations in Power BI, such as issue frequency by dimension, bank, or platform.

Section D is included to provide a more concrete layer to the perception-based measures in Sections B and C. Ratings in those sections are useful, but they can stay a bit broad, because two users can give the same score for very different reasons. Section D asks for feedback on problems in more detail. Respondents are asked to say what the problem was, when it occurred, and which platform they were using. This moves the research away from general opinions and toward easier patterns of faults. For example, the data might reveal recurring transfer failures in mobile applications or slow responses from a customer support center at a particular time. This information also strengthens the Power BI dashboard with a stronger footing because it provides practical evidence that can help guide the answer to which problems should be fixed first.

3.1.3 Data Collection

The questionnaire is shared online using Google Forms. It begins with a short introduction that explains the purpose of the study and the consent terms. Participation is voluntary, and respondents may stop at any time if they feel uncomfortable or do not wish to continue. No identifying details are collected, and the responses are used only for academic purposes. To reduce privacy risk, the form does not ask for sensitive identifiers, so there are no items requesting names, IC numbers, or bank account details. That said, online surveys are never perfectly risk-free, so the form is designed to collect only the minimum information needed for the analysis and to avoid any questions that could directly identify a respondent.

Skip logic is implemented in Section D so that issue follow-up questions are only displayed and expected when `issue_experience` is answered “Yes”. Data collection took place from 19 November to 22 December 2025, covering 33 days (approximately five weeks), and yielded 491 responses by the end of the fieldwork period. After the survey was closed for the initial analysis, responses were exported from Google Forms in Excel format for data preparation and analysis.

3.1.4 Data Preprocessing (refer Appendix C)

Data preprocessing refers to the set of planned steps used to convert raw survey exports into an analysis-ready dataset (Joshi & Patel, 2021). In this project, Google Forms responses are processed in Python through a reproducible pipeline that includes exploration, recoding, cleaning, validation, and (where needed) reshaping for dashboard use. The purpose is to ensure that variables follow the questionnaire design, categories remain consistent for grouping and filtering, and construct items are suitable for computing composite scores and running statistical tests.

Raw survey exports rarely come out “clean,” so preprocessing matters if the analysis is going to be trustworthy. Some problems are simple but persistent, like respondents typing slightly different versions of the same idea in an open-text “Other” field, which later creates fragmented categories. Some problems also come from the way the form is built. For example, skip logic can hide follow-up questions, so the export ends up with blank cells. Those blanks are not always “missing.” Sometimes they just mean the question was never shown, so the respondent could not answer it. If that rule is not stated clearly, the blanks can be treated as missing data by mistake. Besides, exports can look different after download, especially when the file is opened in one program, then saved again in another one. That is when strange blanks or invalid values may appear. If these issues are not cleaned, the analysis can be quietly distorted. The dataset may produce wrong averages, unfair comparisons across banks or demographic groups, or misleading links in correlation and regression. Therefore, preprocessing sets clear rules early, keeps the dataset consistent for statistical tests and dashboard reporting, and makes it easier to update later when new responses are added. Figure 3.2 summarises the workflow used to turn raw questionnaire data into an analysis-ready dataset.

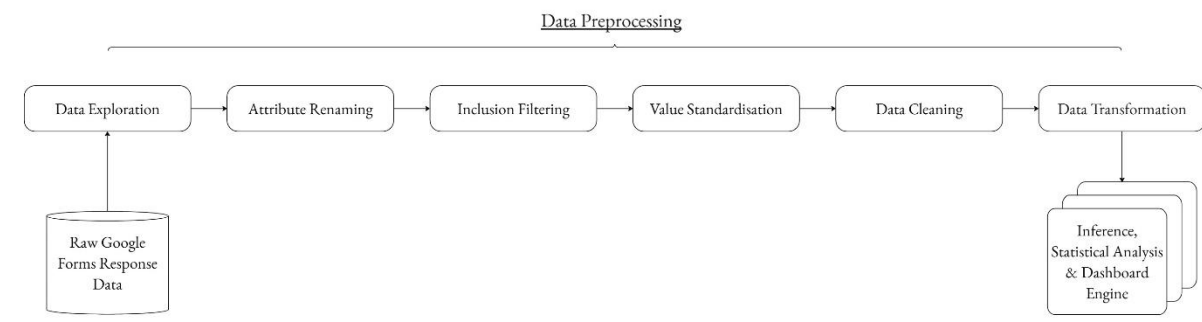


Figure 3.2: Data preprocessing workflow

Step 1: Data Exploration

After downloading the Google Forms responses as an Excel file, the dataset is imported into Python using pandas and a first data audit is carried out before any transformations are applied. The audit includes: (1) previewing sample records; (2) inspecting data types to detect numeric fields stored as text and mixed formats; (3) reviewing missing-value patterns; and (4) detecting duplicate rows to distinguish between likely data issues versus routing/export effects. These checks guide the cleaning rules that follow and reduce the risk of unnecessary or incorrect transformations.

```
#Setup
import pandas as pd
import numpy as np
import re
from IPython.display import display

# FILE PATHS
INPUT_XLSX = r"C:/Users/Admin/Downloads/Malaysian E-Banking Service Quality & Customer Satisfaction Survey (Responses).xlsx"

df_raw = pd.read_excel(INPUT_XLSX)

# Step 1: Data Exploration
# Preview
print("\nPreview (first 5 rows)")
display(df_raw.head())

# Dtypes
print("\nInfo:")
df_raw.info()

# Missing values count
print("\nMissing values per column:")
display(df_raw.isnull().sum().sort_values(ascending=False).rename("missing").to_frame())

# Duplicate responses (common check: full-row duplicate)
print("\nDuplicate rows:", df_raw.duplicated().sum())
```

Figure 3.3: Python Setup and Initial Exploration of the Raw Survey Dataset

Figure 3.3 shows the Python setup used to import the required libraries, load the raw survey file, and perform the initial dataset checks. The raw dataset was previewed using `df_raw.head()` to confirm that the file was imported correctly and to verify the overall structure of the survey variables. As shown in Figure 3.4, the first five responses include questions in section A, B, C, and D.

Preview (first 5 rows)

	Timestamp	I have read the information above and agree to participate.	Age Group	Nationality	Ethnicity	Gender	Employment Status	Monthly Income (RM)	Number of Bank Accounts	Experience Using E-Banking	The layout and menus are well organized and easy to follow.	Visual elements such as colors, icons, and fonts make the interface appealing.	I am satisfied with the overall quality of my bank's e-banking services.	My experience using the e-banking system meets my expectations.	I would recommend my bank's e-banking services to others.	Have you faced any issues when using your bank's e-banking services?	When did the issue occur (within the last 3 months)?	Which e-banking platform did you experience the issue on?	Which bank did you encounter the problem with?	What issue(s) did you experience? (Select all that apply)
0	2025-11-26 19:54:01.767	Yes, I agree	18-20 years	Malaysian	Chinese	Female	Student	Below 2,000	One	1 - 3 years	3.0	2.0	1.0	2.0	2.0	Yes	1-4 weeks ago	Web / Online Banking	Al Rajhi Bank Malaysia	[REL] Transaction failure, [REL] Delayed confi...
1	2025-11-26 19:54:46.725	Yes, I agree	31-40 years	Malaysian	Chinese	Female	Employed	4,001 - 6,000	One	More than 6 years	4.0	5.0	5.0	4.0	5.0	Yes	1-4 weeks ago	Both	AEON Bank	[REL] Scheduled payment not processed as plann...
2	2025-11-26 19:58:50.045	Yes, I agree	31-40 years	Malaysian	Malay	Female	Employed	Above 8,000	More than two	1 - 3 years	5.0	4.0	4.0	5.0	4.0	Yes	Within the last 7 days	Web / Online Banking	AEON Bank	[REL] Transaction failure, [REL] Delayed confi...
3	2025-11-26 20:01:06.017	Yes, I agree	31-40 years	Malaysian	Malay	Male	Employed	6,001 - 8,000	More than two	More than 6 years	5.0	4.0	4.0	5.0	4.0	Yes	1-4 weeks ago	Web / Online Banking	Bank Rakyat	[REL] Transaction failure, [REL] Delayed confi...
4	2025-11-26 20:08:17.798	Yes, I agree	41-50 years	Malaysian	Indian	Female	Employed	2,001 - 4,000	One	1 - 3 years	4.0	5.0	5.0	4.0	5.0	Yes	Within the last 7 days	Both	Citibank Malaysia	[REL] Transaction failure, [REL] Transfer cred...

5 rows x 43 columns

Figure 3.4: Preview of the raw survey dataset (first five rows)

`df_raw.info()` was used to summarise the dataset structure, size, and data types. Figure 3.5 indicates that the dataset contains 491 responses (rows) and 43 variables (columns). The variables include 1 datetime column (Timestamp), 18 categorical/object columns (mainly demographics and issue-monitoring fields), and 24 numeric columns stored as float64, which represent Likert-scale ratings.

```

Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 491 entries, 0 to 490
Data columns (total 43 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   Timestamp                                                            491 non-null   datetime64[ns]
1   I have read the information above and agree to participate.          491 non-null   object
2   Age Group                                                            486 non-null   object
3   Nationality                                                          486 non-null   object
4   Ethnicity                                                            486 non-null   object
5   Gender                                                              486 non-null   object
6   Employment Status                                                    486 non-null   object
7   Monthly Income (RM)                                                  486 non-null   object
8   Number of Bank Accounts                                              486 non-null   object
9   Experience Using E-Banking                                           486 non-null   object
10  Frequency of E-Banking Use                                           486 non-null   object
11  Main Device Used for E-Banking                                       486 non-null   object
12  Region / Residence                                                  486 non-null   object
13  Which bank do you use most often for your e-banking services?        486 non-null   object
14  My bank's digital banking services are delivered accurately and without errors. 486 non-null   float64
15  My bank consistently provides the digital services it promises.       486 non-null   float64
16  Transactions are completed correctly the first time.                 486 non-null   float64
17  My bank responds promptly when I need assistance with digital services. 486 non-null   float64
18  Digital services are processed quickly when requested.               486 non-null   float64
19  My bank clearly informs me when online services will be completed.    486 non-null   float64
20  It is easy and quick to complete transactions using my bank's online platform. 486 non-null   float64
21  I can easily find the information or service I need on the platform.   486 non-null   float64
22  The digital platform loads and operates smoothly without delay.       486 non-null   float64
23  I feel safe providing my personal or financial information online.     486 non-null   float64
24  My bank protects my transaction details from unauthorized access.      486 non-null   float64
25  I trust my bank's digital services to handle data securely.           486 non-null   float64
26  The e-banking interface is easy to understand and use.               486 non-null   float64
27  I can navigate the e-banking platform confidently without guidance.    486 non-null   float64
28  Performing any banking task on the website or app requires little effort. 486 non-null   float64
29  I can access the e-banking system whenever I need it.                 486 non-null   float64
30  The pages and functions load quickly and rarely crash.                486 non-null   float64
31  The e-banking platform works smoothly across all my devices.          486 non-null   float64
32  The e-banking website has an attractive and professional design.      486 non-null   float64
33  The layout and menus are well organized and easy to follow.           486 non-null   float64
34  Visual elements such as colors, icons, and fonts make the interface appealing. 486 non-null   float64
35  I am satisfied with the overall quality of my bank's e-banking services. 486 non-null   float64
36  My experience using the e-banking system meets my expectations.        486 non-null   float64
37  I would recommend my bank's e-banking services to others.            486 non-null   float64
38  Have you faced any issues when using your bank's e-banking services?  486 non-null   object
39  When did the issue occur (within the last 3 months)?                 338 non-null   object
40  Which e-banking platform did you experience the issue on?             338 non-null   object
41  Which bank did you encounter the problem with?                       338 non-null   object
42  What issue(s) did you experience? (Select all that apply)            338 non-null   object
dtypes: datetime64[ns](1), float64(24), object(18)
memory usage: 165.1+ KB

```

Figure 3.5: Dataset structure and data types summary

Missing values were assessed using a column-wise null count with `df_raw.isnull().sum().sort_values(ascending=False)`. As shown in Figure 3.6, the missingness pattern is not uniform. The four issue follow-up variables—issue types, issue bank, issue platform, and issue timing—each contain 153 missing values. This pattern is consistent with the survey’s skip logic because these items only apply to respondents who indicated they had faced issues; respondents who answered “No” would not provide values for these follow-up fields. In addition, many other variables (including several demographics and Likert-scale items) each contain 5 missing values, while both the consent/acknowledgement item (“I have read the information above and agree to participate.”) and Timestamp show 0 missing values, indicating that all recorded submissions include complete consent confirmation and a timestamp.

Missing values per column:	
	missing
What issue(s) did you experience? (Select all that apply)	153
Which bank did you encounter the problem with?	153
Which e-banking platform did you experience the issue on?	153
When did the issue occur (within the last 3 months)?	153
The e-banking platform works smoothly across all my devices.	5
My bank protects my transaction details from unauthorized access.	5
I trust my bank's digital services to handle data securely.	5
The e-banking interface is easy to understand and use.	5
I can navigate the e-banking platform confidently without guidance.	5
Performing any banking task on the website or app requires little effort.	5
I can access the e-banking system whenever I need it.	5
The pages and functions load quickly and rarely crash.	5
The layout and menus are well organized and easy to follow.	5
The e-banking website has an attractive and professional design.	5
The digital platform loads and operates smoothly without delay.	5
Visual elements such as colors, icons, and fonts make the interface appealing.	5
I am satisfied with the overall quality of my bank's e-banking services.	5
My experience using the e-banking system meets my expectations.	5
I would recommend my bank's e-banking services to others.	5
Have you faced any issues when using your bank's e-banking services?	5

I feel safe providing my personal or financial information online.	5
I can easily find the information or service I need on the platform.	5
Main Device Used for E-Banking	5
It is easy and quick to complete transactions using my bank's online platform.	5
Ethnicity	5
Gender	5
Employment Status	5
Monthly Income (RM)	5
Number of Bank Accounts	5
Experience Using E-Banking	5
Frequency of E-Banking Use	5
Region / Residence	5
Age Group	5
Which bank do you use most often for your e-banking services?	5
My bank's digital banking services are delivered accurately and without errors.	5
My bank consistently provides the digital services it promises.	5
Transactions are completed correctly the first time.	5
My bank responds promptly when I need assistance with digital services.	5
Digital services are processed quickly when requested.	5
My bank clearly informs me when online services will be completed.	5
Nationality	5
I have read the information above and agree to participate.	0
Timestamp	0

Figure 3.6: Missing values per column

Duplicate responses were checked using `df_raw.duplicated().sum()`. The output in Figure 3.7 shows 0 duplicate rows, suggesting there are no fully repeated submissions. This check is important because duplicate entries can distort summary statistics and bias modelling outcomes, so confirming no duplicates supports the reliability of the dataset for subsequent preprocessing and analysis.

Duplicate rows: 0

Figure 3.7: Duplicate response check

Step 2: Attribute Renaming

Before any modelling or calculations begin, the column headers are renamed into short, clear labels that are easier to work with during analysis. Google Forms exports often produce very long question-style headers, sometimes with punctuation, line breaks, or repeated wording, and that can make coding messy and increase the chance of picking the wrong variable unintentionally altered. A standard naming scheme is used instead, so each item follows the questionnaire structure

and stays consistent across steps. This kind of clean labeling supports repeated tasks such as computing composite scores, running reliability testing using Cronbach's alpha, and building correlation or regression models. Renaming is documented and traceable back to the original questions so that the labels remain auditable when reporting results or refreshing the pipeline. Table 3.6 summarises attribute renaming scheme for survey export.

Table 3.6: Attribute Renaming Scheme for Survey Export

Section	Question / Item (Label)	Renamed Attribute
Pre / Consent	Timestamp	<i>response_time</i>
	Consent agreed	<i>consent_agreed</i>
A	Age Group	<i>age_group</i>
	Nationality	<i>nationality</i>
	Ethnicity	<i>ethnicity</i>
	Gender	<i>gender</i>
	Employment Status	<i>employment_status</i>
	Monthly Income (RM)	<i>monthly_income_rm</i>
	Number of Bank Accounts	<i>num_bank_accounts</i>
	Experience Using E-Banking	<i>ebanking_experience</i>
	Frequency of E-Banking Use	<i>usage_frequency</i>
	Main Device Used for E-Banking	<i>main_device</i>
	Region / Residence (Urban/Rural)	<i>residence_region</i>
	Main Bank Used for E-Banking	<i>main_bank</i>
B	Reliability Likert items	<i>REL1</i>
		<i>REL2</i>
		<i>REL3</i>
	Responsiveness Likert items	<i>RES1</i>
		<i>RES2</i>
		<i>RES3</i>
	Efficiency Likert items	<i>EFF1</i>
		<i>EFF2</i>

		<i>EFF3</i>
	Privacy & Security Likert items	<i>PS1</i>
		<i>PS2</i>
		<i>PS3</i>
	Ease of Use Likert items	<i>EOU1</i>
		<i>EOU2</i>
		<i>EOU3</i>
	System Availability Likert items	<i>SA1</i>
		<i>SA2</i>
		<i>SA3</i>
	Website Design Likert items	<i>WBD1</i>
		<i>WBD2</i>
		<i>WBD3</i>
C	Customer Satisfaction Likert items	<i>CS1</i>
		<i>CS2</i>
		<i>CS3</i>
D	Issue Experience (Yes/No)	<i>issue_experience</i>
	Timing of the Issue (Within the last 7 days / 1–4 weeks ago / 1–3 months ago)	<i>issue_timing</i>
	Platform Used During the Issue (Mobile App / Web / Both)	<i>issue_platform</i>
	Bank Involved in the Issue	<i>issue_bank</i>
	Type of Issue Experienced (checkbox list as exported)	<i>issue_types_raw</i>

```

# 3) Normalize mapping keys
rename_map_norm = {normalize_header(k): v for k, v in rename_map.items()}
likert_map_norm = {normalize_header(k): v for k, v in likert_map.items()}
full_map_norm = {**rename_map_norm, **likert_map_norm}

# 4) Apply renaming
df = df_raw2.rename(columns=rename_map_norm).rename(columns=likert_map_norm).copy()

# 5) Rename table (no Found_in_File)
full_map_norm = {**rename_map_norm, **likert_map_norm}
rename_table = pd.DataFrame(
    [(old, new) for old, new in full_map_norm.items()],
    columns=["Original", "Renamed"]
)

display(rename_table)

```

Figure 3.8: Attribute Renaming in Python

In implementation, header cleaning was performed first to reduce mapping errors caused by formatting inconsistencies in the raw export. A *normalize_header()* function was used to remove line breaks and tabs, collapse repeated spaces, trim whitespace, and standardise quotation marks. After this normalisation, renaming was applied using two dictionaries: (1) a *rename_map* for respondent profile and issue-monitoring fields, and (2) a *likert_map* to encode the Likert-scale measurement items into construct-based labels. To avoid mapping failures caused by minor header differences, both mapping dictionaries were also normalised before application. The renaming was then applied to the dataset using chained *df.rename(columns=...)* operations, producing a cleaned dataframe (df) with standardised names. Figure 3.8 shows the Python code used to normalise raw questionnaire headers and rename respondent, issue-monitoring, and Likert-scale variables into standardised analytical labels. Finally, a renaming reference table was generated to document the transformation from each original question label to its renamed variable, as shown in Figure 3.9.

	Original	Renamed		
0	Timestamp	response_time	22	My bank responds promptly when I need assistan... RES1
1	I have read the information above and agree to...	consent_agreed	23	Digital services are processed quickly when re... RES2
2	Age Group	age_group	24	My bank clearly informs me when online service... RES3
3	Nationality	nationality	25	It is easy and quick to complete transactions ... EFF1
4	Ethnicity	ethnicity	26	I can easily find the information or service I... EFF2
5	Gender	gender	27	The digital platform loads and operates smooth... EFF3
6	Employment Status	employment_status	28	I feel safe providing my personal or financial... PS1
7	Monthly Income (RM)	monthly_income_rm	29	My bank protects my transaction details from u... PS2
8	Number of Bank Accounts	num_bank_accounts	30	I trust my bank's digital services to handle d... PS3
9	Experience Using E-Banking	ebanking_experience	31	The e-banking interface is easy to understand ... EOU1
10	Frequency of E-Banking Use	usage_frequency	32	I can navigate the e-banking platform confiden... EOU2
11	Main Device Used for E-Banking	main_device	33	Performing any banking task on the website or ... EOU3
12	Region / Residence	residence_region	34	I can access the e-banking system whenever I n... SA1
13	Which bank do you use most often for your e-ba...	main_bank	35	The pages and functions load quickly and rarel... SA2
14	Have you faced any issues when using your bank...	issue_experience	36	The e-banking platform works smoothly across a... SA3
15	When did the issue occur (within the last 3 mo...	issue_timing	37	The e-banking website has an attractive and pr... WBD1
16	Which e-banking platform did you experience th...	issue_platform	38	The layout and menus are well organized and ea... WBD2
17	Which bank did you encounter the problem with?	issue_bank	39	Visual elements such as colors, icons, and fon... WBD3
18	What issue(s) did you experience? (Select all ...	issue_types_raw	40	I am satisfied with the overall quality of my ... CS1
19	My bank's digital banking services are deliver...	REL1	41	My experience using the e-banking system meets... CS2
20	My bank consistently provides the digital serv...	REL2	42	I would recommend my bank's e-banking services... CS3
21	Transactions are completed correctly the first...	REL3		

Figure 3.9: Attribute renaming reference table

Step 3: Inclusion Filtering

Only respondents who meet the project's eligibility rules are included in the Objective 1 analysis dataset. This step aligns the final sample with the project's target population, adult users of e-banking services in Malaysia who use e-banking regularly, so the results reflect the group that the questionnaire is intended to represent. Filtering is important because responses from ineligible participants can introduce systematic bias. For instance, respondents under 18 may have a different legal status and more limited access to banking services, so their e-banking use may not reflect typical customer behaviour. In the same way, very infrequent users may judge service quality based on only a few interactions, not on regular use. Because of that, the "service quality perception" score may not represent a stable view, and comparisons across dimensions and customer satisfaction can become less reliable.

In this project, three filters are applied. First, only respondents who agreed to participate are retained, so any record with "No, I do not agree" in the consent field is excluded. Second, only respondents who are 18 years old or above are retained, so any record that selected "Below 18

years” for age group is excluded. Third, respondents must use e-banking at least once a month, so any record that selected “Less than once a month” for usage frequency is removed. After screening, the remaining records become the eligible analysis dataset used for subsequent preprocessing, construct score calculation, and statistical testing.

```
#Step 3: Inclusion Filtering

df_before = len(df)

# Count how many match each exclusion rule
consent_no = (df["consent_agreed"].astype(str).str.strip() == "No, I do not agree").sum()
below_18 = (df["age_group"].astype(str).str.strip() == "Below 18 years").sum()
less_month = (df["usage_frequency"].astype(str).str.strip() == "Less than once a month").sum()

# Apply the filters
consent_no = (df["consent_agreed"] == "No, I do not agree").sum()
below_18 = (df["age_group"] == "Below 18 years").sum()
less_month = (df["usage_frequency"] == "Less than once a month").sum()

df = df[
    (df["consent_agreed"] != "No, I do not agree") &
    (df["age_group"] != "Below 18 years") &
    (df["usage_frequency"] != "Less than once a month")
].copy()

df_after = len(df)
```

Figure 3.10: Respondent inclusion filtering in Python

Figure 3.10 shows the Python filtering rules used to exclude respondents without consent, respondents below 18 years old, and users who accessed e-banking less than once a month. The output in Figure 3.11 shows that 5 records were excluded for no consent, 3 records were excluded for being below 18 years old, and 57 records were excluded for using e-banking less than once a month. In total, 65 responses were removed, reducing the dataset from 491 to 426 records.

```
Excluded: No consent = 5
Excluded: Below 18 years = 3
Excluded: Less than once a month = 57
Rows before filter: 491
Rows after filter: 426
Total removed: 65
```

Figure 3.11: Inclusion filtering results

Step 4: Value Standardisation (Categorical Recoding)

After attribute renaming and inclusion filtering, selected categorical variables are standardised in Python using predefined option lists to ensure stable and repeatable categories for demographic and bank-level breakdowns, for example counts by ethnicity and satisfaction comparisons by bank. This step is required because the *ethnicity*, *main_bank*, and *issue_bank* questions include an “Other (please specify)” option, which can create many text variants in exports (differences in spelling, casing, spacing, abbreviations, or formats such as “Other: <text>”). If left untreated, these variants fragment groupings and clutter Power BI filters. Figure 3.12 shows the Python procedure used to standardise bank and ethnicity values into consistent categories for analysis and dashboard filtering.

```
def standardize_bank(x):
    if pd.isna(x):
        return np.nan
    s = str(x).strip()
    return s if s in approved_banks else "Other"

df["main_bank"] = df["main_bank"].apply(standardize_bank)
df["issue_bank"] = df["issue_bank"].apply(standardize_bank)

# Ethnicity counts (fixed order)
eth_counts = (
    df["ethnicity"]
    .value_counts(dropna=False)
    .reindex(["Malay", "Chinese", "Indian", "Other"], fill_value=0)
    .reset_index()
)
eth_counts.columns = ["ethnicity", "count"]
display(eth_counts)
```

Figure 3.12: Categorical Value Standardisation for Bank and Ethnicity Variables in Python

Figure 3.13 presents the distribution of ethnicity values after the standardisation and recoding procedure. Following cleaning, all responses were successfully mapped into a consistent set of predefined categories: Chinese (191), Malay (143), and Indian (92). Notably, the “Other” category recorded a count of 0, indicating that no entries fell outside the recognised options after standardisation. This confirms that the ethnicity field is now consistently coded and suitable for subsequent descriptive analysis and subgroup comparisons.

	ethnicity	count
0	Malay	143
1	Chinese	191
2	Indian	92
3	Other	0

Figure 3.13: Ethnicity counts after value standardisation

For bank grouping (*main_bank* and *issue_bank*), an approved bank list is used as the reference: if a response matches a listed bank name, it is retained; otherwise it is recoded to “Other”. This standardisation keeps grouped comparisons clean and reproducible for analysis and dashboard filtering. Figure 3.14 visualises the bank distribution after value standardisation to support the interpretation of subsequent analysis.

	main_bank	count
0	Agrobank	30
1	Standard Chartered Bank Malaysia	22
2	OCBC Bank	22
3	GXBank	21
4	RHB Bank	20
5	Affin Bank	20
6	Citibank Malaysia	20
7	HSBC Bank Malaysia	19
8	ICBC Malaysia	19
9	SeaBank Malaysia	18
10	Bank Rakyat	18
11	Bank Islam Malaysia	18
12	Bank of China Malaysia	17
13	AEON Bank	17
14	CIMB Bank	16
15	Alliance Bank	16
16	MBSB Bank	16
17	BSN	14
18	UOB Malaysia	13
19	Boost Bank	13
20	Public Bank	12
21	Hong Leong Bank	11
22	Al Rajhi Bank Malaysia	10
23	Maybank	8
24	Other	16

	issue_bank	count
0	NaN	126
1	Bank of China Malaysia	20
2	HSBC Bank Malaysia	17
3	Affin Bank	16
4	Alliance Bank	15
5	Standard Chartered Bank Malaysia	15
6	CIMB Bank	14
7	SeaBank Malaysia	14
8	UOB Malaysia	14
9	Agrobank	13
10	Bank Rakyat	13
11	GXBank	13
12	MBSB Bank	12
13	AEON Bank	12
14	Public Bank	11
15	BSN	11
16	ICBC Malaysia	11
17	Hong Leong Bank	10
18	Bank Islam Malaysia	10
19	OCBC Bank	10
20	Al Rajhi Bank Malaysia	9
21	RHB Bank	8
22	Maybank	7
23	Citibank Malaysia	7
24	Boost Bank	6
25	Other	12

Figure 3.14: Bank distribution after value standardisation

Step 5: Data Cleaning

Data cleansing is a process for detecting incorrect or noisy data and correcting them or removing them from the dataset (Maharana et al., 2022). In this project, the cleaning step applies explicit rules so the dataset used for analysis contains only valid, consistent, and interpretable responses before any modelling or score computation. Figure 3.15 shows the Python logic used to classify structural missing values, validate follow-up responses, and clean records that conflict with the questionnaire's skip logic.

```

# Step 5: Data Cleaning
# Structural Missing Handling (+ flag inconsistencies + blank out inconsistent follow-ups)

followup_fields = ["issue_timing", "issue_platform", "issue_bank", "issue_types_raw"]

def compute_followup_status(row):
    exp = row.get("issue_experience")
    exp = "" if pd.isna(exp) else str(exp).strip()

    followup_has_any = any(
        pd.notna(row.get(f)) and str(row.get(f)).strip() != ""
        for f in followup_fields
    )

    if exp == "No":
        return "Inconsistent" if followup_has_any else "Not Applicable"
    if exp == "Yes":
        return "Applicable" if followup_has_any else "Missing"
    return "Missing" # unexpected Label

# 1) Create followup_status
df["followup_status"] = df.apply(compute_followup_status, axis=1)

# 2) If "No" but follow-up has values -> flag is already "Inconsistent"
# Now blank out the follow-up entries to align with Google Forms Logic
incon_mask = df["followup_status"].eq("Inconsistent")
df.loc[incon_mask, followup_fields] = pd.NA

# become "Not Applicable" after cleaning
df["followup_status"] = df.apply(compute_followup_status, axis=1)

```

Figure 3.15: Structural Missing-Value Handling and Follow-Up Response Validation in Python

- **Structural Missing Handling**

When `issue_experience = "No"`, Google Forms does not display the issue follow-up questions (`issue_timing`, `issue_platform`, `issue_bank`, `issue_types_raw`), so these fields are expected to be blank/NaN in the exported dataset. Issue experience was summarised using `df["issue_experience"].value_counts(dropna=False)`. The output in Figure 3.16 shows 303 respondents answered “Yes” and 123 answered “No” ($n = 426$). This step is important because the survey uses skip logic: respondents who select “No” should not have follow-up issue details, so this distribution provides the baseline for validating structural missingness in the next step.

	count
issue_experience	
Yes	303
No	123

Figure 3.16: Issue experience distribution

The preprocessing therefore keeps these follow-up fields blank and creates a separate column, `followup_status`, to interpret the missingness correctly for Power BI and subsequent analysis. Records with `issue_experience = "No"` are labelled “Not Applicable”. However, if a record is marked “Not Applicable” but contains any follow-up values, it is flagged as “Inconsistent”; in that case, the follow-up entries are replaced with blanks (NaN) to align with the survey logic. The printed output “Total inconsistent rows: 21” (Figure 3.17) indicates 21 responses contained contradictory patterns relative to Google Forms skip logic.

	issue_experience	issue_timing	issue_platform	issue_bank	issue_types_raw
102	No	<NA>	<NA>	<NA>	<NA>
105	No	<NA>	<NA>	<NA>	<NA>
108	No	<NA>	<NA>	<NA>	<NA>
112	No	<NA>	<NA>	<NA>	<NA>
114	No	<NA>	<NA>	<NA>	<NA>
119	No	<NA>	<NA>	<NA>	<NA>
131	No	<NA>	<NA>	<NA>	<NA>
132	No	<NA>	<NA>	<NA>	<NA>
145	No	<NA>	<NA>	<NA>	<NA>
146	No	<NA>	<NA>	<NA>	<NA>
147	No	<NA>	<NA>	<NA>	<NA>
148	No	<NA>	<NA>	<NA>	<NA>
153	No	<NA>	<NA>	<NA>	<NA>
156	No	<NA>	<NA>	<NA>	<NA>
157	No	<NA>	<NA>	<NA>	<NA>
158	No	<NA>	<NA>	<NA>	<NA>
163	No	<NA>	<NA>	<NA>	<NA>
165	No	<NA>	<NA>	<NA>	<NA>
166	No	<NA>	<NA>	<NA>	<NA>
168	No	<NA>	<NA>	<NA>	<NA>
169	No	<NA>	<NA>	<NA>	<NA>
Total inconsistent rows: 21					

Figure 3.17: Inconsistent cases identified and cleaned

Records with `issue_experience = "Yes"` are labelled “Applicable” when at least one follow-up value is present, and labelled “Missing” only when all follow-up fields are unexpectedly blank. The output in Figure 3.18 shows Applicable = 279, Missing = 24, and Not Applicable = 123,

confirming that most “Yes” responses provided follow-up details, while a smaller subset of “Yes” cases had unexpectedly blank follow-ups.

	count
followup_status	
Applicable	279
Not Applicable	123
Missing	24

Figure 3.18: Follow-up status after structural-missing handling

- **Handle missing values**

All questions shown to respondents are set as mandatory in Google Forms, but blanks can still appear after export/import due to parsing errors, formatting conversion, or abnormal submissions. For Likert-scale items, responses must be numeric and fall within the expected 1–5 range; any value that cannot be parsed as a number or falls outside 1–5 is treated as invalid. Because these Likert items form the core constructs for Objective 1, listwise deletion is applied: if a respondent contains any invalid or missing value in the required analysis variables (Section A–C and required Likert items), the entire record is removed from the analysis dataset. This rule does not apply to skip-logic follow-up fields that are blank because *issue_experience* = “No”. The missingness check returned 0 rows with missing values, so no records were removed. Figure 3.19 shows the missing-value and listwise-deletion check for the Objective 1 variables. The result confirms that all 426 eligible records were retained because no required analysis values were missing.

```
Before deletion:
Total rows: 426
Rows that have missing in Objective 1 columns: 0

After deletion:
Total rows: 426
Rows removed: 0

Objective 1 rows after listwise deletion: 426
```

Figure 3.19: Missing-value check and listwise deletion outcome for Objective 1

Step 6: Data Transformation

This transformation step is performed to support Objective 2 (dashboard development) and is not required for Objective 1, because the SPSS analyses for Objective 1 operate on the cleaned survey dataset where each row already represents one completed questionnaire. However, Power BI visualisations for issue monitoring require a different structure because the checkbox field *issue_types_raw* can contain multiple selections from a single respondent.

To make this data dashboard-friendly, the preprocessing script converts *issue_types_raw* from a comma-separated string into a normalised long-format issue table. In Python, the script splits the string into individual options, trims whitespace, and reshapes the field using an explode operation so that each selected issue becomes one row. To support stable relationships between tables, the script generates a simple surrogate key for dashboard relational modelling. This *response_id* is created after cleaning and is used to link the respondent-level dataset with the long-format issue table.

Each issue option starts with a short tag that signals its dimension, such as [REL], [RES], or [EFF]. During preprocessing, pattern matching is used to identify the tag in each issue so the issue can be linked to the correct service quality dimension. After the tag is found, it is removed from the issue text so the final issue label stays short and easy to read. The issue table then stores this label together with key context fields, such as the bank, platform, time of occurrence, and respondent group, which are needed for Power BI filtering and breakdowns. Each issue is saved as one record only. This avoids double counting, keeps totals accurate, and allows results to be aggregated in a consistent way. Figure 3.20 shows the full workflow used to transform the data before modelling and dashboard development.

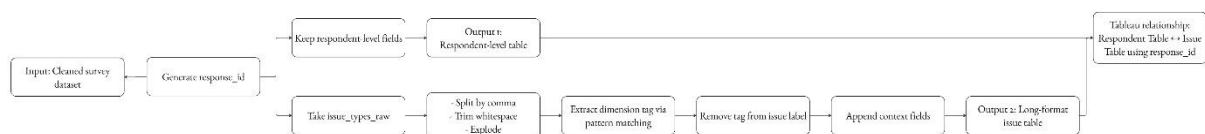


Figure 3.20: Workflow for Data Transformation

Figure 3.21 shows that the final table has 2,956 rows. In other words, the dataset captured 2,956 issue selections from all respondents who answered that section. Some *response_id* values repeat

because one person could tick more than one issue, and each tick is saved as its own row. That layout matters later, because Power BI can then count and group issues in a consistent way. In the last step, the script pulled the dimension code from the issue prefix using regex with str.extract, saved it as *issue_dimension_tag*, and then removed the code from the text using str.replace. This leaves a cleaner *issue_label* that is easier to read in charts and tables.

Issue long table rows: 2956

response_id	issue_experience	followup_status	issue_timing	issue_platform	issue_bank	issue_types_raw	main_bank	ethnicity	age_group	gender	issue_option_raw	issue_dimension_tag	issue_label
0	1	Yes	Applicable	1-4 weeks ago	Both AEON Bank	[REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-40 years	Female	[REL] Scheduled payment not processed as planned	REL	Scheduled payment not processed as planned
0	1	Yes	Applicable	1-4 weeks ago	Both AEON Bank	[REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-40 years	Female	[RES] Unhelpful chatbot / automated replies	RES	Unhelpful chatbot / automated replies
0	1	Yes	Applicable	1-4 weeks ago	Both AEON Bank	[REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-40 years	Female	[PS] Concern about data security	PS	Concern about data security
0	1	Yes	Applicable	1-4 weeks ago	Both AEON Bank	[REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-40 years	Female	[EOU] Hard-to-find functions	EOU	Hard-to-find functions
0	1	Yes	Applicable	1-4 weeks ago	Both AEON Bank	[REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-40 years	Female	[EOU] Difficulty logging in	EOU	Difficulty logging in
0	1	Yes	Applicable	1-4 weeks ago	Both AEON Bank	[REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-40 years	Female	[WBD] Unattractive/confusing colour scheme	WBD	Unattractive/confusing colour scheme
1	2	Yes	Applicable	1-4 weeks ago	Web / Online Banking	[REL] Transaction failure, [REL] Delayed confi...	Citibank Malaysia	Malay	31-40 years	Male	[REL] Transaction failure	REL	Transaction failure
1	2	Yes	Applicable	1-4 weeks ago	Web / Online Banking	[REL] Transaction failure, [REL] Delayed confi...	Citibank Malaysia	Malay	31-40 years	Male	[REL] Delayed confirmation	REL	Delayed confirmation
1	2	Yes	Applicable	1-4 weeks ago	Web / Online Banking	[REL] Transaction failure, [REL] Delayed confi...	Citibank Malaysia	Malay	31-40 years	Male	[REL] Service outage	REL	Service outage
1	2	Yes	Applicable	1-4 weeks ago	Web / Online Banking	[REL] Transaction failure, [REL] Delayed confi...	Citibank Malaysia	Malay	31-40 years	Male	[REL] Transfer credited very late	REL	Transfer credited very late

Figure 3.21: Preview of the Normalised Long-Format Issue Table

3.1.5 Statistical Analysis

Objective 1 focused on examining the relationship between the seven e-banking service-quality dimensions and overall Customer Satisfaction. After the survey responses were cleaned and prepared in Python, the final dataset was imported into IBM SPSS Statistics (Version 30) for analysis. The analysis began with Cronbach's Alpha to assess the internal consistency of the items under each construct. Descriptive statistics were then produced to summarise the respondent profile, service-quality dimension scores, and Customer Satisfaction score. Pearson correlation analysis was used to examine the strength and direction of the relationship between each service-quality dimension and Customer Satisfaction. Multiple linear regression was conducted in the final stage to identify which dimensions made significant unique contributions to Customer Satisfaction after the overlap among the predictors was controlled. Relevant model diagnostics were also reviewed to support the interpretation of the regression results. The findings were reported through a reliability table, descriptive summaries, correlation coefficients, regression coefficients, and

model-diagnostic results. These outputs were interpreted according to Objective 1 and later used to develop the Power BI dashboard and Evidence-to-Action recommendations. Figure 3.22 confirms that the cleaned dataset was successfully saved after preprocessing. This output was then used for the subsequent SPSS analysis and Power BI preparation.

```
import os

OUTPUT_DIR = r"C:/Users/Admin/Downloads/"
os.makedirs(OUTPUT_DIR, exist_ok=True)

issues_long.to_excel(os.path.join(OUTPUT_DIR, "issues_long.xlsx"), index=False)
df_dash.to_excel(os.path.join(OUTPUT_DIR, "respondents_dash.xlsx"), index=False)
df_obj1.to_excel(os.path.join(OUTPUT_DIR, "objective1_cleaned.xlsx"), index=False)

print("Saved to:", OUTPUT_DIR)

Saved to: C:/Users/Admin/Downloads/
```

Figure 3.22: Cleaned data saved

Step 1: Reliability Analysis

After data preprocessing and before examining the relationships between service quality and Customer Satisfaction, Cronbach's Alpha (α) was used to assess the internal consistency of each multi-item construct. It measures whether the items developed to represent the same construct produce reasonably consistent responses. Cronbach's Alpha is commonly used in e-banking service-quality studies to determine whether a scale is sufficiently consistent for further statistical analysis (Khatoon et al., 2020; Ighomereho et al., 2022; Ul Haq & Awan, 2020; Malc et al., 2023).

Cronbach's Alpha was calculated separately for each construct in IBM SPSS Statistics (Version 30) using the original Likert-scale items rather than the composite mean scores. Values around 0.70 or above generally indicate good internal consistency. Values between 0.60 and 0.70 may still be accepted in exploratory or applied social-science research, particularly when a construct contains only a small number of items (Khatoon et al., 2020; Ul Haq & Awan, 2020; Malc et al., 2023). Very high Alpha values, particularly those close to or above 0.90, were interpreted carefully because they may suggest that some items are too similar or repetitive (Ighomereho et al., 2022). Where a construct recorded a lower Alpha value, the item-total statistics and the "Cronbach's Alpha if Item Deleted" results were reviewed to determine whether removing an item would meaningfully improve consistency. Any decision to remove an item also needed to consider whether the construct would still be measured adequately (Ul Haq & Awan, 2020; Malc

et al., 2023). The final reliability table therefore distinguishes acceptable from marginal internal consistency rather than labelling every construct acceptable.

Cronbach's Alpha assesses internal consistency but does not by itself establish construct validity. The questionnaire was adapted from previous studies, and several constructs may overlap conceptually, particularly Efficiency, Ease of Use, Website Design, and System Availability. A factor-based construct-validity assessment, such as exploratory factor analysis with KMO, Bartlett's test, communalities, and factor loadings, was not included in the final analysis. This limits the extent to which the dimensional structure can be confirmed and is acknowledged in Chapter 5.

Step 2: Composite Score Construction

After data preprocessing and Cronbach's Alpha reliability analysis were completed, composite scores were created for each service-quality dimension and for overall Customer Satisfaction. This step converted the item-level responses into construct-level variables that could be used in the subsequent analysis. Each construct was measured using several related Likert-scale items. Analysing every item separately would make the results fragmented and more difficult to interpret because the research objective focuses on service-quality dimensions rather than individual questionnaire statements. Combining related items into one score provided a clearer representation of each construct questionnaire items were measured using a five-point Likert scale, ranging from 1 for Strongly Disagree to 5 for Strongly Agree. Composite scores were calculated in IBM SPSS Statistics (Version 30) by taking the mean of the items belonging to the same construct. This produced one composite score for each of the seven service-quality dimensions and one score for Customer Satisfaction. Mean scores were used because they retained the original scale from 1 to 5, making the results easier to interpret. For example, a score close to 4 indicates general agreement and a relatively positive evaluation of the construct. Table 3.7 presents the composite variables, their corresponding items, and the naming conventions used in the analysis.

Table 3.7: Composite Variables (Mean Scores) and Naming Convention

Constructs	Composite mean
Reliability	<i>Reliability_mean</i>
Responsiveness	<i>Responsiveness_mean</i>

Efficiency	<i>Efficiency_mean</i>
Privacy and Security	<i>PrivacySecurity_mean</i>
Ease of Use	<i>EaseOfUse_mean</i>
System Availability	<i>SystemAvailability_mean</i>
Website Design	<i>WebsiteDesign_mean</i>
Customer Satisfaction	<i>Satisfaction_mean</i>

Construct-level mean scores help stabilise measurement by reducing random item-level noise. Individual Likert items may be influenced by wording sensitivity, temporary respondent mood, or minor misunderstandings; averaging across multiple indicators provides a more stable estimate of the construct. In SPSS, these computed composite variables are stored as new fields in the dataset and therefore remain available for subsequent analysis as long as the SPSS data file is saved after computation. The composite-score columns are then used as the main inputs for Objective 1 statistical testing, descriptive statistics, correlation analysis, and multiple linear regression, where service quality dimensions act as predictors and overall satisfaction acts as the dependent variable. In addition, the same dataset, including the composite score fields is exported from SPSS for use in Power BI, ensuring that the dashboard visualisations and filters are based on the same construct-level measures used in the statistical analysis.

Step 3: Descriptive Analysis

Descriptive analysis summarises and organizes the collected survey data to describe its main characteristics. It answers the question “what does the data look like?” by presenting a clear snapshot of the sample and the key variables before moving on to inferential tests. In quantitative research, descriptive statistics are typically the first stage of analysis because they help researchers understand the distribution of responses, check whether values are reasonable and establish baseline patterns that support later correlation and regression analysis.

Descriptive analysis is conducted after composite scores have been constructed for each dimension of e-banking service quality and overall customer satisfaction in this project. The descriptive analysis is built around three parts. First, the distribution shows how often each response value appears, so it is easier to see whether answers cluster around a few points or are more evenly spread. Second, central tendency gives a simple sense of the “typical” score, such as

the mean or median for a service quality dimension. Third, variability shows how wide the responses are, because a group with mixed opinions will look very different from a group that mostly agrees. Put together, these summaries give an early picture of how respondents rate service quality and overall satisfaction. They also help flag unusual patterns, such as extreme outliers or very uneven spreads, that could affect later tests like correlation and regression.

IBM SPSS Statistics (Version 30) is used to produce descriptive results for two parts of the analysis: the respondent profile and the main study variables. Respondent background is summarised using frequencies and percentages for demographics and e-banking usage, which shows who took part and supports later comparisons across groups in the dashboard. Descriptive statistics are then computed for each construct-level variable, which include the dimension scores and the outcome variable *Satisfaction_mean*. The mean and standard deviation are reported, and when range is useful its minimum and maximum value are shown too. In this paper a higher mean typically represents a more favorable perception, and the standard deviation shows if responses are tightly grouped or are distributed more widely in the responses of respondents.

Simple visual checks are also used to confirm that the output looks reasonable. They help spot obvious data issues early, before they affect the final dashboard. Basic charts are enough for this stage. For example, bar charts can summarise categorical profiles, such as gender, bank, or platform used, while histograms can show how construct scores are distributed. These visuals make it easier to see whether responses are clustered, unusually skewed, or affected by strange values that should be checked before moving to later analysis. These plots can reveal patterns that tables sometimes hide, like most respondents choosing the same scale point repeatedly, or a small number of scores sitting far away from the rest as possible outliers. At the same time, these descriptive visuals give a useful baseline for dashboard design, because they suggest which group comparisons and filters are likely to be meaningful for Objective 2.

Step 4: Correlation Analysis

After descriptive analysis, this project performs correlation analysis to examine the linear relationships between the composite variables. Correlation analysis is a statistical technique used to determine the direction and strength of association between two quantitative variables. It is useful for identifying patterns in the data and for providing an initial understanding of how variables move together, before applying regression modelling.

In this project, Pearson's correlation coefficient (r) is used because the composite variables are treated as continuous scores derived from Likert-scale items. Pearson's r measures the strength of a linear relationship and ranges from -1 to $+1$, where $+1$ indicates a perfect positive linear relationship, -1 indicates a perfect negative linear relationship, and 0 indicates no linear relationship. A positive correlation means that as one construct increases, the other tends to increase, while a negative correlation means that as one increases, the other tends to decrease. To interpret the strength of the correlation, this study follows the correlation interpretation guideline discussed by Papageorgiou (2022), where correlation values are interpreted as follows:

Table 3.8: Interpretation of Pearson Correlation Coefficients

Pearson correlation value	Interpretation
0.00 – 0.19	Very weak relationship
0.20 – 0.39	Weak relationship
0.40 – 0.59	Moderate relationship
0.60 – 0.79	Strong relationship
0.80 – 1.00	Very strong relationship

Table 3.8 summarises the correlation-coefficient ranges used to interpret the strength of Pearson relationships. The absolute value of r determines the strength category, while the sign indicates the direction of the relationship. The interpretation is based on the absolute value of r . Therefore, both positive and negative correlations can be interpreted using the same strength range, while the sign shows the direction of the relationship.

Pearson correlations are computed between each e-banking service quality dimension and customer satisfaction. This provides an initial view of which dimensions are more strongly associated with satisfaction. The correlation matrix also helps identify whether the independent variables are highly related to each other. If predictors are extremely highly correlated, this may indicate possible multicollinearity, which can make regression coefficients unstable or difficult to interpret. Shrestha (2020) explained that multicollinearity occurs when independent variables are highly correlated with one another, which may affect the stability and interpretation of regression results.

In this study, a p-value is used to determine whether the correlation result is statistically significant. According to Kwak (2023), p-values are compared with the significance level to determine whether the result is statistically significant. Therefore, this study uses 0.05 as the significance level. A p-value below 0.05 indicates that the relationship is statistically significant, while $p < 0.001$ indicates very strong statistical evidence.

However, correlation analysis only examines relationships between variables individually. It does not show whether each service quality dimension remains important when the other dimensions are considered together. Therefore, multiple linear regression is conducted in the next step to identify the unique contribution of each dimension to customer satisfaction.

Step 5: Multiple Linear Regression

To test Objective 1, this project applies multiple linear regression to estimate how the service quality dimensions jointly relate to overall customer satisfaction.

Model specification

- Dependent variable (Y): Customer Satisfaction (composite mean score)
- Independent variables (X): the seven service quality dimensions (composite mean scores)

Equation 3.1 presents the multiple linear regression model used in this study as follows:

$$CS_i = \beta_0 + \beta_1 REL_i + \beta_2 RES_i + \beta_3 EFF_i + \beta_4 PS_i + \beta_5 EOU_i + \beta_6 SA_i + \beta_7 WBD_i + \varepsilon_i \quad (3.1)$$

Where:

Table 3.9: Variables and Coefficients in the Multiple Linear Regression Equation

Symbol	Definition
CS_i	Customer satisfaction score for respondent i
β_0	Regression intercept or constant
β_1 to β_7	Regression coefficients for the seven predictors
REL_i	Reliability score for respondent i
RES_i	Responsiveness score for respondent i

EFF_i	Efficiency score for respondent i
PS_i	Privacy and security score for respondent i
EOU_i	Ease-of-use score for respondent i
SA_i	System availability score for respondent i
WBD_i	Website design score for respondent i
ε_i	Error term for respondent i

Table 3.9 defines the symbols and coefficients used in Equation 3.1, including Customer Satisfaction, the seven service-quality predictors, the intercept, and the error term. This analysis is used to estimate the unique contribution of each service quality dimension while controlling for overlap among the other dimensions. The regression results are reported using unstandardized coefficients, standardized beta values, t-values, p-values, and variance inflation factor values. The standardized beta values are used to compare the relative strength of each predictor.

The overall model performance is evaluated using R^2 and adjusted R^2 . R^2 , also known as the coefficient of determination, explains how much variance in the dependent variable is explained by the independent variables in the regression model (Roustaei, 2024). For example, an R^2 value of 0.697 means that 69.7% of the variation in customer satisfaction is explained by the seven service quality dimensions. Adjusted R^2 is also reported because it adjusts the R^2 value based on the number of predictors in the model, making it more suitable when several independent variables are included.

R^2 itself does not have one fixed “significant” value. A higher R^2 indicates stronger explanatory power, but statistical significance is determined using the F-test and p-value. Sureiman and Mangera (2020) explained that the F-test of overall significance checks whether the regression model provides a better fit than a model with no predictor variables. Therefore, if the model p-value is below 0.05, the overall regression model is considered statistically significant. The p-value is also used to determine whether each independent variable is a significant predictor of customer satisfaction. A coefficient p-value below 0.05 means that the predictor has a statistically significant relationship with customer satisfaction after controlling for the other variables. A coefficient p-value above 0.05 means that the predictor is not statistically significant in the regression model.

The ANOVA result was used to determine whether the overall regression model was statistically significant. A significance value below 0.05 indicates that the independent variables, when considered together, significantly predict customer satisfaction. In addition, the Durbin-Watson statistic is used to check the independence of residuals. A value close to 2 suggests that there is no serious autocorrelation problem. VIF values are used to assess multicollinearity among the independent variables. Shrestha (2020) stated that VIF values between 1 and 5 indicate moderate correlation among predictors, while VIF values from 5 to 10 suggest high correlation. Therefore, in this study, VIF values below 5 are considered acceptable, meaning that multicollinearity is not severe.

3.2 Phase 2: Interactive E-Banking Service Quality Dashboard

Phase 2 addresses Objective 2 by developing an interactive Power BI dashboard that visualises and tracks e-banking service quality and Customer Satisfaction results by dimension and segment. The dashboard is based on the cleaned survey dataset and presents the findings through four pages: Executive Overview, Diagnostic Comparison, Issue Monitoring, and Insight Summary. In this project, tracking refers to monitoring the current KPI values, comparing results through filters, and establishing baseline indicators for future review. The dashboard does not automatically receive live complaint, transaction, or system-performance data. However, its data model and KPI structure allow the results to be updated and compared when a new dataset is added in the future. The dashboard follows a decision-first storytelling approach. Rather than presenting every statistic that can be produced from the survey, it focuses on the information that non-technical stakeholders need when deciding what to improve, where problems are concentrated, and which customer groups may require further attention (Cassimally, 2023; Wharton, 2025). This approach reflects business-intelligence research showing that dashboards are more useful when they begin with clear decision needs and avoid overcrowded screens that may weaken understanding or slow decision-making (Lavalle et al., 2025; Ahadi et al., 2024; Sethi, 2023).

3.2.1 Decision Use-Cases

A decision-first structure is used because bank stakeholders typically need direct answers on what to improve, where the gaps are, and which customer groups are most affected (Wharton, 2025). To keep the dashboard focused and usable, the views are organized around a small set of decision questions that translate survey results into clear comparisons. This reflects decision-first guidance that requirements should be captured as decision needs and then translated into a task flow users can follow on the dashboard (Lavalle et al., 2025). The four primary decision use-cases are summarized in Table 3.10, covering improvement prioritisation, platform targeting, segment identification, and issue-monitoring insights where issue-related data is available.

Table 3.10: Decision Use-Case Requirements Summary

Primary user	Decision supported	Dashboard question (requirement)	Dashboard output / view
--------------	--------------------	----------------------------------	-------------------------

Service quality team	Prioritize improvement areas	Which service-quality dimension scores lowest?	• Dimension ranking • Gap summary for quick targeting
Product/e-banking team	Decide where fixes should focus	Do results differ by platform (mobile app vs web), and where are the gaps largest?	• Platform comparison view with filters for bank/segment
Management	Identify vulnerable groups	Which demographic or usage groups report lower satisfaction?	• Segment comparison view (demographics/usage) to support targeted action
Service monitoring / improvement lead	Target issue handling	Which issue types appear most often, and in what context (timing/platform), if captured?	• Issue type frequency • Context breakdown to guide specific fixes

3.2.2 KPI & Measurement Design

The KPI structure follows directly from the decision questions and from the survey's construct design. The main outcome KPI is Overall Customer Satisfaction, computed as a composite mean score. The main driver KPIs are the composite mean scores for each service quality dimension, so stakeholders can see which part of service quality is linked to stronger or weaker satisfaction. The dashboard also includes issue-monitoring KPIs that translate reported problems into measurable signals. These include issue prevalence (%) and the most common issue types. They also include issue context such as when the issue happened and which platform the user was using, if the survey captured these fields. Dashboards become easier to interpret when measures are clearly defined and supported by organized data preparation and integration steps, instead of being added casually during visual design (Gonçalves et al., 2023). Thus, the dashboards

deliver more value when definitions are consistent and governed, because inconsistent measures reduce trust and weaken action.

The project prioritises KPIs, so each dashboard page serves one clear purpose. The dashboard is organised into four pages with a clear sequence. The overview page provides a rapid look at overall satisfaction and primary dimension scores. The diagnostic page aids in gap checks across groups (platform type, bank group, key demographic segments). The action page then redirects to areas needing improvement, such as, the highest-reported type of glitch is ranked and where it happens most often. But the structure alone will not suffice if the layout grows cluttered. Dashboards are prone to go cluttered with additional charts added without a clear directive. Sethi (2023) cautions that views should be kept brief and accessible to scan for non-technical users, as excessive visuals can breed “dashboard hell” and decrease actual deployment. Ahadi et al. (2024) similarly emphasizes clean layouts and mindful chart selection to mitigate clutter and avoid inaccurate portrayal. Table 3.11 summarises the dashboard KPIs, their reporting purposes, and the filter or breakdown options available for each service-quality area.

Table 3.11: KPI Dictionary and Measurement Rules

Construct / Area	KPI name (dashboard label)	Used for (Overview / Diagnostic / Action)	Slice options (filters / breakdowns)
Outcome KPI	Overall Customer Satisfaction (Mean)	Overview; Diagnostic	Bank, Platform, Demographics, Usage group
Driver KPIs	Reliability (Mean)	Overview; Diagnostic	
	Responsiveness (Mean)		
	Efficiency (Mean)		
	Security and Privacy (Mean)		
	Ease of Use (Mean)		
	System Availability (Mean)		

	Website Design (Mean)		
Issue KPIs	Issue Prevalence (%)	Overview; Diagnostic; Action	
	Top Issue Types (Count / %)	Diagnostic; Action	
	Issue Timing Distribution (%)		
	Issue Platform Distribution (%)		
Comparative KPIs	Satisfaction Gap (by group)	Diagnostic	
	Dimension Gap (by group)		

3.2.3 Data Understanding and ETL Preparation

There is no modification of the test constructs on Phase 2 because Phase 1 finished scores of all constructs are already created. Instead, Phase 2 prepares an analysis dataset that will be smoothly handled by a dashboard builder, Power BI. A repeatable ETL workflow reshapes the raw data, preparing the data in a structured form for dashboard development (Hjelle et al., 2024). ETL documentation explains the variables and the KPI's calculation step by step. Data quality validation includes checking missing values and discrepancies. The dataset's final version enables Power BI operations like filtering and drilling while maintaining logical consistency and decision-making utility (Gonçalves et al., 2023; Zamil et al., 2024; Sethi, 2023).

In order to create a repeatable dashboard workflow, the ETL process is designed for reusability. That way, every data refresh generates the same behaviour within the Power BI platform. During extraction, responses to the survey are exported with a consistent layout. Column names are stable and responses are available with the same categories in each cycle. In transformation, the data is standardised. This involves aligning labels, setting data types with a consistent method and performing validation checks to catch the most popular errors in the data

early in the process. This matters because the smallest mistake in input can distort the KPI cards, the trendlines or the comparisons in the group on the dashboard (Zamil et al., 2024; Sethi, 2023). After cleanup, data is reshaped for reporting. The dimension-level composite scores validated in Objective 1 remained intact and the responses to the multi-select issues of each question were converted to a normalised long-format issue table. The latter is a format that supports the accurate counting and grouping of the problems of each type, by channel, bank and time window. As it was developed to improve the reliability of drill-down procedures, it is also the best format for reporting the results of each KPI (Gonçalves et al., 2023). During the loading step, the workflow generates reports to the dashboard in the form of two tables: a respondent-level table and an issue fact table. The first keeps the totals more stable throughout the pages, whereas the latter is less likely to result in a duplicated result when a filter is applied (Sethi, 2023; Zamil et al., 2024). Figure 3.23 illustrates the ETL pipeline used to extract the survey data, standardise and transform the variables, and load dashboard-ready tables into Power BI. The workflow also includes validation checks to maintain consistent KPI results.

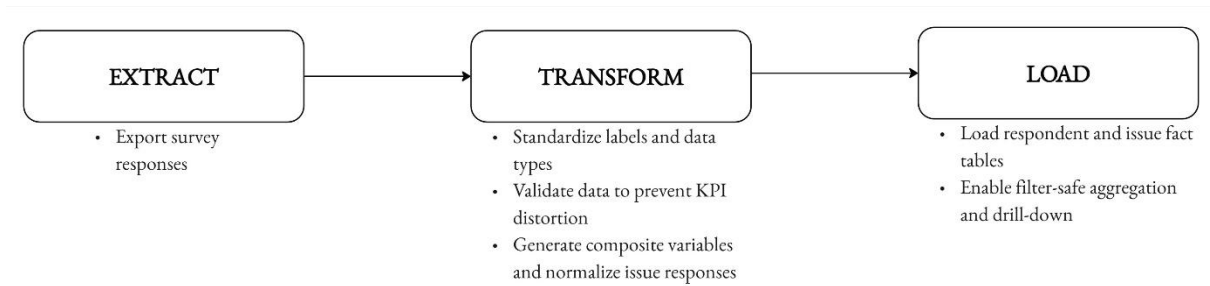


Figure 3.23: ETL Pipeline for Power BI Dashboard Preparation

Figure 3.24 shows how the dashboard-ready respondent and issue tables were loaded into Power Query and validated before being used in the Power BI data model.

The screenshot shows the Power Query interface with a table loaded from a source. The table has columns: response_time, consent_agreed, age_group, nationality, and ethnicity. The data is as follows:

	1.2 response_time	AB consent_agreed	AB age_group	AB nationality	AB ethnicity
1	45987.82971	Yes, I agree	31–40 years	Malaysian	Chinese
2	45987.8341	Yes, I agree	31–40 years	Malaysian	Malay
3	45987.83909	Yes, I agree	41–50 years	Malaysian	Indian
4	45987.84076	Yes, I agree	31–40 years	Malaysian	Indian
5	45987.84098	Yes, I agree	41–50 years	Malaysian	Malay

Figure 3.24: Loading and validating the dashboard-ready tables in Power Query

3.2.4 Data Model Design

The raw export of ETL survey data is not directly imported into Power BI, and it is not maintained in the original shape. The raw export data is generally wide and unstructured, which may increase errors in filtering and summarisation. Instead, it needs to be prepared into a cleaner analytics model with separate tables and clearer field names, making grouping and comparisons easier to manage in the dashboard and support more consistent interpretation (Gonçalves et al., 2023; Zamil et al., 2024). It will also help avoid reporting troubles, such as totals changing unexpectedly when filters are applied or different charts displaying conflicting values for the same KPI. By preparing the data model before building the dashboard, the results should be more stable across pages and across user segments.

The data are organised in a star scheme to keep the analysis and reporting in the dashboard well-structured. In a star schema, descriptive information, such as profiles of respondents and labels for categories, are kept in separate tables, and the results measured are located in a fact table. This is important because filtering and summary calculations are kept stable when users explore the dashboard. Figure 3.25 illustrates the schema of the project. Linked tables allow totals to remain constant when filtering and drilling down for data. The main table in this project is *FactResponses*. It has one record for each completed questionnaire. It contains a *response_id* that was created in Python as the key for linking and a *submission_date* field that can be used for filtering by date and viewing trends. It also contains profile fields, fields related to issues being raised, and the resulting compositemean score that will be used as one of the main KPIs (Gonçalves et al., 2023; Zamil et al., 2024).

Issue monitoring is kept in a separate table because one respondent can tick more than one issue type. If all of those choices were forced into the same survey row, the dataset would become cluttered very fast and harder to analyse. Issue data is stored in a separate table called *FactIssues_Long* to keep the structure clean. The table has one row for each issue selected by a respondent. Each row links back to the main survey record using *response_id* as a foreign key. The table also keeps the main reporting fields, such as *issue_dimension*, *issue_type_raw*, *issue_timing*, *issue_platform*, and *issue_bank*. If total counts are needed in the dashboard, an *issue_count* field can also be added.

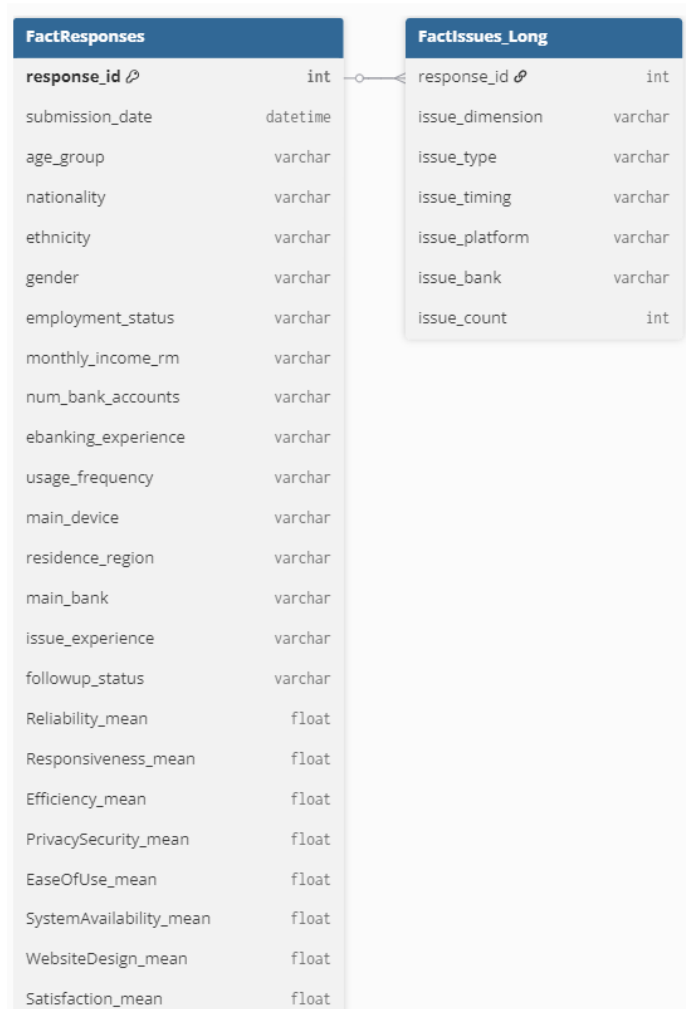


Figure 3.25: Star-schema style dashboard model diagram

The relationship between the two tables relies on a single respondent identifier. Power BI uses this link to move from high-level patterns in satisfaction and dimensions to detailed issue views without changing the underlying means or double-counting respondents. When users select a specific issue type, Power BI filters the issue table first and then highlights the linked respondents in the main table. Mean scores for satisfaction and each service quality dimension still come from the respondent table, so they stay accurate even when the dashboard focuses on a narrow subset. This separation of roles supports consistent KPI calculations and keeps drill-down views responsive and interpretable for managers and analysts (Zamil et al., 2024; Sethi, 2023; Gonçalves

et al., 2023). Figure 3.26 shows the one-to-many relationship between the respondent-level table and the issue-level table, linked through the response identifier to support filtering without double-counting respondents.

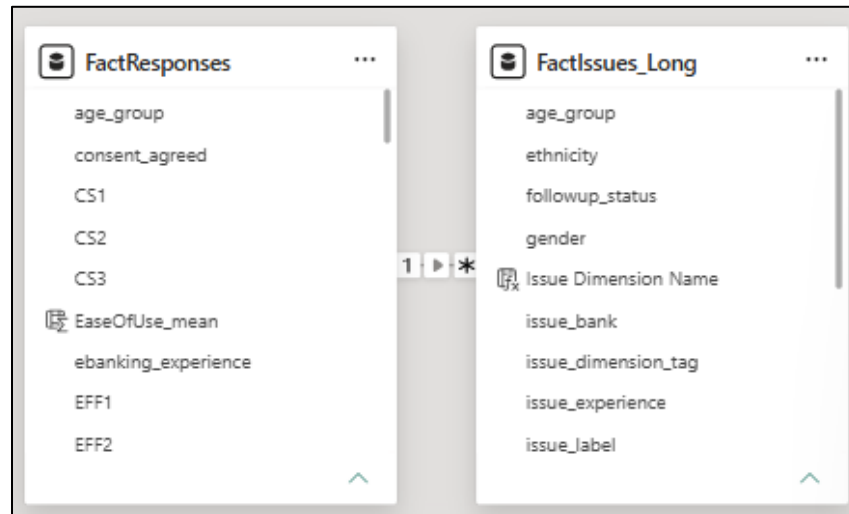


Figure 3.26: One-to-many relationship between the respondent and issue tables in Power BI

3.2.5 Dashboard Mock-up and Storyline

Figure 3.27 visualizes the planned dashboard storyline as a page-by-page blueprint, showing how the user moves from an executive overview to diagnostic views and action-focused issue monitoring. Before the Power BI dashboard is built, the project prepares a small mock-up (wireframe) to test both the data and the story. The mock-up checks whether the cleaned dataset can produce the intended KPIs, and whether a non-technical user can understand the page flow without extra explanation (Gonçalves et al., 2023).

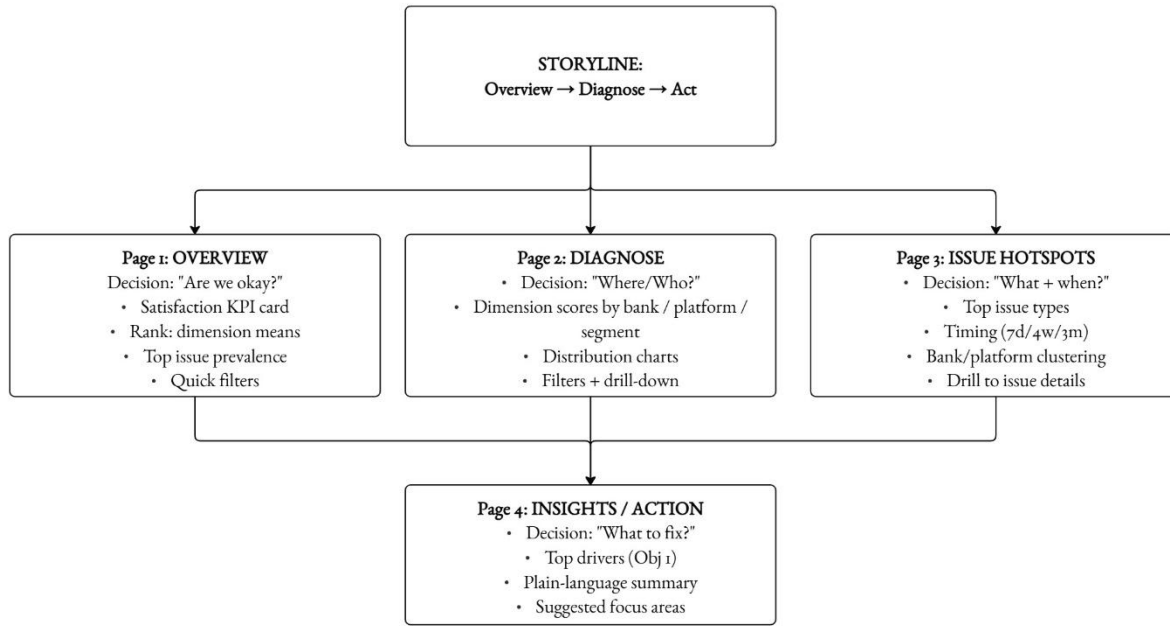


Figure 3.27: Dashboard Mock-up Storyline Blueprint

Figure 3.28 converts the blueprint into a thumbnail storyboard, so the intended page flow and content emphasis can be understood at a glance before any dashboard is implemented. The storyline uses a simple “Overview → Diagnose → Act” path. This page order follows decision-first storytelling by moving from a decision signal (overview), to structured comparisons (diagnosis), and then to action-oriented details (issue hotspots), supported by short explanations and consistent definitions (Cassimally, 2023; Wharton, 2025; Lavallo et al., 2025; Liu et al., 2024).

The first page gives a fast executive scan. It uses one overall satisfaction KPI card, a ranked view of service quality dimension mean scores and a compact view of issue prevalence, such as the top issue types. This layout supports quick monitoring, which is a common dashboard purpose in performance management settings (Tantipisut & Yenradee, 2020).

The second page is designed for diagnosis. It compares service-quality dimension scores across banks, platforms, and key respondent segments. Rather than relying on a single average score, the page employs distribution style visuals to show ratings' dispersion. This helps in ascertaining whether feedback is predominantly identical or vastly scattered, as well as making discrepancies among groups more obvious when opinions are noticeably higher or lower. As per a straightforward principle, dashboards should "focus on patterns that emerge quickly" rather than having users peruse lengthy tables line by line (Ahadi et al., 2024).

The third page is for issue monitoring and hotspot checking. It shows which problems happen most often and where they concentrate. The visuals highlight the top issue categories, when they usually happen, and whether they are more common for certain banks or platforms. Filters are available so users can narrow the view to a specific segment when deeper checking is needed. Navigation is kept straightforward as well. Users start with the overall picture and then move into the details that support follow-up action, which matches how banking dashboards are commonly used (Rachansa & Meditya, 2024).

The final page is the insight page. It summarises the Objective 1 results into clear points that users can act on. For example, the page points out which service-quality dimensions seem to matter more, based on the correlation patterns and the regression results. It also includes a short summary box that suggests a few planning priorities, so users do not need to guess what the numbers are trying to tell them. This step is not only about presentation. Decision quality improves when users can understand and use the information, not simply when the analysis has been completed (Hjelle et al., 2024).

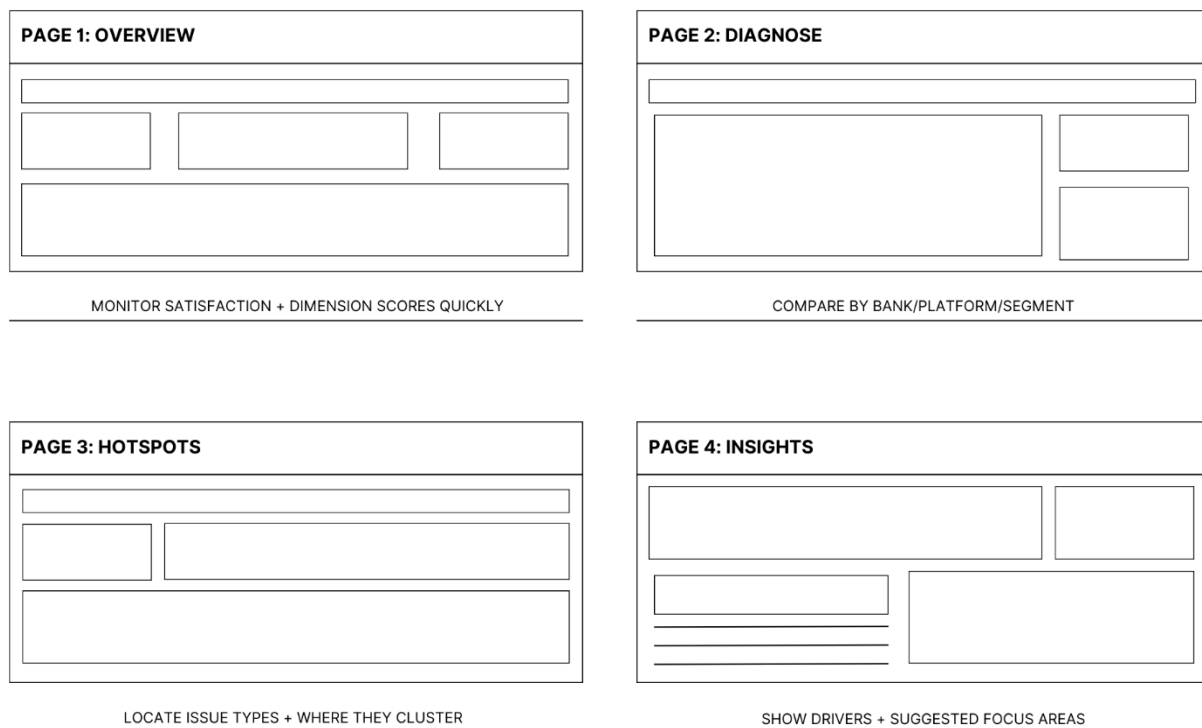
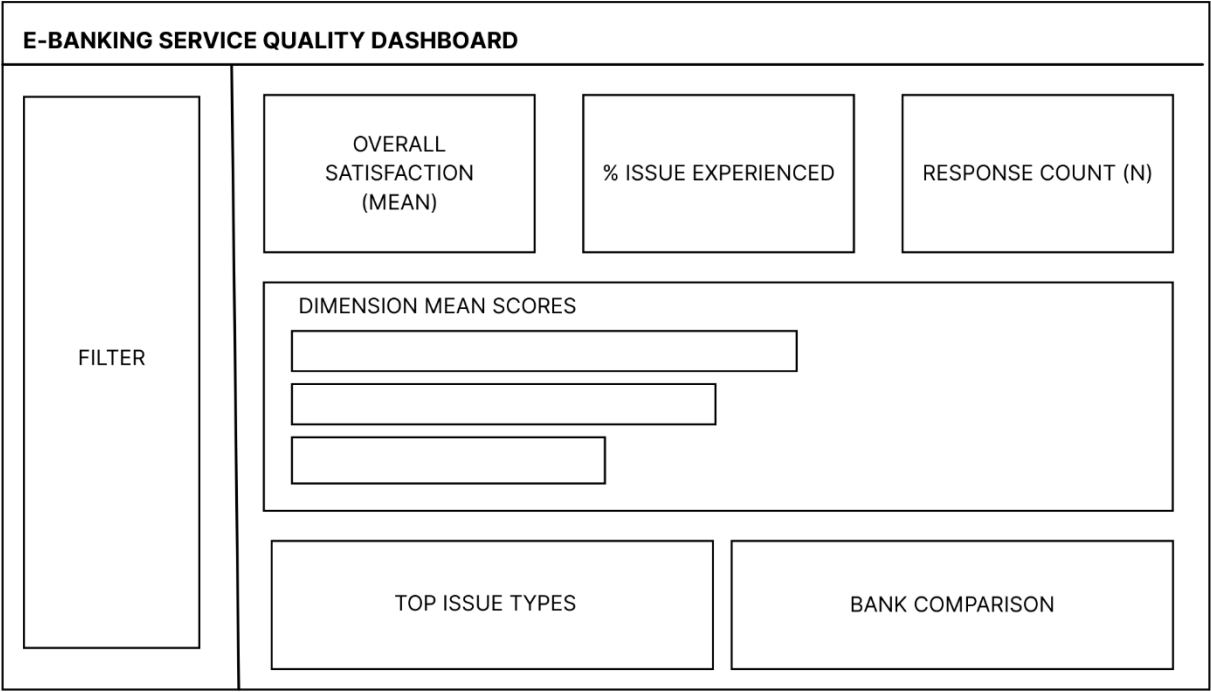


Figure 3.28: Thumbnail storyboard showing dashboard storyline

The mock-up follows a few practical ideas from the dashboard literature about adoption and readability. Each page is limited to key KPIs and consistent layouts, while filters and drill-down interactions handle deeper detail (Zamil et al., 2024; Sethi, 2023). To illustrate how the dashboard will present results to decision makers, Figure 3.29 shows a mock-up of the landing page (Page 1). Page 1 is used as the example because it provides the overall summary view and acts as the main entry point for interpreting the findings.



The first set of visuals supports distribution and comparison. The dashboard shows how satisfaction and service-quality scores vary across common groupings such as bank, platform and selected demographic segments. The design uses KPI cards for headline values and ranked bar charts to compare mean scores across dimensions, because these formats help users see “who is higher” and “who is lower” at a glance (Gonçalves et al., 2023; Ahadi et al., 2024).

The second set of visuals focuses on relationships and allows for straightforward follow-up checks. Service quality scores appear with issue reports and respondent segments, making practical questions an easy task to do from a single view. For example, a drop in reliability can be compared with alterations in issue categories, and the same pattern can be analyzed across channels like the mobile app and web version. This layout facilitates this process through crosstab style breakdowns, grouped comparisons, and a guided drill down path. Users start with a summary KPI view and then drill into smaller segments only when they need deeper detail (Rachansa & Meditya, 2024).

The interaction design is focused on non-technical users, with simple wording and commonly used labels for key metrics rather than technical terms. (Fernandez Nieto et al., 2022). The filters for bank, platform, and respondent segment help users to narrow down the view quickly without having to separately consult files. Additionally, the drill down is included within the same dashboard; thus, transitioning between overview to detailed breakdowns does not require additional reports or tools. To ensure clarity, the dashboard uses accessible terminology such as “drivers of satisfaction” and “higher impact dimensions”, instead of complex statistical terminology. Short tooltips are added to explain each KPI and its calculation (Sethi, 2023), while a KPI glossary is added to the workbook to ensure consistency in definitions and minimize miscommunication across teams (Ahadi et al., 2024). Performance is considered part of usability as well. If the dashboard becomes slow, Power BI extracts or more aggregated tables can be used to keep filtering and drill-down responsive.

3.2.7 Dashboard Implementation

The final interactive dashboard is developed in Power BI only after the mock-up confirms that the selected KPIs, visuals, and page flow can support the intended storyline. It connects to the cleaned, dashboard-ready tables produced by the ETL pipeline (Gonçalves et al., 2023). In Power BI, calculated fields are created to match the agreed KPI definitions, including overall satisfaction

(composite mean), mean scores for each service quality dimension, issue prevalence measures and subgroup indicators for comparisons by bank, platform, and respondent segments. The design is meant for non-technical stakeholders, so each page keeps only a small number of top-level metrics (Ahadi et al., 2024; Sethi, 2023). Users can start from overall satisfaction, move to the service quality dimension scores, and then review issue-level details using filters and drill-down controls. This navigation follows the decision questions in Table 3.10 and supports a consistent task flow across pages (Wharton, 2025; Lavallo et al., 2025). The final outputs are a Power BI workbook that is published for review and demonstration, and a short user guide that explains each page, the KPI meanings, and the key filters and drill paths needed to answer the decision questions set earlier. Figure 3.30 shows an example DAX measure used in Power BI to calculate a dashboard KPI consistently across filters and report pages.

```
1 Issue Prevalence % =  
2 DIVIDE(  
3     CALCULATE(  
4         DISTINCTCOUNT(FactResponses[response_id]),  
5         FactResponses[issue_experience] = "Yes"  
6     ),  
7     DISTINCTCOUNT(FactResponses[response_id])  
8 )
```

Figure 3.30: Example of DAX Measure in Power BI

Dashboard evaluation in this project focused on data reconciliation and functional checks, including respondent totals, KPI consistency, filters, and reset behaviour. The dashboard was not tested through sustained use by bank managers, analysts, IT staff, customer-service teams, or UX specialists. Objective 2 therefore demonstrates successful dashboard development and baseline decision support, but it does not establish that the dashboard improves decision quality, reduces response time, or produces operational benefits in an actual banking organisation.

3.3 Phase 3: Recommendation and Insight

Phase 3 turns the outputs from Objective 1 and the dashboard patterns from Phase 2 into concrete, evidence-linked suggestions for improving Malaysian e-banking service quality. This phase treats the findings as decision-support inputs rather than adding new interpretation. In practical terms, Phase 3 applies an Evidence-to-Action mapping approach that frames evidence use as a structured process for closing the research–practice gap, rather than a one-off reporting exercise (Ejaz et al., 2025; Das et al., 2023). Each recommendation is traceable to a specific empirical signal and translated into an implementable action with ownership and monitoring (Chamathi Kossinna et al., 2025; Nørgaard et al., 2025).

No additional data are collected at this stage. Phase 3 extends beyond reporting by applying a structured mapping process that translates each key finding into actionable steps. Every action is explicitly linked to one or more results, assigned to the stakeholder team responsible for implementation, and tied to KPIs that can be monitored through the Power BI dashboard. This strategy is in harmony with implementation guidance focused on local adaptation and participative consultation, as interventions are more effective if teams can tailor them to the context to their needs and measure outcomes along the way in iterative cycles (Ejaz et al., 2025; Nørgaard et al., 2025). In this project, Phase 3 complements the support provided in Objective 3, creating a prioritized list of recommendations that are owned by teams, quantifiable, and still accessible for long-term review (Neumann et al., 2023).

3.3.1 Evidence-to-Action Mapping Framework

To make the recommendations defensible, Phase 3 uses a simple mapping rule. Each action should be related to an identifiable “know–do gap” or an observable problem signal and then translate it into something the team can realistically do (Chamathi Kossinna et al., 2025; Nørgaard et al., 2025). The pathway is led using the E2A evidence-building and learning cycle, which is employed as an overlay to maintain consistency in the pathway and trackability of the decisions (Das et al., 2023). For this work, the E2A steps are implemented as follows:

Table 3.12: Evidence-to-Action Mapping Stages and Operational Roles in Phase 3

E2A steps	Operational Role in Phase 3
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Co-design	Use Objective 1 + Phase 2 outputs to prioritise the main problems and define what the signals mean in context.
Co-produce	Build the Recommendation Mapping Matrix by linking evidence triggers to actions and defining the target KPIs.
Co-act	Assign ownership and entry points by specifying which stakeholder team is responsible for implementing each action.
Co-review	Use dashboard-based monitoring to support learning and revision by tracking KPIs and uptake indicators over time.

Table 3.12 summarises the stages of the Evidence-to-Action mapping process and identifies the operational role of each stage. It shows how evidence is converted into traceable recommendations and monitoring responsibilities. This reflects how work commonly starts from problem identification and barrier-aware planning, rather than generic solution lists (Chamathi Kossinna et al., 2025; Ejaz et al., 2025). It also matches structured decision approaches that make the pathway from evidence to recommendation explicit and auditable, which improves transparency when recommendations are reviewed by stakeholders (Neumann et al., 2023).

In this step, every proposed action must be tied to at least one survey result or one specific dashboard view. When issue-monitoring variables are not available, the “evidence trigger” is defined in a different way. It is based on the satisfaction of drivers that are supported by the statistical results, together with clear gaps in dimension performance that can be seen in the dashboard. In these cases, the “Affected scope” is defined using the segment filters available in the dataset. Scope is determined based on which groups are affected. For example, it could be customer type, channel, region, or product group. Scope is not determined by the frequency with which a given issue category occurs.

The Recommendation Mapping Matrix captures relationships between the findings, the actions, the unit responsible, and the KPIs that are used to monitor the plan. This system is based on three practical principles. First, actions are adapted to the bank’s local context, so the plan is

based on actual constraints and internal processes, rather than a one-size-fits-all template. Second, each recommendation has a defined owner, which ensures that it is delivered, along with accountability across departments. Third, progress is measured using KPIs such as turnaround time, error rates, customer complaint volume, and service quality scores, which provide empirical evidence for monitoring instead of relying on assumptions (Ejaz et al., 2025; Nørgaard et al., 2025).

3.3.2 Prioritisation Method

Once the recommendations have been mapped, Phase 3 ranks them using two main criteria: impact and prevalence. Impact refers to how strongly each service-quality dimension is related to Customer Satisfaction. A dimension is considered to have high impact when it makes a statistically significant unique contribution in the multiple linear regression model. The standardised beta coefficient is used to compare the relative strength of the significant predictors. A dimension may still have a strong correlation with Customer Satisfaction but become non-significant in the regression model. In this case, it is treated as having moderate impact because part of its relationship with satisfaction may overlap with other service-quality dimensions.

Prevalence refers to how clearly and widely a service weakness appears in the descriptive statistics and dashboard findings. It is assessed using the dimension mean score, improvement gap, issue frequency, number of affected respondents, and whether the weakness appears consistently across dashboard segments. Issue counts are treated as supporting evidence rather than the only basis for prioritisation. This is because respondents could select more than one issue, and the number of issue options was not the same for every dimension.

When issue-monitoring information is unavailable for a particular subgroup, prevalence can still be assessed using other indicators. These include the size of the service-quality gap, the percentage of respondents who gave ratings below a defined threshold, and whether the same weakness appears across different dashboard segments.

Looking at both impact and prevalence gives a more balanced basis for setting priorities. A dimension may have a strong statistical relationship with satisfaction but show relatively few operational problems. Another dimension may not remain significant in the regression model but may still have a low mean score or a large number of reported issues. Considering both forms of evidence helps the project identify actions that are relevant to satisfaction and meaningful to users.

This follows structured evidence-to-decision thinking, where recommendations are based on clear criteria and transparent judgement rather than a single result (Neumann et al., 2023). The recommendations are divided into three practical priority tiers.

P1 Immediate includes areas where both impact and prevalence are high. These dimensions make a significant unique contribution to Customer Satisfaction and also show a clear performance weakness or widespread operational problem. They should receive immediate attention because improvement may support satisfaction while addressing problems experienced by a meaningful number of users.

P2 Next includes areas where either impact or prevalence is high, but not both, or where the evidence is mixed. This may include a significant predictor that currently has a relatively positive mean score or lower issue frequency. It may also include a dimension with a clear performance weakness or high issue frequency that does not make a significant unique contribution in the regression model. These areas remain important and may be addressed after P1 actions or alongside them when resources and operational risks allow.

P3 Monitor includes areas where both impact and prevalence are relatively low or where the available evidence is not strong enough to support immediate action. These areas are not ignored. They are monitored through the dashboard so that any emerging changes can be identified when more data become available.

This method prevents priorities from being based only on statistical significance or issue frequency. The final ranking considers the dimension's statistical contribution, current service performance, and the operational problems shown in the dashboard.

Table 3.13: Evidence-to-Action Mapping Matrix Components

Recommendation Mapping Matrix Components	Explanation	References
Evidence trigger	The specific empirical result or dashboard signal that indicates a problem or improvement opportunity.	Chamathi Kossinna et al. (2025), Nørgaard et al. (2025), Das et al. (2023)

Target KPI / dimension	The service-quality construct or KPI that the recommendation aims to improve.	Chamathi Kossinna et al. (2025)
Affected scope	The context where the issue is concentrated, defined using statistical analysis results and dashboard filters.	Ejaz et al. (2025), Nørgaard et al. (2025), Neumann et al. (2023)
Proposed action	The concrete improvement action designed to address the evidence trigger and improve the target KPI/dimension.	Chamathi Kossinna et al. (2025), Das et al. (2023)
Owner team	The stakeholder group responsible for implementing the action.	Ejaz et al. (2025), Chamathi Kossinna et al. (2025), Das et al. (2023)
Expected KPI movement	The expected direction of KPI change if the action is implemented.	Chamathi Kossinna et al. (2025)
Success metrics & monitoring reference	The KPI(s) used to judge success and the Power BI page/view/filter used to monitor progress.	Chamathi Kossinna et al. (2025), Nørgaard et al. (2025)
Priority tier and confidence	The recommendation's priority level (P1–P3) and the strength of evidence supporting it, based on the consistency and clarity of the empirical and dashboard signals.	Neumann et al. (2023)

Table 3.13 defines the components recorded in the Evidence-to-Action mapping matrix, including the evidence trigger, affected scope, recommended action, owner team, expected KPI movement,

and monitoring view. This structure prevents recommendations from becoming generic suggestions, because every item must be traceable and measurable.

3.3.3 Monitoring and Deliverables

Phase 3 produced the following outputs:

1. A completed Recommendation Mapping Matrix that links each evidence trigger to a proposed action, the target KPI/dimension, expected KPI movement, ownership, and a Success metrics & monitoring reference (Power BI page/view/filter).
2. A prioritised recommendation list (P1–P3) that assigns clear ownership to the relevant stakeholder teams.
3. The project provided an insight narrative that synthesised the empirical findings and dashboard insights into data-driven recommendations to improve e-banking service quality, highlighting the highest-impact service-quality dimensions, the most affected customer groups, and the priority actions for immediate attention.

Hence, the final set works as a practical guide for decision making rather than just a general discussion of the project results.

3.4 Summary

This chapter described the three-phase methodology used to examine e-banking service quality and Customer Satisfaction among users in Malaysia. In Phase 1, the raw survey responses were cleaned and prepared in Python before being imported into IBM SPSS Statistics (Version 30). Cronbach's Alpha was used to assess the internal consistency of each construct. Descriptive statistics, Pearson correlation, and multiple linear regression were then conducted to identify the service-quality dimensions associated with Customer Satisfaction and determine which dimensions made significant unique contributions. Phase 2 focused on preparing the data for the Power BI dashboard. An ETL process was used to produce dashboard-ready tables and maintain consistent KPI calculations. The data model included a respondent-level table and a separate long-format issue table. This structure allowed the dashboard to calculate service-quality and satisfaction scores accurately while supporting detailed analysis of reported issues. Data-reconciliation and functional checks were also carried out to ensure that totals, filters, and drill-down results worked as intended. Phase 3 translated the statistical results and dashboard findings into practical recommendations through an Evidence-to-Action mapping process. Each recommendation was linked to a specific finding, assigned to a responsible team, connected to a measurable KPI, and prioritised according to statistical impact, service performance, issue prevalence, and potential severity. Together, the three phases provided a structured path from data collection and analysis to dashboard visualisation and service-improvement planning.

Chapter 4: Results and Discussion

4.0 Overview

This chapter presents the findings in three phases, each linked to a research objective. Phase 1 describes the respondents and reports the reliability, descriptive, Pearson correlation, and multiple linear regression results used to identify the service-quality dimensions associated with customer satisfaction. Phase 2 examines the Power BI dashboard outputs, including overall performance, segment comparisons, issue patterns, and operational hotspots. Phase 3 brings the statistical and dashboard evidence together to develop prioritised recommendations and a monitoring plan. Numerical results are reported in their relevant subsections, while the discussion focuses on interpretation, comparisons with earlier studies, practical implications, and study limitations.

4.1 Phase 1: Key Service Quality Drivers

4.1.1 Respondents' Demographic Profile

The analysis was based on 426 valid responses. The respondent profile covers both demographic and banking-related characteristics: age, gender, nationality, ethnicity, employment status, monthly income, number of bank accounts, e-banking experience, usage frequency, main device, residence region, and main e-banking provider.

Table 4.1: Respondents' Age Group

Variable	Category	Frequency	Percentage (%)
Age Group	18–20 years	39	9.2
	21–30 years	172	40.4
	31–40 years	111	26.1
	41–50 years	70	16.4
	Above 50 years	34	8.0

Table 4.1 summarises the age distribution of the 426 respondents. The largest group was aged 21–30 years, followed by respondents aged 31–40 years. The largest age group was 21–30 years, with 172 respondents (40.4%), followed by those aged 31–40 years, with 111 respondents (26.1%). Together, these groups show that the sample was mainly made up of young and middle-aged adults who are likely to have regular exposure to digital services. Respondents aged 18–20

years accounted for 39 cases (9.2%), while 70 respondents (16.4%) were aged 41–50 and 34 (8.0%) were above 50. Although older users formed a smaller share of the sample, their views remain useful because their expectations of ease of use, system availability, and security may differ from those of younger users.

Table 4.2: Respondents' Gender

Variable	Category	Frequency	Percentage (%)
Gender	Male	176	41.3
	Female	250	58.7

Table 4.2 presents the gender distribution of the respondents. Female respondents represented 58.7% of the sample, while male respondents represented 41.3%. Female respondents made up 58.7% of the sample (250 respondents), while 176 respondents (41.3%) were male. Both groups were therefore represented, although the sample included more women. Their inclusion provides a broader view of e-banking satisfaction because usage patterns, preferences, and concerns may vary between users.

Table 4.3: Respondents' Nationality

Variable	Category	Frequency	Percentage (%)
Nationality	Malaysian	384	90.1
	Non-Malaysian	42	9.9

Table 4.3 summarises the respondents' nationality. Most respondents were Malaysian, which is appropriate for the study's focus on the Malaysian e-banking context. Most respondents were Malaysian, with 384 respondents (90.1%), while 42 respondents (9.9%) were non-Malaysian. Since the study focuses on Malaysia's e-banking context, the high proportion of Malaysian respondents is appropriate and relevant to the research objective.

Table 4.4: Respondents' Ethnicity

Variable	Category	Frequency	Percentage (%)
Ethnicity	Malay	143	33.6
	Chinese	191	44.8

Variable	Category	Frequency	Percentage (%)
	Indian	92	21.6

Table 4.4 presents the ethnic composition of the sample. Chinese respondents formed the largest group, followed by Malay and Indian respondents. Chinese respondents formed the largest ethnic group, with 191 participants (44.8%). Malay respondents accounted for 143 participants (33.6%), while 92 participants (21.6%) were Indian. The sample therefore included users from the three ethnic groups measured in the questionnaire, giving the study a more varied view of e-banking experiences in Malaysia.

Table 4.5: Respondents' Employment Status

Variable	Category	Frequency	Percentage (%)
Employment Status	Student	192	45.1
	Employed	234	54.9

Table 4.5 summarises the respondents' employment status. The sample included both employed users and students, providing perspectives from two important e-banking user groups. The sample included 234 employed respondents (54.9%) and 192 students (45.1%). These groups may use e-banking in different ways. Employed users may rely on it for salary management, bill payments, transfers, and other financial commitments, whereas students may use it more often for daily spending, online purchases, and receiving or sending funds. Their combined responses provide useful perspectives on service quality and satisfaction.

Table 4.6: Respondents' Monthly Income

Variable	Category	Frequency	Percentage (%)
Monthly Income	Below 2,000	113	26.5
	2,001 – 4,000	122	28.6
	4,001 – 6,000	106	24.9
	6,001 – 8,000	47	11.0
	Above 8,000	38	8.9

Table 4.6 presents the respondents' monthly-income distribution. The largest group earned RM2,001–RM4,000 per month, followed by respondents earning below RM2,000. The largest income group earned RM2,001–RM4,000 per month, with 122 respondents (28.6%). This was followed by respondents earning below RM2,000, with 113 participants (26.5%). A substantial

part of the sample therefore came from lower- and middle-income groups. Including different income levels is relevant because users may have different expectations of transaction reliability, efficiency, and security when managing their daily finances online.

Table 4.7: Respondents' Number of Bank Accounts

Variable	Category	Frequency	Percentage (%)
Number of Bank Accounts	One	113	26.5
	Two	206	48.4
	More than two	107	25.1

Table 4.7 summarises the number of bank accounts held by respondents. Nearly half of the respondents had two bank accounts, indicating that many had experience with more than one banking relationship. Nearly half of the respondents had two bank accounts (206 respondents, 48.4%). Another 113 respondents (26.5%) had one account, while 107 (25.1%) had more than two. Many participants therefore had experience with more than one banking relationship and may have been able to compare the e-banking services offered by different providers.

Table 4.8: Respondents' E-Banking Experience

Variable	Category	Frequency	Percentage (%)
E-Banking Experience	Less than 1 year	83	19.5
	1 – 3 years	145	34.0
	4 – 6 years	130	30.5
	More than 6 years	68	16.0

Table 4.8 presents the respondents' length of e-banking experience. Most respondents had used e-banking for at least one year, giving them a reasonable basis for evaluating service quality. Most respondents had used e-banking for at least one year. The largest group reported one to three years of experience (145 respondents, 34.0%), followed by four to six years (130 respondents, 30.5%). Their views were therefore more likely to reflect repeated use rather than a first impression. This experience gave respondents a reasonable basis for evaluating reliability, responsiveness, efficiency, privacy and security, ease of use, system availability, and website design.

Table 4.9: Respondents' E-Banking Usage Frequency

Variable	Category	Frequency	Percentage (%)
Usage Frequency	Daily	64	15.0
	Weekly	116	27.2
	Several times a month	133	31.2
	About once a month	113	26.5

Table 4.9 summarises how frequently respondents used e-banking. Several-times-a-month and weekly usage were the most common patterns, indicating regular exposure to digital banking services. The most common usage pattern was several times a month, reported by 133 respondents (31.2%), followed by weekly use among 116 respondents (27.2%). This indicates that many participants used e-banking regularly. Regular users are more likely to have encountered a range of situations, including successful transactions, delays, support interactions, and security concerns, which makes their feedback relevant to the study.

Table 4.10: Respondents' Main Device Used for E-Banking

Variable	Category	Frequency	Percentage (%)
Main Device Used	Smartphone / Mobile App	72	16.9
	Computer / Laptop (Web/Online Banking)	129	30.3
	Both Mobile and Web	68	16.0
	Tablet	157	36.9

Table 4.10 presents the main devices used by respondents to access e-banking, including mobile, web or computer, tablet, and combined channels. The distribution provides context for later platform-based comparisons. Respondents accessed e-banking through mobile apps, computers or laptops, tablets, or a combination of mobile and web channels. Device choice matters because the experience can differ across platforms. Mobile users may place more value on speed and convenience, while web users may be more sensitive to page layout, navigation, and system performance. Including device information therefore helps explain how customers interact with e-banking services in different settings.

Table 4.11: Respondents' Residence Region

Variable	Category	Frequency	Percentage (%)
Residence Region	Urban	222	52.1
	Rural	204	47.9

Table 4.11 summarises the respondents' residence regions. The sample was almost evenly divided between urban and rural users, providing a broader view of e-banking experiences across Malaysia. The sample was almost evenly divided by residence region: 222 respondents (52.1%) lived in urban areas and 204 (47.9%) lived in rural areas. This balance is useful because internet access, digital familiarity, and access to physical bank branches may differ between locations. Responses from both groups provide a broader view of e-banking satisfaction across Malaysia.

Table 4.12: Respondents' Main E-Banking Provider

Variable	Category	Frequency	Percentage (%)
Main E-Banking Provider	Agrobank	30	7.0
	OCBC Bank	22	5.2
	Standard Chartered Bank Malaysia	22	5.2
	GXBank	21	4.9
	Affin Bank	20	4.7
	Citibank Malaysia	20	4.7
	RHB Bank	20	4.7
	HSBC Bank Malaysia	19	4.5
	ICBC Malaysia	19	4.5
	Bank Islam Malaysia	18	4.2
	Bank Rakyat	18	4.2
	SeaBank Malaysia	18	4.2
	AEON Bank	17	4.0
	Bank of China Malaysia	17	4.0
	Alliance Bank	16	3.8
	CIMB Bank	16	3.8
	MBSB Bank	16	3.8
	Other	16	3.8
	BSN	14	3.3
	Boost Bank	13	3.1

Variable	Category	Frequency	Percentage (%)
	UOB Malaysia	13	3.1
	Public Bank	12	2.8
	Hong Leong Bank	11	2.6
	Al Rajhi Bank Malaysia	10	2.3
	Maybank	8	1.9

Table 4.12 summarises the respondents' main e-banking providers and shows that the sample was distributed across traditional and digital banks. Agrobank had the largest representation, although each bank subgroup remained relatively small. Respondents used a wide range of e-banking providers. Agrobank had the largest representation, with 30 users (7.0%). OCBC Bank and Standard Chartered Bank Malaysia each had 22 users (5.2%), followed by GXBank with 21 (4.9%). Affin Bank, Citibank Malaysia, and RHB Bank each accounted for 20 respondents (4.7%). At the other end of the distribution, Maybank had 8 respondents (1.9%), Al Rajhi Bank Malaysia had 10 (2.3%), and Hong Leong Bank had 11 (2.6%). The spread across traditional and digital banks gives the study insight into customer perceptions across different types of e-banking provider, although the number of respondents for each bank remains relatively small.

Overall, the sample consisted mainly of Malaysian, young to middle-aged users with previous e-banking experience. It also included variation in gender, ethnicity, employment, income, device choice, and residence region. This profile provides an appropriate basis for examining e-banking service quality and customer satisfaction in Malaysia, particularly because most respondents had enough experience to comment on actual service use.

4.1.2 Reliability Analysis

Cronbach's Alpha was used to assess the internal consistency of the items in each construct. Every construct contained three items, and all 426 responses were included. Table 4.13 reports the Alpha values for the seven service-quality dimensions and Customer Satisfaction.

Table 4.13: Reliability Analysis Results

Construct	Number of Items	Cronbach's Alpha	Interpretation
Reliability	3	0.744	Acceptable

Construct	Number of Items	Cronbach's Alpha	Interpretation
Responsiveness	3	0.683	Marginal but retained
Efficiency	3	0.636	Marginal but retained
Privacy and Security	3	0.709	Acceptable
Ease of Use	3	0.612	Marginal but retained
System Availability	3	0.686	Marginal but retained
Website Design	3	0.606	Marginal but retained
Customer Satisfaction	3	0.670	Marginal but retained

The Cronbach's Alpha values ranged from 0.606 to 0.744. Reliability recorded the highest value of 0.744, followed by Privacy and Security at 0.709. These two constructs demonstrated acceptable internal consistency. Responsiveness, System Availability, Customer Satisfaction, Efficiency, Ease of Use, and Website Design recorded values ranging from 0.606 to 0.686. These constructs demonstrated marginal internal consistency but were retained because their values exceeded the minimum threshold of 0.60 adopted for this exploratory study.

No item was removed because the "Cronbach's Alpha if Item Deleted" results did not indicate a clear or meaningful improvement in the internal consistency of the respective constructs. Therefore, all items were retained, and a composite mean was calculated for each construct. The eight composite variables were subsequently used in the descriptive, correlation, and multiple-regression analyses.

4.1.3 Descriptive Statistics

Descriptive statistics were calculated for the composite variables using their means and standard deviations. All questionnaire items used a five-point Likert scale, so a higher mean represents a more positive evaluation of the service-quality dimension or Customer Satisfaction.

Table 4.14: Descriptive Statistics of Service Quality Dimensions and Customer Satisfaction

Variable	N	Minimum	Maximum	Mean	Standard Deviation
Reliability	426	1.33	5.00	4.1002	0.91488
Responsiveness	426	1.00	5.00	4.0767	0.85336
Efficiency	426	1.67	5.00	4.1025	0.77556
Privacy and Security	426	1.33	5.00	4.1197	0.86259
Ease of Use	426	1.67	4.67	4.0055	0.75751
System Availability	426	1.67	5.00	4.0227	0.85268
Website Design	426	1.33	4.67	3.9632	0.78812
Customer Satisfaction	426	1.33	5.00	4.0829	0.84446

Table 4.14 summarises the mean and standard deviation for the seven service-quality dimensions and Customer Satisfaction. Privacy and Security recorded the highest mean, while Website Design recorded the lowest. Mean scores for the seven service-quality dimensions ranged from 3.9632 to 4.1197, showing generally positive evaluations. Privacy and Security had the highest mean ($M = 4.1197$, $SD = 0.86259$), followed by Efficiency ($M = 4.1025$, $SD = 0.77556$) and Reliability ($M = 4.1002$, $SD = 0.91488$). Website Design received the lowest mean ($M = 3.9632$, $SD = 0.78812$), although the score was still close to the agreement level. Customer Satisfaction was also rated positively ($M = 4.0829$, $SD = 0.84446$).

Respondents generally saw Malaysian e-banking services as secure, efficient, and reliable. Reliability had the largest standard deviation among the service-quality dimensions, suggesting that transaction experiences varied more widely for this area. Website Design, while not rated poorly, was the weakest relative dimension and points to possible concerns with interface clarity

and presentation. This pattern is consistent with earlier studies that identify privacy and security, efficiency, reliability, and interface organisation as important parts of the digital-banking experience (Khatoon et al., 2020; Raza et al., 2020; Bankuoru Egala et al., 2021; Kah Boon Lim et al., 2023).

Table 4.15: Respondents' Issue Experience

Variable	Category	Frequency	Percentage (%)
Issue Experience	No	123	28.9
	Yes	303	71.1

Table 4.15 shows that 303 respondents (71.1%) had experienced at least one e-banking issue, while 123 (28.9%) reported no issue. This result adds important context to the positive mean scores: a favourable overall rating does not necessarily mean that users had a problem-free experience. Correlation and regression analyses were therefore needed to move beyond the descriptive ratings and identify which service-quality dimensions were most closely associated with Customer Satisfaction.

4.1.4 Pearson Correlation Analysis

Pearson correlation analysis was used to examine the relationship between each e-banking service-quality dimension and Customer Satisfaction. The results are presented in Table 4.16.

Table 4.16: Pearson Correlation between Service Quality Dimensions and Customer Satisfaction

Service Quality Dimension	Pearson Correlation with Customer Satisfaction	Significance	Interpretation
Reliability	0.719	$p < 0.001$	Strong positive relationship
Responsiveness	0.729	$p < 0.001$	Strong positive relationship

Service Quality Dimension	Pearson Correlation with Customer Satisfaction	Significance	Interpretation
Efficiency	0.747	$p < 0.001$	Strong positive relationship
Privacy and Security	0.699	$p < 0.001$	Strong positive relationship
Ease of Use	0.708	$p < 0.001$	Strong positive relationship
System Availability	0.735	$p < 0.001$	Strong positive relationship
Website Design	0.690	$p < 0.001$	Strong positive relationship

All seven service-quality dimensions were positively and statistically significantly correlated with Customer Satisfaction ($p < 0.001$). The coefficients ranged from $r = 0.690$ to $r = 0.747$, indicating consistently strong positive relationships.

Efficiency recorded the strongest correlation ($r = 0.747$), followed by System Availability ($r = 0.735$), Responsiveness ($r = 0.729$), Reliability ($r = 0.719$), Ease of Use ($r = 0.708$), Privacy and Security ($r = 0.699$), and Website Design ($r = 0.690$). This ranking suggests that functional performance and service accessibility were most closely associated with satisfaction, although every dimension remained relevant when examined separately.

Correlation does not show whether a dimension contributes uniquely after the overlap among service-quality dimensions is controlled. Multiple linear regression was therefore used to identify the independent contribution of each predictor.

4.1.5 Multiple Linear Regression Analysis

Multiple linear regression was used to determine which service-quality dimensions significantly predicted Customer Satisfaction when all seven were included in one model. Customer Satisfaction

was the dependent variable. Reliability, Responsiveness, Efficiency, Privacy and Security, Ease of Use, System Availability, and Website Design were entered as the independent variables.

Table 4.17: Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	0.835	0.697	0.692	0.46871	1.758

Table 4.17 shows that the regression model had an R value of 0.835, indicating a strong overall relationship between the seven service quality dimensions and customer satisfaction. The R Square value was 0.697, meaning that 69.7% of the variance in customer satisfaction was explained by the seven service quality dimensions. The Adjusted R Square value was 0.692, which is very close to the R Square value. This suggests that the model has good explanatory power and is not heavily inflated by the number of predictors. The Durbin-Watson value was 1.758, which is close to 2. This suggests that there is no serious autocorrelation problem in the residuals. Therefore, the model is suitable for interpretation.

Table 4.18: ANOVA Result

Model	Sum of Squares	df	Mean Square	F	Sig.
Regression	211.239	7	30.177	137.361	p < 0.001
Residual	91.831	418	0.220		
Total	303.069	425			

Table 4.18 reports the ANOVA test for the regression model was statistically significant, $F(7, 418) = 137.361$, $p < 0.001$. This means that the seven service quality dimensions, when analysed together, significantly explain customer satisfaction. Therefore, the overall regression model is valid and relevant to the research objective.

Table 4.19: Collinearity Statistics

Predictor	Tolerance	VIF
Reliability	0.289	3.455
Responsiveness	0.306	3.273
Efficiency	0.364	2.749
Privacy and Security	0.291	3.431
Ease of Use	0.349	2.867
System Availability	0.269	3.716
Website Design	0.364	2.748

Before interpreting the regression coefficients, multicollinearity and residual independence were assessed. The tolerance values ranged from 0.269 to 0.364, while the VIF values ranged from 2.748 to 3.716. As all tolerance values were above 0.20 and all VIF values were below 5, severe multicollinearity was not evident. The Durbin–Watson statistic was 1.758, indicating no serious autocorrelation among the residuals. Therefore, these diagnostic results supported the interpretation of the regression model.

Table 4.20: Regression Coefficients

Predictor	B	Standardized Beta	t	Sig.	Result
Constant	0.043				
Reliability	0.081	0.088	1.759	0.079	Not significant
Responsiveness	0.157	0.158	3.248	0.001	Significant
Efficiency	0.314	0.289	6.468	$p < 0.001$	Significant
Privacy and Security	0.021	0.021	0.425	0.671	Not significant
Ease of Use	0.159	0.143	3.130	0.002	Significant
System Availability	0.173	0.175	3.368	$p < 0.001$	Significant
Website Design	0.091	0.085	1.899	0.058	Not significant at 0.05

Four dimensions made significant unique contributions to Customer Satisfaction: Efficiency ($\beta = 0.289$, $p < 0.001$), System Availability ($\beta = 0.175$, $p < 0.001$), Responsiveness (β

= 0.158, $p = 0.001$), and Ease of Use ($\beta = 0.143$, $p = 0.002$). Efficiency was the strongest predictor after the other dimensions were controlled.

Reliability ($p = 0.079$), Privacy and Security ($p = 0.671$), and Website Design ($p = 0.058$) were not statistically significant unique predictors at the 0.05 level. Their strong bivariate correlations indicate that they remained associated with satisfaction, but their explanatory contributions overlapped with the other dimensions in the full model. Equation 4.1 presents the estimated multiple linear regression model based on the unstandardized coefficients obtained from SPSS.

$$CS_i = 0.043 + 0.081REL_i + 0.157RES_i + 0.314EFF_i + 0.021PS_i + 0.159EOU_i + 0.173SA_i + 0.091WBD_i \quad (4.1)$$

The intercept of 0.043 represents the predicted Customer Satisfaction score when all seven service-quality dimensions are equal to zero. However, because zero lies outside the five-point measurement scale used in this study, the intercept is required mathematically but has limited practical interpretation.

Holding the other predictors constant, a one-unit increase in Responsiveness was associated with a 0.157-unit increase in predicted Customer Satisfaction. Similarly, one-unit increases in Efficiency, Ease of Use, and System Availability were associated with increases of 0.314, 0.159, and 0.173 units in predicted Customer Satisfaction, respectively. These four predictors made statistically significant unique contributions to Customer Satisfaction.

Although the estimated coefficients for Reliability, Privacy and Security, and Website Design were positive at 0.081, 0.021, and 0.091, respectively, these coefficients were not statistically distinguishable from zero after controlling for the other service-quality dimensions. Therefore, their individual coefficients should not be interpreted as significant unique predictors of Customer Satisfaction in the full regression model.

4.1.6 Discussion in Relation to Objective 1

Objective 1 examined how the main e-banking service-quality dimensions relate to customers' overall satisfaction. The Pearson results showed that all seven dimensions were positively and significantly related to Customer Satisfaction. In other words, respondents who rated reliability, responsiveness, efficiency, privacy and security, ease of use, system availability, and website design more favourably also tended to report higher satisfaction.

The regression analysis provided a more specific picture. When the seven dimensions were considered together, only Efficiency, System Availability, Responsiveness, and Ease of Use remained significant. These four dimensions therefore offered the strongest unique explanation of Customer Satisfaction in this study.

Table 4.21: Summary of Correlation and Regression Findings

Dimension	Correlation with Satisfaction	Regression Result	Interpretation
Reliability	$r = 0.719, p < 0.001$	Not significant	Strong relationship, but effect overlaps with other dimensions
Responsiveness	$r = 0.729, p < 0.001$	Significant	Important unique predictor of satisfaction
Efficiency	$r = 0.747, p < 0.001$	Significant	Strongest predictor of satisfaction
Privacy and Security	$r = 0.699, p < 0.001$	Not significant	Related to satisfaction, but not a unique predictor

Dimension	Correlation with Satisfaction	Regression Result	Interpretation
Ease of Use	$r = 0.708, p < 0.001$	Significant	Important unique predictor of satisfaction
System Availability	$r = 0.735, p < 0.001$	Significant	Second strongest predictor of satisfaction
Website Design	$r = 0.690, p < 0.001$	Not significant	Related to satisfaction, but weaker unique effect

Efficiency

Efficiency was the strongest unique predictor of Customer Satisfaction ($\beta = 0.289, p < 0.001$). This result is understandable because e-banking is intended to save customers time and effort. Users expect to transfer funds, pay bills, check balances, and receive confirmation without unnecessary delays or complicated steps. A platform that supports these routine tasks smoothly is more likely to meet the practical reason customers use digital banking.

The result is consistent with Hafiz et al. (2024), who reported that Efficiency, Ease of Use, and Responsiveness significantly influenced mobile-banking satisfaction among Malaysian users. Raza et al. (2020) also found positive relationships between internet-banking service quality and electronic customer satisfaction.

System Availability

System Availability was the second-strongest unique predictor ($\beta = 0.175, p < 0.001$). Customers expect e-banking to be available whenever they need it, without the time and location limits of a physical branch. App crashes, downtime, unresponsive pages, and failed logins can prevent users from completing urgent payments or transfers and are likely to lower satisfaction. Earlier studies

have linked technical glitches, slow access, weak system performance, and instability with access difficulties, transaction failures, and poorer user experiences (Khatoon et al., 2020; Ayinaddis et al., 2023). Banks should therefore monitor system stability, strengthen server performance, minimise crashes and downtime, and maintain access during periods of high demand.

Responsiveness

Responsiveness also made a significant unique contribution ($\beta = 0.158$, $p = 0.001$). Financial transactions often require immediate and clear feedback. When confirmation is delayed, customers may be unsure whether money has been deducted, transferred, or received. Failed transactions, OTP problems, and suspicious activity also create situations in which users need timely support and practical instructions. Khatoon et al. (2020) identified responsiveness and communication as important parts of electronic-banking service quality, supporting its continued relevance to satisfaction.

Ease of Use

Ease of Use was another significant predictor ($\beta = 0.143$, $p = 0.002$). Clear navigation and straightforward transaction procedures reduce the effort needed to use an e-banking platform. Users may become frustrated when functions are difficult to find, instructions are unclear, or a simple transaction requires too many steps. Because mistakes in banking can have financial consequences, customers are likely to value an interface that helps them complete tasks accurately and with confidence. This result agrees with Hafiz et al. (2024), who found that perceived Ease of Use significantly influenced mobile-banking satisfaction in Malaysia.

Reliability

Reliability had a strong bivariate relationship with satisfaction ($r = 0.719$) but was not a significant unique predictor in the regression model ($p = 0.079$). This does not make Reliability unimportant. Rather, part of its relationship with satisfaction appears to be shared with other dimensions. A stable and fast platform may also be viewed as efficient and available, while a failed transaction can be experienced at the same time as a Reliability, Efficiency, and System Availability problem.

Evidence from Malaysian studies is also mixed. Hafiz et al. (2024) found that Reliability, Privacy and Security, and interoperability were not significant after other mobile-banking factors were considered, whereas Ease of Use, Efficiency, and Responsiveness were significant. Kah Boon Lim et al. (2023), however, reported a significant Reliability effect. The different findings suggest that Reliability's unique contribution may depend on the service setting and the other variables included in the model.

Privacy and Security

Privacy and Security had the highest descriptive mean and a strong correlation with satisfaction, but it was not a significant unique predictor in the regression model ($p = 0.671$). Security may operate as a basic expectation: customers generally assume that regulated banks will protect their personal and financial information. It may support trust without strongly distinguishing satisfaction unless a serious failure occurs. Aboobucker and Bao (2018) similarly found that security and privacy were not significant factors in internet-banking acceptance, while Apaua & Lallie (2022) stressed that security, privacy controls, and trust remain important for mobile-banking adoption and use.

Website Design

Website Design had the lowest descriptive mean and the weakest correlation of the seven dimensions, although the correlation was still strong and significant. Its unique regression effect was not significant at the 0.05 level ($p = 0.058$). Customers therefore appeared to value layout, appearance, and interface clarity, but functional dimensions such as Efficiency and System Availability explained more of the unique variation in satisfaction. Website Design may also overlap with Ease of Use and Efficiency because a well-organised interface helps users find functions and complete tasks with less effort. Sudirjo et al. (2024) found high task-completion success after applying a user-centred design approach to a mobile-banking application. Alfaridzi et al. (2023) also reported improvements in learnability, efficiency, error rates, and satisfaction following an interface redesign. Website Design should therefore remain a practical concern even though it was not a significant unique predictor in this model.

Overall Interpretation

The correlation and regression results answer related but different questions. Correlation showed that all seven dimensions were positively associated with satisfaction when examined separately. Regression identified Efficiency, System Availability, Responsiveness, and Ease of Use as the dimensions that made significant unique contributions when considered together. Reliability, Privacy and Security, and Website Design still formed part of the wider e-banking experience, but much of their explanatory value overlapped with the significant predictors.

Objective 1 was therefore achieved. The findings point most strongly to practical service performance: users value e-banking platforms that complete tasks efficiently, remain accessible, respond promptly, and are easy to use. Reliability, Privacy and Security, and Website Design should not be neglected, as they were still strongly related to satisfaction and may affect customers through their connection with other service-quality dimensions.

4.2 Phase 2: Interactive E-Banking Service Quality Dashboard Results

Phase 2 reports the outputs of the interactive Power BI dashboard developed for Objective 2. The dashboard methodology, data preparation, and KPI design were explained in Chapter 3, so this section focuses on what the visual pages reveal rather than repeating the development process.

The dashboard turns the survey and statistical results into a practical visual tool. Users can review overall satisfaction, compare service-quality dimensions, examine differences between segments, and explore reported e-banking issues. Its four pages—Executive Overview, Diagnostic Comparison, Issue Monitoring, and Insight Summary—guide the user from a broad performance view to detailed investigation and recommendation planning.

4.2.1 Executive Overview Results

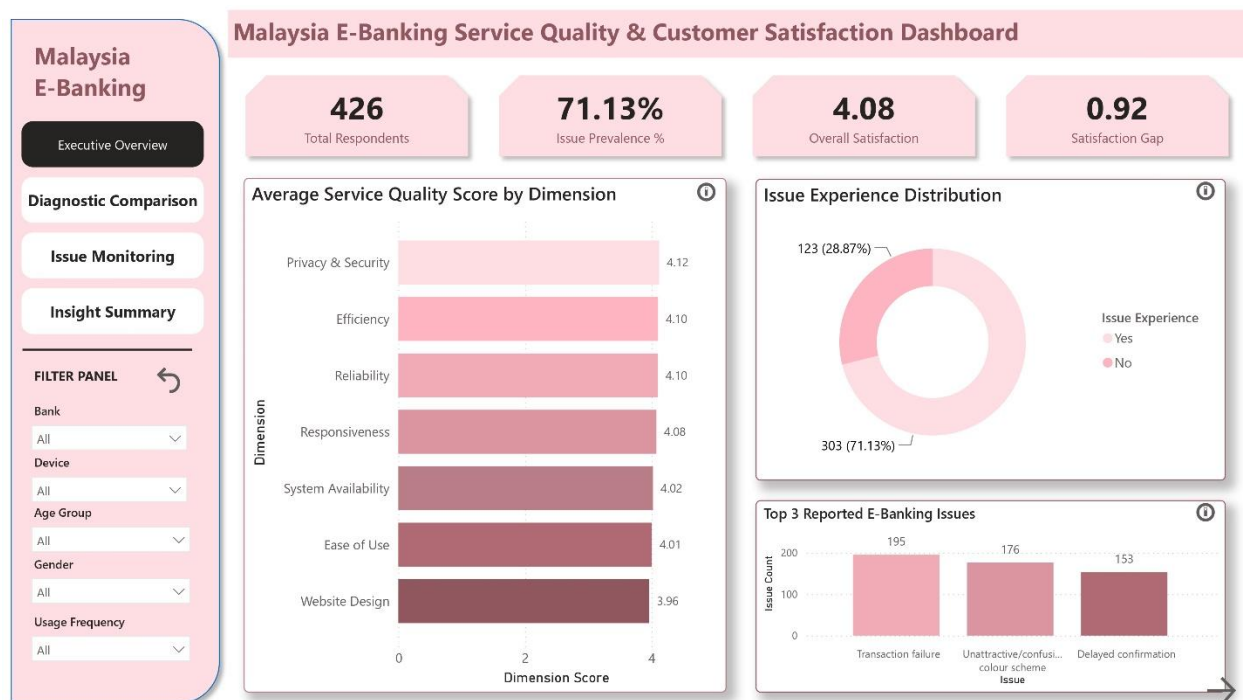


Figure 4.1: Executive Overview Dashboard Page

The Executive Overview page summarises the main service-quality and Customer Satisfaction results for all 426 respondents. Overall satisfaction was 4.08 out of 5, indicating a generally positive evaluation of Malaysian e-banking services. The remaining distance from the maximum score, shown as a satisfaction gap, was 0.92. The score is favourable, but it still leaves room for improvement.

At the same time, 71.13% of respondents had experienced at least one e-banking issue, compared with 28.87% who reported none. The result shows why the average satisfaction score should not be read as evidence of a problem-free service. Many users remained broadly satisfied despite encountering difficulties, although repeated or serious problems could weaken satisfaction over time.

Privacy and Security had the highest average service-quality score at 4.12. Efficiency and Reliability followed at 4.10 each, while Responsiveness scored 4.08. System Availability and Ease of Use recorded 4.02 and 4.01, respectively. Website Design was lowest at 3.96, making it the weakest relative dimension on the overview page.

Transaction failure was the most frequently reported issue, with 195 selections. An unattractive or confusing colour scheme followed with 176, while delayed confirmation recorded 153. The main concerns therefore covered both operational performance, such as failed transactions and slow confirmation, and interface design, particularly colour clarity.

The Executive Overview presents a positive overall picture but also identifies clear areas for attention. Website Design had the lowest dimension score, while transaction failure and delayed confirmation were among the most common reported problems. These results provide an initial direction for more detailed investigation on the remaining dashboard pages.

4.2.2 Diagnostic Comparison Results



Figure 4.2: Diagnostic Comparison Dashboard Page

The Diagnostic Comparison page allows users to compare satisfaction, satisfaction gap, respondent distribution, and the seven service-quality scores across selected segments. The available categories are bank, platform, age group, gender, and usage frequency. This makes it possible to see whether weaker perceptions are concentrated within a particular customer group or service context.

In the default bank view, Bank Islam Malaysia had the highest overall satisfaction score at about 4.40. BSN showed the largest satisfaction gap, at about 1.70, which corresponds to an overall satisfaction score of roughly 3.30. The descriptive difference between the highest and lowest values was therefore around 1.10 points on the five-point scale. This variation is noticeable, but it was not tested using inferential statistics and should not be treated as a statistically confirmed difference between banks.

Bank-level results also need to be read alongside the number of respondents. Bank Islam Malaysia was represented by 18 respondents and BSN by 14, while sample sizes were similarly small and uneven for other banks. A few especially positive or negative responses could therefore

shift an individual bank's score. These comparisons are best treated as early signals for further investigation, not as definitive performance rankings.

The dimension matrix adds detail by showing the average score for each service-quality dimension within every selected bank. Bank Islam Malaysia, for example, had an overall dimension average of about 4.40 and comparatively high scores for Efficiency, Reliability, System Availability, and Ease of Use. Other banks showed lower values in areas such as Website Design, Efficiency, or Ease of Use. This view helps users identify the specific dimensions that may sit behind a lower satisfaction score instead of assuming that every service area performs equally.

Users can switch the page to compare platforms, age groups, gender groups, and usage-frequency groups. These views are useful for exploring whether certain groups report lower satisfaction or weaker service-quality experiences. The values discussed here come from the default bank view. Findings for other segments should only be reported after the relevant category is selected and its values are checked.

The Diagnostic Comparison page helps locate where descriptive differences may be occurring. Phase 1 identifies the dimensions that contribute uniquely to satisfaction across the whole sample, while this page shows the banks or customer segments in which weaker perceptions may be concentrated. Its results can guide further analysis, but bank-level values must be considered together with subgroup size and should not be presented as final rankings.

4.2.3 Issue Monitoring and Hotspot Results

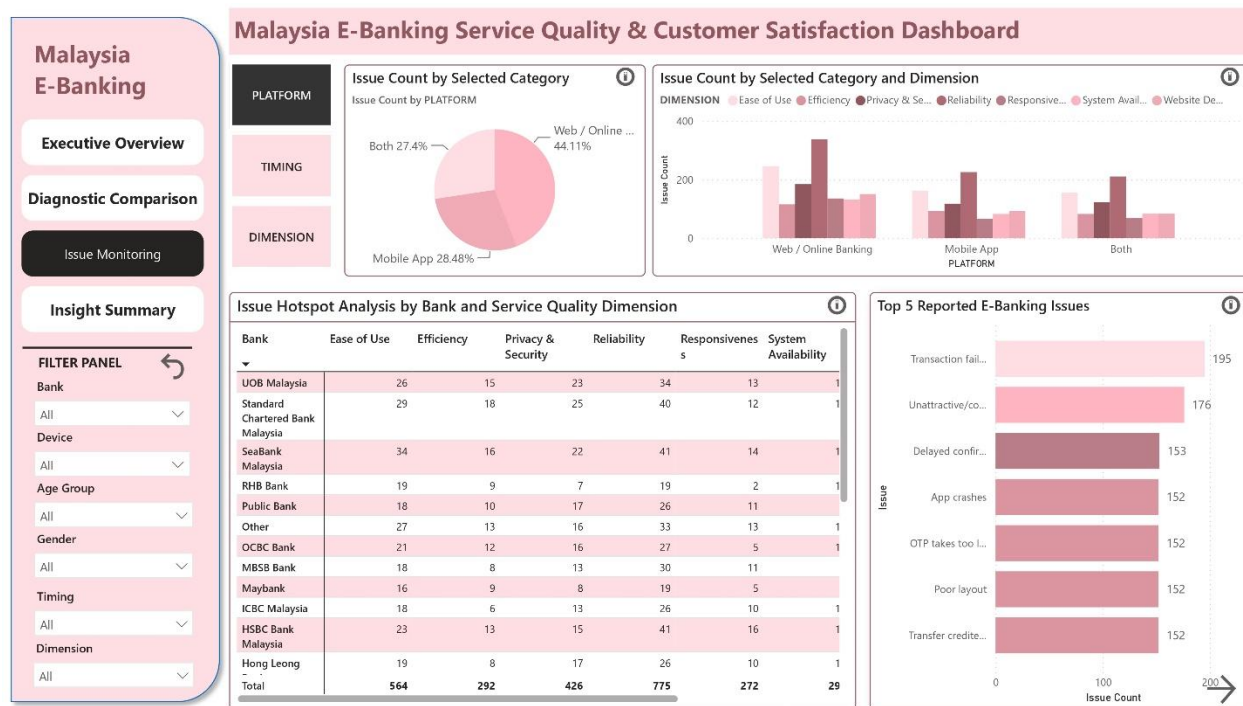


Figure 4.3: Issue Monitoring Dashboard Page

The Issue Monitoring page examines the problems reported by respondents in more detail. Instead of relying only on average satisfaction, it shows issue counts by category and dimension, a bank-by-dimension hotspot table, and the most frequently selected individual issues. This view helps reveal the operational problems behind the overall scores.

Reliability recorded the highest number of issue selections, with 775. Ease of Use followed with 564, Privacy and Security with 426, Website Design with 328, System Availability with 299, Efficiency with 292, and Responsiveness with 272.

This result is noteworthy because Reliability was not a significant unique predictor in the regression model, yet it produced the largest issue count. The two findings are not inconsistent. Reliability may share statistical variation with other dimensions while still representing a common practical concern. Related problems may include failed transactions, delayed confirmation, inaccurate status updates, or uncertainty over whether a transaction was completed.

Transaction failure was the leading individual issue, with 195 reports. An unattractive or confusing colour scheme was selected 176 times, and delayed confirmation 153 times. App crashes, slow OTP delivery, poor layout, and transfer-related difficulties were also frequently reported.

Web or online banking accounted for 44.11% of reported issue selections, compared with 28.48% for mobile apps and 27.40% for respondents using both platforms. The distribution suggests that web banking deserves as much attention as mobile services. Areas worth reviewing include navigation and layout, loading performance, transaction confirmation, and system responsiveness.

By showing both the type and concentration of reported problems, the Issue Monitoring page supports more focused decisions. Banks can investigate specific issue categories and service areas rather than relying only on a general satisfaction average.

4.2.4 Insight Summary Results

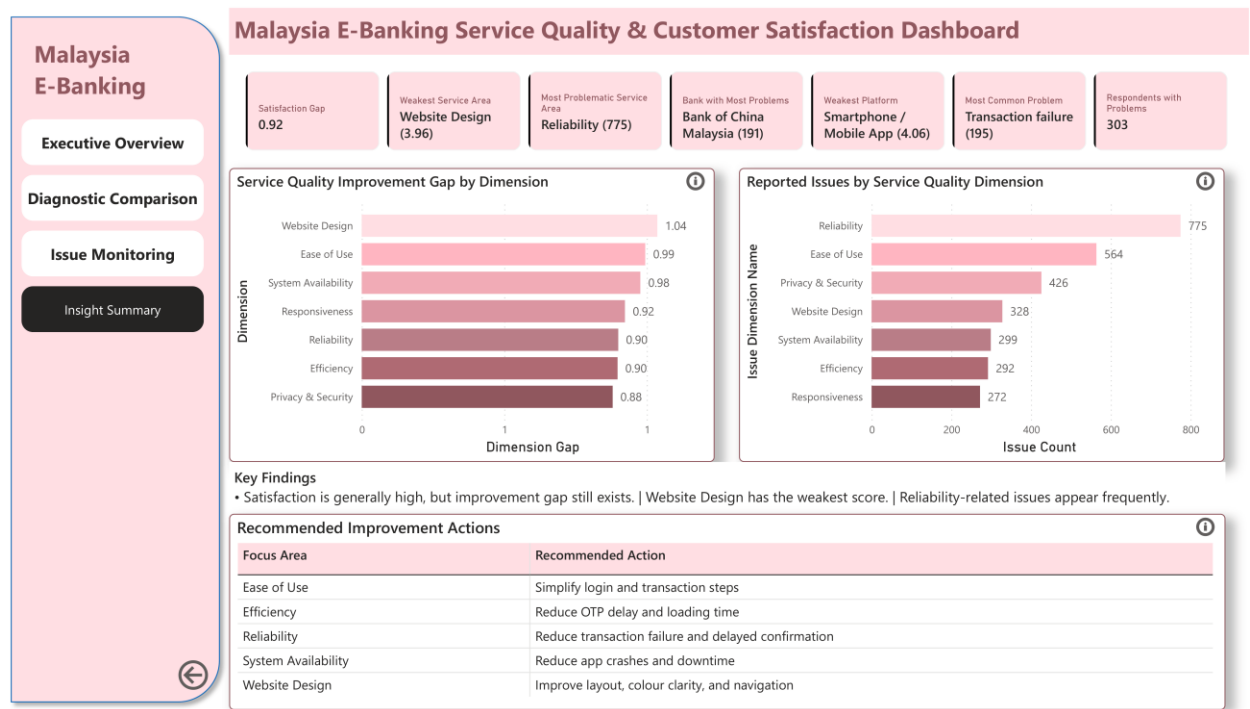


Figure 4.4: Insight Summary Dashboard Page

The Insight Summary brings the main statistical and dashboard results into one decision-support view. It highlights the weakest service-quality dimension, the dimension with the most issue selections, the bank with the highest raw issue count, the weakest platform, the most common individual issue, and the number of respondents who experienced at least one problem.

Website Design was the weakest service-quality dimension, with an average of 3.96. Because the distance from the maximum score is calculated as five minus the observed mean, it also had the largest distance at 1.04. Ease of Use followed with a distance of 0.99 and System Availability with 0.98. The Website Design mean and distance indicator describe the same underlying result rather than two separate findings. Together, they point to layout, visual clarity, colour use, and navigation as the weakest relative parts of the e-banking experience.

Reliability had the highest number of issue selections, with 775, followed by Ease of Use with 564 and Privacy and Security with 426. These are multiple-response counts, not numbers of individual respondents, because one person could select several problems. They should not be compared directly with the sample size of 426. Even so, the large Reliability count indicates that transaction consistency, status updates, confirmation, and successful completion require attention.

Transaction failure was the most frequently reported individual problem, with 195 selections. In total, 303 respondents had experienced at least one e-banking issue. The result again shows that high overall satisfaction can exist alongside recurring operational problems. Banks should track specific service failures as well as the general satisfaction mean.

Smartphone/Mobile App had the lowest platform satisfaction score, at 4.06. The value was still positive and only slightly below the other platform categories, so it should be interpreted as a relative weakness rather than poor performance. Mobile-app usability, loading speed, login procedures, transaction flow, and stability may nevertheless deserve closer review.

Bank of China Malaysia recorded the highest raw issue count, with 191 selections. This does not establish it as the “worst bank” or the provider with the highest failure rate. Only 17 respondents represented the bank, and each could select more than one issue. The result may therefore reflect both multiple selections and a small subgroup. It should be treated as a hotspot for further investigation, not a final judgement of the bank’s overall service quality.

The improvement actions shown on the page follow the combined evidence. Website Design is addressed through clearer layout, colour use, and navigation. Reliability actions focus on reducing failed transactions and delayed confirmation. Efficiency improvements target OTP and loading delays, while System Availability and Ease of Use focus on fewer crashes and outages and simpler login and transaction steps. The Insight Summary links the dashboard findings to the recommendations in Phase 3. It condenses the main service-quality signals and turns them into initial improvement priorities.

4.2.5 Discussion of Phase 2 Findings

The Phase 2 dashboard adds a practical visual layer to the statistical analysis. Phase 1 identified the service-quality dimensions with unique statistical relationships to Customer Satisfaction. Phase 2 shows where weaker results appear, which problems users report most often, and which service areas or customer groups may need attention. The dashboard therefore complements rather than replaces the statistical analysis. Because the study is based on cross-sectional survey data, the findings should be read as associations and possible explanations, not as proof of cause and effect.

Overall Customer Satisfaction and Issue Prevalence

Respondents reported an overall Customer Satisfaction score of 4.08 out of 5, reflecting a generally positive view of Malaysian e-banking services. The dashboard also displayed a satisfaction gap of 0.92, calculated as the difference between the observed mean and the maximum possible score:

$$\text{Satisfaction Gap} = 5 - \text{Average Satisfaction Score} = 5 - 4.08 = 0.92$$

This value is not a traditional SERVQUAL expectation–performance gap because the questionnaire did not measure expectations separately. It simply shows that the average satisfaction score was 0.92 points below the maximum of five. The remaining gap is modest, but it confirms that the experience can still be improved.

The satisfaction score needs to be considered alongside issue prevalence. Although the average was positive, 71.13% of respondents had experienced at least one e-banking problem, while 28.87% reported none. Users may judge the platform based on their overall experience rather than a single failure. For instance, someone who encounters one temporary OTP delay may remain satisfied if the service is usually convenient, accessible, and useful for daily banking. Raza et al. (2020) similarly found that several aspects of internet-banking service quality contribute to customers' overall electronic satisfaction.

Occasional problems may also be tolerated when they are resolved quickly, occur rarely, and do not result in financial loss. Repeated, unresolved, or serious issues are more likely to reduce confidence over time. Banks should therefore monitor satisfaction together with the frequency, severity, and recurrence of service failures. Since respondents were not asked why they remained satisfied after experiencing problems, user tolerance is only a possible interpretation of the pattern.

Service-Quality Performance

The seven service-quality means were close, ranging from 3.96 to 4.12, but the issue data revealed different priorities. Privacy and Security had the highest mean, Website Design the lowest, and Reliability the largest number of issue selections. These measures provide different information: the means reflect general perceptions, while issue selections show the specific difficulties respondents encountered.

The Website Design result is consistent with reports of confusing colour schemes and poor layout. Clearer visual hierarchy, stronger contrast, consistent navigation, and simpler page organisation may make tasks easier to complete (Serrah & Binti Haja Maideen, 2022; Sudirjo et al., 2024). Reliability concerns were more closely linked to failed transactions, delayed

confirmation, and unclear status information. Banks should strengthen transaction stability and clearly tell users whether a transaction is successful, pending, or failed, including whether deductions or reversals have occurred (Abdul Aziz et al., 2025).

Reported Issues

Transaction failure was the most common individual issue, with 195 reports. Since completing transactions is a core function of e-banking, repeated failures can quickly weaken customer confidence. The reported experience may relate to technical, processing, information, or service failures, as discussed by Abdul Aziz et al. (2025). Banks should improve transaction stability and provide clear status messages so that users know whether a transaction is successful, pending, failed, or being reversed.

An unattractive or confusing colour scheme was reported 176 times, making visual clarity a noticeable concern. Weak contrast, inconsistent colour use, and crowded screens can make buttons, warnings, and transaction statuses harder to recognise. Sudirjo et al. (2024) found that users preferred a clearer and more modern mobile-banking interface and completed more tasks successfully after user-centred improvements. Colour, layout, and visual hierarchy should therefore be treated as usability and accessibility concerns, not merely aesthetic choices.

Delayed confirmation received 153 reports. A slow or unclear response can leave users unsure whether a payment or transfer has gone through. Possible explanations include processing delays, peak-hour congestion, timeouts, and unclear post-transaction communication. Earlier research links long loading times, timeouts, and weak system feedback with lower confidence, while responsiveness and communication help users understand what is happening during a digital-banking transaction (Rosdiana Sijabat, 2024; Khatoon et al., 2020; Ayinaddis et al., 2023). An immediate on-screen status, followed where appropriate by push, SMS, or email confirmation, would reduce this uncertainty.

Other commonly selected issues included app crashes, delayed OTPs, poor layout, and transfer-related problems. App crashes may be connected to software, device compatibility, or traffic load. OTP delays may involve authentication systems or mobile networks, while poor layout may reflect unclear navigation or too many transaction steps. These are plausible explanations, but the survey identifies what users experienced rather than the verified technical cause of each problem.

Platform and Segment Comparisons

Web or online banking represented 44.11% of issue selections, followed by mobile apps at 28.48% and users of both platforms at 27.40%. These percentages show how reported issues were distributed across platform groups; they are not normalised failure rates. A larger share for web banking does not automatically mean that it performed worse, because the number of users, frequency of use, and type of transactions may differ across platforms.

Even with this limitation, web banking should not be overlooked in digital-service improvement plans. Garzaro et al. (2020) showed that customer experience and engagement can differ between banking websites and mobile applications, so the channels should be evaluated separately. Banks should review web-specific navigation, information layout, page responsiveness, transaction flows, and confirmation messages. A stronger comparison would calculate the proportion of users reporting problems within each platform group before judging which channel performs worse.

Demographic comparisons may also reveal meaningful differences in experience. Trabelsi-Zoghalmi et al. (2018) found that age and gender moderated some links between mobile-banking service quality and trust. Rajaobelina et al. (2020) further showed that older customers are not one uniform group, as different age ranges may respond differently to aspects of mobile banking. Patterns across age and gender in this dashboard may therefore reflect differences in service expectations, perceived effort, digital familiarity, or trust, although subgroup sizes need to be considered.

Improvement Priorities

The combined evidence highlights several important service areas for different reasons. Ease of Use and System Availability were classified as P1 Immediate because both showed high statistical impact and clear service weaknesses. Efficiency, Reliability and Website Design also require attention because Efficiency was the strongest significant predictor, Reliability recorded the highest number of issue selections, and Website Design had the lowest mean score. Previous research likewise shows that usability, reliability, and efficiency work together to shape digital-banking experiences and future usage intentions (Bankuoru Egala et al., 2021; Kamboj et al., 2021).

These areas should be improved as part of one coordinated programme. A cleaner interface will not prevent failed transactions, and faster processing will not fully help users when navigation or status messages remain unclear. Banks need to combine understandable design, stable transaction processing, timely feedback, and efficient task completion.

Overall Interpretation

Phase 2 shows that Malaysian e-banking users were generally satisfied even though service problems were common. No single measure is enough to describe this experience. Mean scores capture general perceptions, regression coefficients identify the dimensions with unique statistical relationships to satisfaction, and issue counts reveal the operational problems users report most often.

Efficiency was the strongest unique predictor of Customer Satisfaction, System Availability was second, Website Design was the lowest-rated dimension, and Reliability had the most issue selections. These results are not contradictory; each answers a different question. Together, they suggest that banks should improve transaction speed and system access while also reducing reliability failures and making interfaces clearer. Recent research continues to identify efficiency, responsiveness, reliability, security, usability, system availability, and interface design as important parts of digital-banking evaluation (Raza et al., 2020; Khatoon et al., 2020; Serrah & Binti Haja Maideen, 2022; Kah Boon Lim et al., 2023; Hafiz et al., 2024).

The dashboard supports Objective 2 by converting the survey and statistical findings into practical improvement signals. It helps decision-makers understand overall customer views, identify the dimensions most strongly related to satisfaction, recognise frequently reported problems, and locate possible hotspots. These insights form the evidence base for the recommendations in Phase 3.

4.3 Phase 3: Recommendation and Insight

Phase 3 translates the statistical and dashboard findings into prioritised recommendations. The current dashboard indicators are treated as baseline measures rather than evidence of improvement already achieved. The regression results identified Efficiency, System Availability, Responsiveness, and Ease of Use as significant unique contributors to Customer Satisfaction. The dashboard added complementary evidence by showing an overall Customer Satisfaction mean of 4.08, a distance of 0.92 from the maximum score, and an issue-experience prevalence of 71.13%. Website Design had the lowest relative mean, while transaction failure was the most frequently selected individual problem. Reliability produced the highest raw dimension total, but that total is interpreted cautiously because Reliability contained more selectable issue types than the other dimensions. Phase 3 therefore combines statistical contribution, relative performance, raw issue-selection evidence interpreted with caution, and potential severity when setting priorities.

4.3.1 Evidence Summary for Recommendations

Table 4.22 brings together the main evidence used to form the recommendations. Linking each proposed action to a statistical result or dashboard signal makes the reasoning behind the priorities clear.

Table 4.22: Evidence Summary for Phase 3 Recommendations

Evidence Source	Key Finding	Implication for Recommendation
Multiple Linear Regression	Efficiency was the strongest significant predictor of customer satisfaction.	Efficiency should be treated as a major improvement priority because faster and smoother transactions are strongly linked to satisfaction.

	System Availability, Responsiveness, and Ease of Use also significantly predicted customer satisfaction.	Banks should improve system stability, response speed, and platform usability.
Descriptive Statistics and Dashboard Insight Summary	Website Design recorded the weakest score, with a mean score of 3.96.	Website/app layout, visual clarity, and navigation should be improved.
	Website Design had the largest improvement gap at 1.04.	Website Design should be prioritised because it has the largest distance from the maximum score.
Issue Monitoring Dashboard	Reliability recorded the highest number of issue selections, with 775 reported issues.	Reliability should not be ignored even though it was not significant in regression.
	Transaction failure was the most common reported issue, with 195 reports.	Banks should focus on transaction stability and clearer transaction confirmation.
Executive Overview Dashboard	303 respondents, or 71.13%, experienced at least one e-banking issue.	Recommendations should focus on reducing actual service problems, not only increasing satisfaction scores.

The priorities should not be based on a single result. Regression alone would place Efficiency, System Availability, Responsiveness, and Ease of Use at the centre of the plan. The dashboard, however, shows that Website Design and Reliability are also visible weaknesses. The

Phase 3 recommendations therefore consider statistical significance, mean performance, improvement gaps, and issue frequency together.

4.3.2 Evidence-to-Action Recommendation Mapping Matrix

Table 4.23 links each service area to a practical action, the affected scope, the responsible team, the expected KPI movement, and the relevant dashboard view. This evidence-to-action structure makes the recommendations traceable and allows their progress to be assessed through measurable indicators. The following paragraphs explain the reasoning behind each action.

Table 4.23: Evidence-to-Action Recommendation Mapping Matrix

Priority	Evidence Trigger	Focus Area	Affected Scope	Recommended Action	Owner Team	Expected KPI Movement	Dashboard Monitoring
P1 Immediate	Ease of Use was a significant predictor and had the second largest improvement gap.	Ease of Use	Users facing confusing navigation, login difficulty, unclear instructions, or complicated transaction flow.	Simplify login flow, reduce unnecessary steps, improve labels and error messages, and make key functions easier to find (Alfaridzi et al., 2023).	UX/UI Design Team	Ease of Use score increases; navigation-related issue count decreases; satisfaction gap decreases.	Executive Overview; Diagnostic Comparison; Insight Summary
P1 Immediate	System Availability was the second strongest significant predictor and had an improvement gap.	System Availability	Users affected by app crashes, downtime, freezing screens, failed login, or system interruptions.	Improve app stability, reduce downtime, monitor crash patterns, and strengthen system backup during high-usage periods (Kamboj et al., 2021).	IT Operations Team	System Availability score increases; app crash and downtime-related issue counts decrease.	Executive Overview; Diagnostic Comparison; Issue Monitoring
P2 Next	Efficiency was the strongest significant predictor of customer satisfaction.	Efficiency	All e-banking users, especially users affected by slow transactions, OTP delay, or unnecessary transaction steps.	Reduce transaction steps, improve page loading speed, shorten OTP waiting time, and simplify common banking tasks such as transfers and bill payments (Hafiz et al., 2024).	Digital Banking / Product Team	Efficiency score increases; overall satisfaction increases; satisfaction gap decreases.	Executive Overview; Diagnostic Comparison; Insight Summary
P2 Next	Reliability recorded the highest issue count, and transaction failure was the most common reported issue.	Reliability	Users who experienced transaction failure, delayed confirmation, unclear transaction status, or transfer-related problems.	Improve transaction stability, provide clearer transaction status messages, reduce failed transactions, and strengthen confirmation notifications (Kamboj et al., 2021).	IT Operations / Digital Banking Team	Reliability-related issue count decreases; transaction failure count decreases; customer confidence improves.	Issue Monitoring; Insight Summary
P2 Next	Website Design recorded the weakest mean score and largest improvement gap.	Website Design	Users affected by confusing layout, unclear colour scheme, poor visual hierarchy, or difficult navigation.	Improve layout, colour clarity, menu structure, button placement, information presentation, and visual consistency across platforms (Alfaridzi et al., 2023).	UX/UI Design Team	Website Design mean score increases; Website Design gap decreases; ease of use may improve.	Executive Overview; Diagnostic Comparison; Insight Summary
P2 Next	Responsiveness significantly predicted customer satisfaction.	Responsiveness	Users needing help with failed transactions, OTP issues, delayed confirmation, or customer enquiries.	Improve support response time, provide clearer in-app guidance, strengthen chatbot support, and improve complaint follow-up (Jiang et al., 2022).	Customer Service / Operations Team	Responsiveness score increases; complaint resolution time decreases; issue experience decreases.	Diagnostic Comparison; Issue Monitoring
P3 Monitor	Privacy and Security recorded the highest mean score but still had reported issues.	Privacy and Security	Users concerned with authentication, scam risk, secure login, or data protection.	Continue scam alerts, secure login reminders, privacy notices, and user education on safe e-banking behaviour (Tan & Chung, 2023).	IT Security / Risk Team	Privacy and Security score remains high; security-related issue count decreases.	Executive Overview; Issue Monitoring
P3 Monitor	Some bank, platform, and segment differences were shown in the dashboard, but subgroup results require cautious interpretation.	Segment Monitoring	Banks, platforms, age groups, gender groups, and usage-frequency groups selected through dashboard filters.	Continue monitoring satisfaction and service quality by segment before making highly specific segment-based decisions (Trabelsi-Zoghalmi et al., 2018).	Service Quality / Analytics Team	Segment-level satisfaction differences become clearer over time; unstable results are reduced through future data collection.	Diagnostic Comparison

Efficiency

Efficiency should be addressed by the Digital Banking or Product Team. The team can begin by reviewing the customer journeys used most often, including fund transfers, bill payments, balance enquiries, and other routine transactions. Hafiz et al. (2024) identified efficiency as an important predictor of mobile-banking satisfaction among Malaysian users. Repeated data entry, unnecessary screens, and extra confirmation steps should be removed when this can be done without reducing security. The team should also examine loading and processing times to identify where delays usually occur. These changes would help customers complete common banking tasks more quickly and with less effort.

Reliability

Transaction reliability requires close attention from the IT Operations or Digital Banking Team. Kamboj et al. (2021) explain that functional, system, information, and service failures are often connected and may affect both satisfaction and future use. Customers should be able to see immediately whether a transaction is successful, pending, failed, or reversed. When a problem occurs, the platform should clearly state whether funds have been deducted, whether an automatic reversal will take place, and whether the customer needs to do anything further. Transaction records and notifications should also show the same information across the mobile app, website, SMS, and email.

Website Design

Website Design should be improved through a user-centred process rather than decisions based only on internal preferences. Alfaridzi et al. (2023) found that users had difficulty locating unfamiliar functions and menus, while the redesigned prototype improved learnability, efficiency, error performance, and satisfaction. The UX/UI Design Team should therefore involve users through interviews, usability testing, prototype evaluation, and direct observation of task completion. Particular attention should be given to the menu structure, visual hierarchy, colour contrast, font readability, button labels, warning messages, and consistency between the mobile and web interfaces.

System Availability

The IT Operations Team should strengthen system monitoring and service-continuity planning. Kamboj et al. (2021) recognise system interruption as a form of mobile-banking failure, while Raza et al. (2020) highlight the importance of reliable access. Performance should be monitored during normal and peak usage periods to identify repeated crashes, failed logins, timeouts, and outages. Planned maintenance should be announced early so customers can make other arrangements. During unexpected disruptions, the platform should display clear service-status information and provide regular updates until the system has recovered.

Ease of Use

Improving ease of use means making login, navigation, and transaction procedures simpler without weakening essential security controls. Hafiz et al. (2024) found that ease of use was significantly associated with mobile-banking satisfaction. Alfaridzi et al. (2023) also showed that clearer menus and better task flows can improve learnability and reduce errors. Frequently used functions should appear in predictable locations, while labels and instructions should use familiar language. Error messages also need to be more helpful. Instead of displaying technical codes or unclear failure notices, they should explain what went wrong and tell the customer what to do next.

Responsiveness

Customer enquiries should be acknowledged and resolved more quickly. Shankar et al. (2020) identify responsiveness and customer support as important parts of mobile-banking service quality. The Customer Service or Service Operations Team can use automated messages to confirm that a case has been received, provide a reference number, state the expected response time, and update the customer as the case progresses. Jiang et al. (2022) also found that responsive chatbot communication can support satisfaction. Chatbots are suitable for routine questions, but complex, sensitive, or unresolved cases should be transferred to trained staff without unnecessary delay.

Privacy and Security

Privacy and security controls should continue to be reviewed by the IT Security or Risk Team. Tan and Chung (2023) recommended stronger authentication factors, login controls,

transaction checks, and password policies in the Malaysian FinTech setting. However, strong protection should still be understandable to users. Security alerts need to explain the risk clearly, state the action required, and direct customers to a legitimate support channel. Rosle Mohidin et al. (2025) also emphasised the importance of technical safeguards, usable security features, and customer education. Awareness materials should therefore cover phishing, suspicious links, credential sharing, device protection, and safe transaction-verification practices.

Segment Monitoring

The Service Quality or Analytics Team should continue comparing satisfaction and service-quality indicators across customer groups, platforms, usage patterns, and banking providers. Trabelsi-Zoghalmi et al. (2018) found that age and gender may influence how users evaluate certain mobile-banking dimensions. However, each segment result must be interpreted together with its number of respondents. A pattern based on a small group should be treated as an early signal rather than a confirmed difference. Continued monitoring can help the bank identify whether the same service design creates different experiences for users with different needs or levels of digital familiarity.

These improvements require cooperation across product design, system operations, customer service, cybersecurity, and analytics. A change in one area may affect another. For example, simplifying a transaction should not weaken verification, while stronger security controls should not make routine banking unnecessarily difficult. The teams should therefore review related dashboard indicators together and monitor whether each operational change is followed by better customer evaluations.

4.3.3 Prioritised Improvement Roadmap

The recommendations were grouped into three levels: P1 Immediate, P2 Next, and P3 Monitor. Each service area was prioritised by considering both its impact and prevalence, based on the assessment approach described in Section 3.3.2.

Impact refers to how strongly a service-quality dimension was related to Customer Satisfaction. Efficiency, System Availability, Responsiveness, and Ease of Use were considered high-impact dimensions because each made a statistically significant unique contribution in the regression model. Reliability, Privacy and Security, and Website Design

were classified as moderate-impact dimensions. Although they were strongly correlated with Customer Satisfaction, they were no longer statistically significant after the overlap among the seven dimensions was taken into account.

Prevalence was assessed using the descriptive mean, improvement gap, and issue-monitoring results. A dimension was considered highly prevalent when it showed a relatively large improvement gap or appeared as a major operational hotspot in the dashboard. The issue counts were interpreted with care because respondents could select more than one issue, and the number of issue types was not the same for every dimension.

P1 Immediate

Ease of Use and System Availability were assigned to P1 because both showed high impact and high prevalence.

Ease of Use was a significant predictor of Customer Satisfaction, with $\beta = 0.143$ and $p = 0.002$. It also recorded the second-largest improvement gap at 0.99 and the second-highest number of issue selections, with 564 selections. Taken together, these results show that ease-of-use problems were not only associated with satisfaction but were also frequently experienced by users. Difficult login procedures, confusing navigation, functions that are hard to locate, and unnecessary transaction steps should therefore be addressed immediately.

System Availability also requires immediate attention. It was the second-strongest significant predictor of Customer Satisfaction, with $\beta = 0.175$ and $p < 0.001$. Its improvement gap was 0.98, the third largest among the seven dimensions. Problems such as app crashes, unresponsive pages, failed logins, and service interruptions can prevent customers from completing urgent banking activities. Improving system stability and keeping the platform accessible should therefore be treated as a P1 priority.

P2 Next

P2 Next includes Efficiency, Responsiveness, Reliability and Website Design. Each area showed either high statistical impact or high operational prevalence, but not both criteria at the same time. These areas remain important but should be addressed after or alongside the P1 Immediate actions, depending on available resources and operational requirements.

Efficiency was the strongest significant predictor of Customer Satisfaction, with $\beta = 0.289$ and $p < 0.001$. However, it also recorded a relatively high mean score of 4.10 and did not show the same level of operational prevalence as Ease of Use or Reliability. This suggests that Efficiency is highly important to satisfaction, but its weaknesses were less widespread in the dashboard results. Banks should continue improving transaction speed, loading performance, and task completion, although the prioritisation rules place this dimension under P2 rather than P1.

Responsiveness was another significant predictor, with $\beta = 0.158$ and $p = 0.001$. At the same time, it recorded the lowest number of issue selections among the seven dimensions. This indicates strong statistical impact but lower operational prevalence. Improvements are still needed in areas such as transaction confirmation, enquiry acknowledgement, and problem resolution, but the available evidence supports a P2 classification.

Reliability showed the opposite pattern. It recorded the highest number of issue selections, with 775 selections, while transaction failure was the most frequently reported individual issue. However, Reliability was not a statistically significant unique predictor in the regression model, with $p = 0.079$. Its strong correlation with Customer Satisfaction shows that it remains important, but part of its effect may overlap with Efficiency and System Availability. Reliability is therefore classified as P2 because it has high operational prevalence but only moderate unique statistical impact. Even so, transaction failures and unclear transaction status may require immediate escalation when they involve possible financial loss or risk.

Website Design recorded the lowest mean score at 3.96 and the largest improvement gap at 1.04. Users also reported problems such as confusing colour schemes and poor layout. However, Website Design was not statistically significant at the 0.05 level after the other dimensions were controlled, with $p = 0.058$. It was therefore placed under P2 because it represents a clear performance weakness, even though its unique statistical effect was weaker.

P3 Monitor

P3 Monitor includes Privacy and Security and Segment Differences. These areas should not be ignored, but the current evidence does not support immediate or next-stage intervention. They should be monitored continuously so that changes can be identified when additional survey or operational data become available.

Privacy and Security was placed under P3 Monitor because it recorded the highest mean score of 4.12 and the smallest improvement gap of 0.88. It was also not a statistically significant unique predictor of Customer Satisfaction in the regression model. Although security failures can have serious consequences, the current performance evidence is weaker than that of the P1 and P2 areas. Banks should therefore maintain existing safeguards, review authentication procedures, provide scam-awareness information and monitor security-related indicators continuously.

Segment Differences were placed under P3 Monitor because the dashboard comparisons provide useful early signals, but the evidence is not strong enough to justify immediate targeted action. Results for individual banks and customer groups were based on unequal and sometimes small subgroup sizes. The observed differences were also descriptive and were not tested using inferential statistics. These patterns should therefore be monitored rather than treated as confirmed differences.

Table 4.24: Prioritised Improvement Roadmap

Priority Tier	Focus Area	Impact Evidence	Prevalence Evidence	Justification	Timing
P1 Immediate	Ease of Use	Significant unique predictor ($\beta = 0.143$, $p = 0.002$).	Second-largest gap (0.99) and second-highest raw issue count (564).	Both impact and prevalence are high. Prioritise login, navigation, hard-to-find functions, and excessive transaction steps.	Short term
	System Availability	Second-strongest significant unique predictor ($\beta = 0.175$, $p < 0.001$).	Third-largest gap (0.98); crashes, unresponsive pages, downtime, and access problems were reported.	High impact is accompanied by a clear performance gap. Improve system stability and accessibility for time-sensitive tasks.	Short term
P2 Next	Efficiency	Strongest significant unique predictor ($\beta = 0.289$, $p < 0.001$).	High mean (4.10), smaller gap (0.90), and lower raw issue count (292) than the main hotspots.	Impact is high, but the weakness is less widespread than the P1 areas. Improve loading speed, OTP flow, and task completion.	Short to medium term
	Responsiveness	Significant unique predictor ($\beta = 0.158$, $p = 0.001$).	Lowest raw issue count (272) and a moderate gap (0.92).	Impact is high, but prevalence is lower. Improve acknowledgement, confirmation, response time, and escalation.	Medium term
	Reliability	Strong correlation ($r = 0.719$, $p < 0.001$), but not a significant unique predictor ($p = 0.079$).	Highest raw issue count (775); transaction failure was the top individual issue (195).	Prevalence is high, but the regression contribution overlaps with other dimensions. Severe failures still require immediate escalation.	Short to medium term
	Website Design	Strong correlation ($r = 0.690$, $p < 0.001$), but not significant at 0.05 in regression ($p = 0.058$).	Lowest mean (3.96), largest gap (1.04), and 328 raw issue selections.	A clear performance weakness exists, but its unique regression contribution is weaker. Improve layout, colour clarity, hierarchy, and navigation.	Short to medium term
P3 Monitor	Privacy and Security	Strong correlation ($r = 0.699$, $p < 0.001$), but not a significant unique predictor ($p = 0.671$).	Highest mean (4.12) and smallest gap (0.88), although 426 raw issue selections were recorded.	Immediate performance evidence is weaker. Continue monitoring safeguards, authentication procedures, and customer education because security failures may be serious.	Continuous
	Segment Differences	No inferential tests were reported for subgroup differences.	Unequal and sometimes small subgroup sizes; raw issue counts were not normalised into failure rates.	Treat subgroup patterns as preliminary signals. More data and formal testing are needed before targeted intervention or definitive ranking.	Continuous

Table 4.24 prioritises improvements into immediate, next-stage, and monitoring actions based on their impact, prevalence, and urgency. The roadmap combines two kinds of evidence. Regression indicates which dimensions contribute uniquely to satisfaction, while dashboard issue counts show the problems users encounter most often. A practical improvement plan needs both perspectives rather than relying on only one result.

4.3.4 Monitoring Plan

The monitoring plan explains how the current dashboard indicators can be used as baselines for reviewing future improvement. The dashboard already tracks the KPI values contained in the present survey dataset. However, evaluating change requires a later dataset collected after the recommended actions are implemented. Future values can then be compared with the current baselines to determine whether the indicators are moving in the expected direction. For example, overall Customer Satisfaction has a current baseline of 4.08, and the expected direction is an increase when new data are collected. The current satisfaction gap of 0.92 should decrease, while the Website Design mean of 3.96 should increase. The monitoring table therefore records each current baseline, its expected direction of change, and the dashboard page or view that can be used for later comparison.

Table 4.25: Monitoring Plan for Phase 3 Recommendations

Recommendation Area	KPI to Monitor	Current Baseline	Expected Direction	Dashboard Page / View
Overall Satisfaction	Overall satisfaction score	4.08	Increase	Executive Overview
Satisfaction Gap	Satisfaction gap	0.92	Decrease	Executive Overview; Insight Summary
Website Design	Website Design mean score	3.96	Increase	Executive Overview; Diagnostic Comparison; Insight Summary

Website Design Gap	Website Design improvement gap	1.04	Decrease	Insight Summary
Reliability	Reliability issue count	775	Decrease	Issue Monitoring; Insight Summary
Transaction Failure	Transaction failure issue count	195	Decrease	Issue Monitoring
Efficiency	Efficiency mean score	4.10	Increase	Executive Overview; Diagnostic Comparison
System Availability	System Availability mean score	4.02	Increase	Executive Overview; Diagnostic Comparison
Ease of Use	Ease of Use mean score	4.01	Increase	Executive Overview; Diagnostic Comparison
Responsiveness	Responsiveness mean score	4.08	Increase	Executive Overview; Diagnostic Comparison
Privacy and Security	Privacy and Security mean score	4.12	Maintain or increase	Executive Overview; Issue Monitoring

Table 4.25 presents the monitoring plan for the Phase 3 recommendations by linking each improvement area to a KPI baseline, expected direction of change, and dashboard view. It provides a structured basis for comparing future results after the recommendations are implemented. The indicators allow banks to review progress over time. After a UX/UI redesign, for example, the Website Design mean should rise and its improvement gap should fall. If transaction processing becomes more reliable, both the Reliability issue count and transaction-failure count should decline. Improvements to task speed should be reflected in a higher Efficiency mean and, ideally, a higher overall satisfaction score.

4.3.5 Discussion of Objective 3 Findings

Objective 3 was to develop data-driven recommendations for improving e-banking service quality. It was addressed through a structured process that converted the findings into an evidence summary, recommendation mapping matrix, prioritised roadmap, and monitoring plan. The proposed actions were connected to measurable results and assigned to relevant operational teams rather than being based only on general managerial judgement. This approach is consistent with business-intelligence research, which argues that analytical findings become more useful when they are linked to clear performance indicators, responsibilities, and decision needs (Biagi et al., 2021).

Phase 3 also combines different forms of evidence. Statistical analysis identifies the dimensions with stronger unique relationships with satisfaction, while the dashboard shows how weaknesses and reported problems appear in practice. An area may still require attention because it occurs frequently or has serious consequences even when its unique statistical effect is limited. The recommendations therefore reflect both customer-perception patterns and reported service issues instead of depending on one analytical measure.

The monitoring plan adds practical value by establishing clear baseline indicators and expected directions of change. The dashboard currently tracks the results within the completed dataset, while future data refreshes would allow banks to compare later performance with these baselines. Therefore, the dashboard provides a decision-support and monitoring structure, although the present study does not test whether the recommendations have already produced improvements.

The dashboard's value depends on relevant indicators, accurate and consistent data, and information that stakeholders can interpret correctly. Research on dashboard evaluation also stresses task performance, usability, implementation, user engagement, and practical utility. A visually attractive dashboard has limited value unless it supports actual decisions and work processes (Zhuang et al., 2022).

Phase 3 moves the project beyond reporting statistical results. It establishes a clear path from evidence to action and then to future monitoring. Its main contribution is a practical decision-support structure that shows how findings can be prioritised, assigned to responsible teams, and reviewed through measurable indicators.

4.4 Summary

This chapter presented the findings and discussion in three phases, with each phase linked to one research objective. The first phase analysed data from 426 valid respondents to identify the e-banking service-quality factors associated with Customer Satisfaction. All eight constructs achieved acceptable reliability, with Cronbach's Alpha values ranging from 0.606 to 0.744. Respondents generally viewed Malaysian e-banking services positively. Privacy and Security received the highest mean score of 4.12, whereas Website Design recorded the lowest mean of 3.96. Overall Customer Satisfaction was also favourable at 4.08. Even so, 71.1% of respondents had experienced at least one e-banking issue. This shows that customers can remain generally satisfied while still facing service problems. The correlation results showed that all seven service-quality dimensions had strong and significant positive relationships with Customer Satisfaction. The regression model explained 69.7% of the variation in satisfaction. Efficiency emerged as the strongest significant predictor, followed by System Availability, Responsiveness, and Ease of Use. Reliability, Privacy and Security, and Website Design were not significant unique predictors once the overlap among the dimensions was considered. However, they still had strong relationships with satisfaction. Based on these findings, Objective 1 was achieved.

Phase 2 achieved Objective 2 by developing a four-page Power BI dashboard consisting of the Executive Overview, Diagnostic Comparison, Issue Monitoring, and Insight Summary pages. The dashboard visualises and tracks service-quality and Customer Satisfaction indicators by dimension and segment. In the current project, tracking refers to monitoring KPI values within the completed survey dataset, comparing results through filters, and establishing baseline measures. Future datasets would be required to assess changes over time. The dashboard reported an overall satisfaction score of 4.08, a satisfaction gap of 0.92, and an issue prevalence rate of 71.13%. Website Design showed the largest improvement gap at 1.04. Reliability recorded the highest number of issue selections at 775, while transaction failure was the most frequently reported individual problem, with 195 selections. Comparisons across banks and customer segments were treated cautiously because the subgroup sizes were unequal and the differences were not tested using inferential statistics. Even with this limitation, the dashboard helped identify where weaker service performance and reported issue hotspots were concentrated.

The final phase translated the statistical and dashboard findings into practical priorities and a monitoring plan. Ease of Use and System Availability were classified as P1 Immediate

because both showed high statistical impact and clear service weaknesses. Efficiency, Responsiveness, Reliability and Website Design were placed under P2 Next because each showed either high impact or high prevalence, but not both. Privacy and Security and differences across customer segments were grouped under P3 Monitor because the current evidence did not support immediate intervention. The proposed KPIs provide current baselines and expected directions for future comparison. New survey or operational data would be required to determine whether Customer Satisfaction and service-quality scores increase or whether improvement gaps and issue selections decrease. Together, these recommendations provide a practical basis for improving e-banking service quality and reviewing future progress.

Chapter 5: Conclusion and Future Work

5.0 Overview

This chapter brings together the main findings of the study and explains how they address the three research objectives. It begins with the overall conclusion before reviewing the statistical analysis, the interactive Power BI dashboard, and the data-driven recommendations developed in the project. The Malaysian findings are then considered alongside recent mobile-banking research involving Singapore to identify practices that may be useful for Malaysian banks. The chapter ends by outlining the study's academic and practical contributions, its limitations, and several directions for future research and system development.

5.1 Overall Conclusion of the Study

The growing use of e-banking has changed the way customers interact with banks and judge the quality of their services. Many routine activities, such as checking balances, transferring funds, and paying bills, are now completed through digital platforms. Customers therefore expect these services to work quickly, remain available when needed, respond clearly, protect their information, and provide procedures that are easy to follow. This project examined these expectations through statistical analysis, an interactive Power BI dashboard, and evidence-based recommendations linked to responsible teams and measurable indicators.

Respondents generally had a positive view of Malaysian e-banking services. The mean score for Customer Satisfaction was 4.08 out of 5. Even so, 303 respondents, or 71.13% of the sample, reported experiencing at least one e-banking issue. This shows that a positive satisfaction score does not mean that customers had a completely problem-free experience. Some users may still consider e-banking useful and convenient even after facing an occasional delayed OTP, application crash, or transaction failure. However, repeated problems or issues that are not resolved properly may gradually reduce their confidence and satisfaction.

The results also suggest that practical service performance played the strongest role in Customer Satisfaction. Efficiency, System Availability, Responsiveness, and Ease of Use made significant unique contributions in the regression model. However, these were not the only areas that mattered. Website Design recorded the lowest average score, while Reliability had

the highest number of raw issue selections. This means that regression results, mean scores, and issue counts each provide a different view of the customer experience.

For this reason, banks should not rely on only one type of evidence when deciding what to improve. Regression analysis helps identify the dimensions that make significant unique contributions to satisfaction. Mean scores show which areas received relatively weaker evaluations, while issue selections highlight problems that customers reported frequently. Considering these findings together provides a more balanced basis for improving e-banking service quality.

5.2 Achievement of the Research Objectives

Table 5.1 summarises the evidence used to assess each research objective and distinguishes project completion from untested organisational outcomes.

Table 5.1: Achievement of the Research Objectives

Objective	Evidence of Achievement	Status	Important Limitation
Objective 1: Examine relationships between seven service-quality dimensions and Customer Satisfaction.	Reliability analysis, descriptive statistics, Pearson correlation, and multiple linear regression using 426 valid responses.	Achieved with methodological caution.	Six constructs had marginal Alpha values; construct-validity and full residual diagnostics were not reported; cross-sectional results are associative.
Objective 2: Develop an interactive Power BI dashboard by dimension and segment.	Four-page dashboard, respondent-level and long-format issue model, KPI reconciliation, filter checks, and baseline comparisons.	Achieved as dashboard development and baseline decision support.	No live banking data and no stakeholder usability or decision-impact evaluation.
Objective 3: Propose data-driven service-quality recommendations.	Evidence-to-Action matrix, criteria-based priority classification, owner teams, time frames, KPI baselines, target directions, and monitoring plan.	Achieved at the recommendation-development level.	Recommendations were not implemented, experimentally tested, or evaluated through before-and-after data.

5.2.1 Achievement of Objective 1

The first objective was:

To examine the relationship between key e-banking service-quality dimensions and customers' overall satisfaction using statistical analysis.

Reliability analysis, descriptive statistics, Pearson correlation, and multiple linear regression were used to address this objective.

The Cronbach's Alpha values for the eight constructs ranged from 0.606 to 0.744. Several values were below 0.70, but the constructs were retained because the study was exploratory and each construct contained only three measurement items. The results were considered adequate for the following analyses. At the same time, the lower reliability values suggest that some questionnaire items should be reviewed and refined in future research.

The descriptive results showed that respondents generally evaluated all seven service-quality dimensions positively. Privacy and Security received the highest mean score, whereas Website Design recorded the lowest. The Pearson correlation results also showed that all seven dimensions had strong, positive, and statistically significant relationships with Customer Satisfaction when they were examined separately.

Multiple linear regression provided a clearer view of the contribution made by each dimension after the overlap among the predictors was controlled. The model explained 69.7% of the variance in Customer Satisfaction. Efficiency was the strongest significant predictor, followed by System Availability, Responsiveness, and Ease of Use. Reliability, Privacy and Security, and Website Design were not significant unique predictors in the full model. However, all three remained strongly correlated with satisfaction.

Objective 1 was achieved because the analysis identified both the overall relationships between the seven service-quality dimensions and Customer Satisfaction and the dimensions that made significant unique contributions when all predictors were considered together.

5.2.2 Achievement of Objective 2

The second objective was:

To develop an interactive Power BI dashboard that visualises and tracks e-banking service quality and customer satisfaction results by dimension and segment.

This objective was achieved through the development of a four-page Power BI dashboard consisting of the Executive Overview, Diagnostic Comparison, Issue Monitoring, and Insight Summary pages. The dashboard visualised and tracked the service-quality and Customer Satisfaction indicators contained in the completed survey dataset. Users could monitor KPI values, compare the seven service-quality dimensions, apply segment filters, and investigate reported issue patterns.

The Executive Overview showed that Customer Satisfaction was generally positive, although several service areas still had room for improvement. Website Design recorded the lowest mean and the largest improvement gap, followed by Ease of Use and System Availability. These findings highlighted interface clarity, navigation, login procedures, and service accessibility as areas that may require further attention. The Issue Monitoring page added more detail by showing the problems reported by respondents. Reliability recorded the highest number of raw issue selections, while transaction failure was the most frequently selected individual issue. The dashboard also supported comparisons across platforms, banks, age groups, gender groups, and usage-frequency categories. These subgroup findings were interpreted carefully because the group sizes were unequal and the observed differences were not tested using inferential statistics.

Tracking in the current dashboard refers to reviewing and comparing the existing indicators rather than receiving live information from banking systems. The dashboard also established baseline values that could support future monitoring when new survey or operational data are added. It therefore fulfilled Objective 2 within the defined scope of the project, although it did not provide real-time incident tracking. Rather than replacing the statistical analysis, the dashboard made the findings easier to interpret and helped users identify weaker performance areas, commonly reported problems, and possible operational hotspots.

5.2.3 Achievement of Objective 3

The third objective was:

To propose data-driven recommendations to improve e-banking service quality based on the empirical findings and dashboard insights.

A structured Evidence-to-Action process was used to meet this objective. Its main outputs were an evidence summary, an Evidence-to-Action Recommendation Mapping Matrix, a prioritised improvement roadmap, and a monitoring plan. Every recommendation was linked to a

statistical result or dashboard finding, assigned to an appropriate operational team, and connected to a measurable KPI.

The recommendations were not prioritised using statistical significance alone. The process also considered mean scores, improvement gaps, issue selections and the possible seriousness of each problem. Ease of Use and System Availability were placed under P1 Immediate because both showed significant statistical importance and clear operational concerns. Recommended actions included simplifying login and transaction procedures, improving navigation, reducing application crashes and maintaining better system accessibility.

Efficiency, Responsiveness, Reliability and Website Design were classified as P2 Next. Suggested actions included improving transaction speed, reducing OTP delays, strengthening service communication, providing clearer transaction-status information and improving interface design.

Privacy and Security and differences across banks and customer segments were placed under P3 Monitor. Privacy and Security should continue to be monitored through authentication reviews, security controls and customer education. Segment differences should also be monitored because the subgroup sizes were unequal and the observed differences were not tested using inferential statistics.

5.3 Comparison with Singapore and Lessons for Malaysian E-Banking

This section uses Singapore-related research as an international reference to identify digital-banking practices that may support improvements in Malaysia. The comparison focuses on transferable lessons rather than direct numerical performance differences between the two countries.

5.3.1 Comparison of the Malaysian and Singapore-Related Findings

Singapore was chosen as the international comparison because it offers a relevant neighbouring setting for studying digital-banking customer experience. The purpose was to identify practices that may be transferable to Malaysia, not to decide whether Malaysian or Singaporean customers were more satisfied. Direct numerical comparison would not be appropriate because the present study and the selected previous research used different samples, constructs, measurement instruments, and statistical methods.

Ogunmola et al. (2026) examined 1,656 mobile-banking users from Singapore, South Korea, Uzbekistan, and Nigeria. Their findings showed that emotional design and user-centred design predicted satisfaction and perceived value, which later influenced customer loyalty. The study also highlighted localised user-experience strategies, intuitive layouts, and empathetic messaging. Since the research involved four countries, its overall coefficients should not be presented as findings that represent Singapore alone.

The results of Ogunmola et al. (2026) are broadly aligned with the present Malaysian study. Ease of Use made a significant unique contribution to Customer Satisfaction, while Website Design had the lowest mean and the largest improvement gap. Both studies suggest that customers judge digital banking not only by whether a transaction can be completed, but also by how easily they can understand, navigate, and use the platform.

Earlier usability research reached a similar conclusion. User-centred interface improvements were associated with better learnability, task efficiency, error performance, and satisfaction (Alfaridzi et al., 2023; Sudirjo et al., 2024). Interface quality is therefore more than a visual concern. It can shape how quickly customers find services, understand information, and complete banking tasks.

The two studies did not place equal emphasis on the same issues. The present Malaysian study focused more strongly on functional service performance. Efficiency was the strongest unique predictor, followed by System Availability, Responsiveness, and Ease of Use. The

dashboard also showed that transaction failure, delayed confirmation, app crashes, and OTP delays were common concerns.

The cross-national study involving Singapore gave greater attention to emotional design, user-centred interaction, co-creation, and perceived value (Ogunmola et al., 2026). Malaysian banks should therefore continue solving basic operational problems such as slow transactions, poor accessibility, transaction failures, and delayed confirmations. At the same time, they should develop a more customer-centred and reassuring digital experience.

5.3.2 Lessons from Cross-National Research Involving Singapore

Continuous user-centred design

One clear lesson is that user-centred design should be continuous rather than limited to a single redesign project. Malaysian banks should not rely only on the assumptions of managers or developers when changing mobile applications and websites. Customers need to be involved through interviews, prototype reviews, usability testing, and task-based observation. This process can reveal unclear terms, confusing navigation, and unnecessary transaction steps from the user's point of view (Ogunmola et al., 2026; Alfaridzi et al., 2023; Sudirjo et al., 2024).

This is especially relevant because Website Design was the weakest-performing dimension in the present study, while Ease of Use significantly influenced Customer Satisfaction. Testing should focus on frequently used activities such as fund transfers, bill payments, login, password recovery, and transaction-history checks. These sessions may uncover repeated data entry, unclear buttons, crowded screens, or instructions that require too much effort from users.

Empathetic and informative communication

A second lesson involves the way digital platforms communicate during a problem. Ogunmola et al. (2026) identified empathetic messaging as part of an emotionally intelligent and user-centred banking experience. A message such as "Transaction Failed" is not enough. Customers should be told whether the transaction went through, whether money was deducted, whether the payment is still pending, whether a reversal is expected, and what they should do next.

This recommendation is directly connected to the present results because transaction failure and delayed confirmation were among the most frequently reported issues. Kamboj et

al. (2021) also found that mobile-banking failures may involve functional, system, information, and service dimensions. Clear status messages can reduce uncertainty by showing whether a transaction is successful, pending, failed, or reversed.

Effective service recovery

Banks also need to improve what happens after a service problem, not only the experience when transactions work normally. A failed transfer, delayed OTP, app crash, or account-access issue should be followed by immediate acknowledgement, a clear expected resolution time, transaction-status tracking, and access to support.

Kamboj et al. (2021) found that mobile-banking failures can affect service use, satisfaction, and customer engagement. Automated tools may answer routine questions, but complicated, sensitive, or unresolved cases should be handed over to trained staff without unnecessary delay. Effective recovery can stop a temporary technical problem from becoming a wider confidence or trust issue.

Designing for different customer groups

Another lesson is that the same interface may not work equally well for every customer. Previous studies suggest that age and gender can influence how users evaluate certain mobile-banking service-quality dimensions and trust. Older customers should also not be treated as one identical group because their levels of digital familiarity, confidence, and support needs may differ (Trabelsi-Zoghalmi et al., 2018; Rajaobelina et al., 2020).

Malaysian banks could consider adjustable text size, stronger colour contrast, multilingual guidance, simpler navigation, and extra support for customers with lower digital confidence. However, the subgroup results from the current dashboard should not yet be used to create different services for specific groups. The subgroup sizes were unequal, and the observed differences were not tested inferentially. More representative evidence is needed first.

Continuous feedback and performance monitoring

The final lesson is that customer involvement should continue after a system is launched. Malaysian banks should move beyond one-time surveys by combining short post-transaction feedback, complaint records, usability-test findings, and anonymised operational indicators. These data could be added to the Power BI dashboard to show whether redesigns, system improvements, and service-recovery actions lead to measurable changes.

Interactive dashboards can help managers organise KPIs, compare results, identify unusual changes, and check whether indicators are moving in the intended direction (Gonçalves et al., 2023). Their value, however, should be judged through usability, task performance, stakeholder engagement, and practical usefulness rather than appearance alone (Zhuang et al., 2022).

5.3.3 Implications for Malaysian E-Banking Improvement

The comparison suggests that Malaysian banks need to improve both functional performance and the wider customer experience. The current findings place Ease of Use and System Availability under P1 Immediate. Efficiency, Responsiveness, Reliability and Website Design are placed under P2 Next, while Privacy and Security and Segment Differences are placed under P3 Monitor. Research involving Singapore adds another important point: these actions should be supported by intuitive design, empathetic communication and regular customer involvement (Ogunmola et al., 2026).

Customer satisfaction will not improve simply by making transactions faster or changing the visual appearance of an application in isolation. A platform can be technically functional and still frustrate users through confusing navigation or unclear failure messages. In the same way, an attractive interface cannot compensate for repeated transaction failures, slow OTP delivery, or frequent system interruptions.

Improvement therefore requires coordination among Product, IT Operations, UX/UI, Customer Service, Security, and Analytics teams. The aim should be a digital-banking experience that is fast and stable, but also clear, reassuring, secure, and responsive to customer needs.

5.4 Contributions and Practical Implications

This section explains how the project contributes to academic knowledge and provides practical value for Malaysian banking stakeholders.

5.4.1 Academic Contribution

This study adds local evidence on e-banking service quality and Customer Satisfaction in Malaysia. It examined seven dimensions in the same model and separated their bivariate relationships from their unique regression contributions. This distinction matters because a dimension can have a strong relationship with satisfaction when studied alone but may no longer remain significant after related dimensions are considered at the same time.

The study also shows the value of using several forms of evidence together. Statistical significance identifies the dimensions that uniquely explain satisfaction. Descriptive means reveal areas with relatively weaker performance, while issue selections show the problems customers encounter most often. Combining these measures led to a more balanced interpretation than relying on only one type of result.

A further contribution is the connection between service-quality research, data analytics, and business intelligence. The project did not stop after the statistical analysis. Its findings were presented in an interactive dashboard and then translated into an Evidence-to-Action Recommendation Mapping Matrix. This extended the work from identifying relationships to supporting practical decisions and future monitoring.

The study also contributes by showing how statistical findings can be converted into a dashboard structure for visualisation and KPI tracking. The dashboard tracks the indicators within the current survey dataset and provides baseline values for later comparison. This demonstrates how service-quality research can be extended beyond statistical tables into a reusable decision-support format.

5.4.2 Practical Contribution

The main practical contribution is a repeatable decision-support structure that Malaysian banks could adapt. The Power BI dashboard brings together service-quality scores, overall Customer Satisfaction, improvement gaps, reported issues, and segment comparisons in one place. This makes the findings easier to understand for stakeholders who may not be comfortable interpreting detailed statistical tables.

The dashboard gives stakeholders one place to review and track the current service-quality scores, Customer Satisfaction, improvement gaps, issue selections, and segment patterns. Its filters allow users to examine where weaker results or reported problems are concentrated. Although the current dashboard is not connected to live banking systems, its KPI definitions and data structure can support future updates when new data are collected.

Gonçalves et al. (2023) noted that interactive dashboards can support management by organising performance indicators into views that help users monitor patterns and compare results. Their usefulness still depends on accurate data, suitable KPIs, stakeholder involvement, and whether the dashboard supports real tasks and decisions (Zhuang et al., 2022).

The recommendation matrix also makes responsibility clearer. Product or Digital Banking teams can focus on task efficiency and transaction journeys. IT Operations can address system stability and transaction failures. UX/UI teams can improve navigation and interface clarity. Customer Service can strengthen responsiveness and service recovery. IT Security or Risk teams can maintain security controls and customer education, while Analytics teams can monitor segment patterns and KPI movement.

The project also establishes a way to assess recommended changes when future data become available. Higher satisfaction and dimension scores, together with lower improvement gaps and issue counts, would show movement in the expected direction. The current results serve as baseline values rather than evidence that improvement has already occurred. The project therefore supports a shift from one-time reporting towards structured monitoring, ownership, and follow-through.

5.5 Limitations of the Current Study

This section identifies the methodological and practical limitations that should be considered when interpreting the findings and assessing the project's wider applicability.

1. Cross-Sectional Research Design

The survey data were collected during a single period. The analysis could therefore identify relationships between service-quality dimensions and Customer Satisfaction, but it could not show whether changes in service quality caused later changes in satisfaction. The results should be interpreted as associations rather than proof of cause and effect.

The design also did not capture how customer expectations may change over time. Perceptions can shift after a system update, security incident, platform redesign, or the introduction of a new digital-banking feature.

2. Reliance on Self-Reported Perceptions

The study depended on questionnaire responses and therefore measured perceived service quality and remembered experiences. These responses may be influenced by memory, personal expectations, response style, or the respondent's most recent banking experience. The survey could identify the type of problem customers believed they experienced, but it could not confirm the technical cause. An app crash may be linked to software, server capacity, device compatibility, or network conditions. An OTP delay may involve the bank, an authentication service, or a telecommunications provider. Internal transaction records, backend architecture, server-performance data, and technical incident logs were not included in the analysis. The dashboard can therefore highlight and prioritise perceived problems, but it cannot diagnose technical root causes or detect incidents automatically. These limits were already defined in the project scope.

3. Unequal Representation Across Banks and Customer Groups

Although the study included 426 valid responses, the number of respondents was not evenly distributed across banks, age groups, and other customer segments. Several banks had relatively small samples, so their results may be strongly affected by a small number of especially positive or negative responses. For this reason, the bank-level findings should not be treated as final rankings of institutional performance. Differences across customer groups

should also be viewed as preliminary because they were descriptive and were not tested using inferential statistics. The questionnaire was distributed online and participation was voluntary. This may have encouraged stronger participation from users who were already more active or confident in digital environments. Older customers and people with lower digital confidence may therefore have been less well represented.

4. Limited Organisational Evaluation of the Dashboard

The Power BI dashboard was developed, checked, and refined as part of an academic project. It was not evaluated through long-term use by bank managers, analysts, IT staff, customer-service teams, or UX specialists in an actual banking environment. The study therefore cannot confirm that the dashboard would reduce decision time, improve coordination, or lead to measurable operational benefits. Its usefulness would also depend on regular data updates, consistent KPI definitions, and accurate ETL procedures.

5. Absence of Real-Time and Implementation Evidence

The dashboard used the collected survey dataset rather than live complaint, transaction, or system data. It supported analysis and prioritisation but could not detect real-time incidents automatically. The proposed recommendations were also not implemented during the study. The project identified what banks should improve and how progress could be monitored, but it did not demonstrate that the actions would directly increase satisfaction or reduce issue counts.

5.6 Suggestions for Future Work

This section proposes future research and system-development activities that could address the current study's limitations and strengthen the practical value of the project.

1. Conduct Longitudinal Research

Future studies should collect service-quality and satisfaction data at several points in time. Repeated measurement would allow researchers to examine whether customer perceptions change after platform updates, system interruptions, or improvement programmes. A longitudinal design would also help banks assess whether recommended actions are followed by higher satisfaction, smaller improvement gaps, and fewer reported issues. The questionnaire wording, variable definitions, and data-cleaning procedures should remain consistent across each collection period so that the results can be compared reliably.

2. Integrate Perception Data with Operational Evidence

Future work should combine survey responses with anonymised operational measures such as transaction-success rates, system downtime, page-loading time, OTP delivery time, complaint records, and incident-resolution time. This would reduce the study's dependence on customer perceptions alone. Banks could compare reported problems with technical evidence and investigate likely root causes more accurately. Strong privacy, security, and access controls would be essential because operational banking data are sensitive.

3. Improve Sample Representation

Stratified or quota sampling could be used in future research to achieve a more balanced distribution across banks, age groups, residence regions, devices, and usage-frequency categories. Larger samples for individual banks would make institutional comparisons more reliable and reduce the influence of single responses. More effort should also be made to include older customers and users with lower digital confidence, as their experiences may reveal different usability and accessibility needs.

4. Evaluate the Dashboard with Banking Stakeholders

Future work should test the dashboard with bank managers, analysts, IT staff, customer-service teams, and UX specialists. The evaluation should examine clarity, task-completion time, ease of interpretation, usefulness, and influence on decision-making. Zhuang et al. (2022) recommended evaluating dashboards through task performance, usability, implementation, engagement, and practical value rather than visual appearance alone. Feedback from stakeholders could then be used to improve navigation, filters, tooltip descriptions, and recommendation views.

5. Test the Recommendations Through Implementation

Selected recommendations should be implemented and assessed through a before-and-after design. A bank could, for example, simplify the transfer process, redesign transaction-status messages, or reduce service downtime. The related KPIs should then be compared before and after the change. A higher Ease of Use mean, fewer transaction-failure selections, and a higher overall satisfaction score would provide stronger evidence that the action was effective.

6. Develop Automated and More Frequent Data Updates

A future dashboard could use an automated ETL process connected to regularly updated survey, complaint, and operational data. This would reduce manual preparation work and help decision-makers identify changes sooner. Threshold-based alerts could notify responsible teams when satisfaction gaps, issue rates, downtime, or transaction failures exceed agreed limits. Reliable automation would still require consistent data definitions, validation rules, and access controls.

5.7 Summary

This project achieved all three research objectives. It identified the e-banking service-quality dimensions related to Customer Satisfaction, presented the findings through an interactive Power BI dashboard, and converted the evidence into prioritised recommendations supported by responsible teams and measurable KPIs.

The findings show that e-banking improvement should not be based on one statistical result alone. Banks need to consider the dimensions that have significant relationships with satisfaction, the areas that receive weaker evaluations, the problems reported most often, and the possible seriousness of those problems. Looking at these forms of evidence together creates a stronger link between customer feedback, analysis, dashboard monitoring, and service-improvement planning.

The study still had several limitations, including its cross-sectional design, reliance on self-reported responses, unequal subgroup sizes, and lack of live operational data. Future work could strengthen the framework through repeated data collection, integration with operational records, stakeholder evaluation, and direct testing of the proposed actions. These improvements would provide stronger evidence on whether changes to e-banking services lead to better customer experiences.

Chapter 6. Work Plan

6.0 Overview

This chapter outlines the overall work plan for both Capstone 1, which focuses on the proposal, and Capstone 2, which covers implementation and reporting. It explains how the timeline is managed using three linked planning tools:

1. Gantt charts to visualise the schedule and task overlaps.
2. Milestone timeline chart and tables to translate the schedule into clear checkpoints, stating each deliverable, its target date, completion criteria, and prerequisite tasks.
3. A phase-based flowchart to summarise the end-to-end workflow at a process level.

These tools help the project move in a structured way from early scoping through to the final submission.

6.1 Gantt Chart

The Gantt charts give a timeline view of the planned tasks and the order in which they should happen, so progress can be checked against expected dates and overlaps between tasks can be seen clearly. Separate charts are used for Capstone 1 and Capstone 2, since each phase has its own goals, main outputs, and working period. The detailed task breakdown is provided in the Work Breakdown Table (Appendix D). The Gantt chart is derived by converting the Appendix D work breakdown table into a timeline-based chart. For consistency, all task durations in this chapter are calculated using calendar days, including both the start and end dates.

6.1.1 Capstone 1

Figure 6.1 presents the Capstone 1 work plan (Gantt chart), highlighting key activities and the planned timeline.

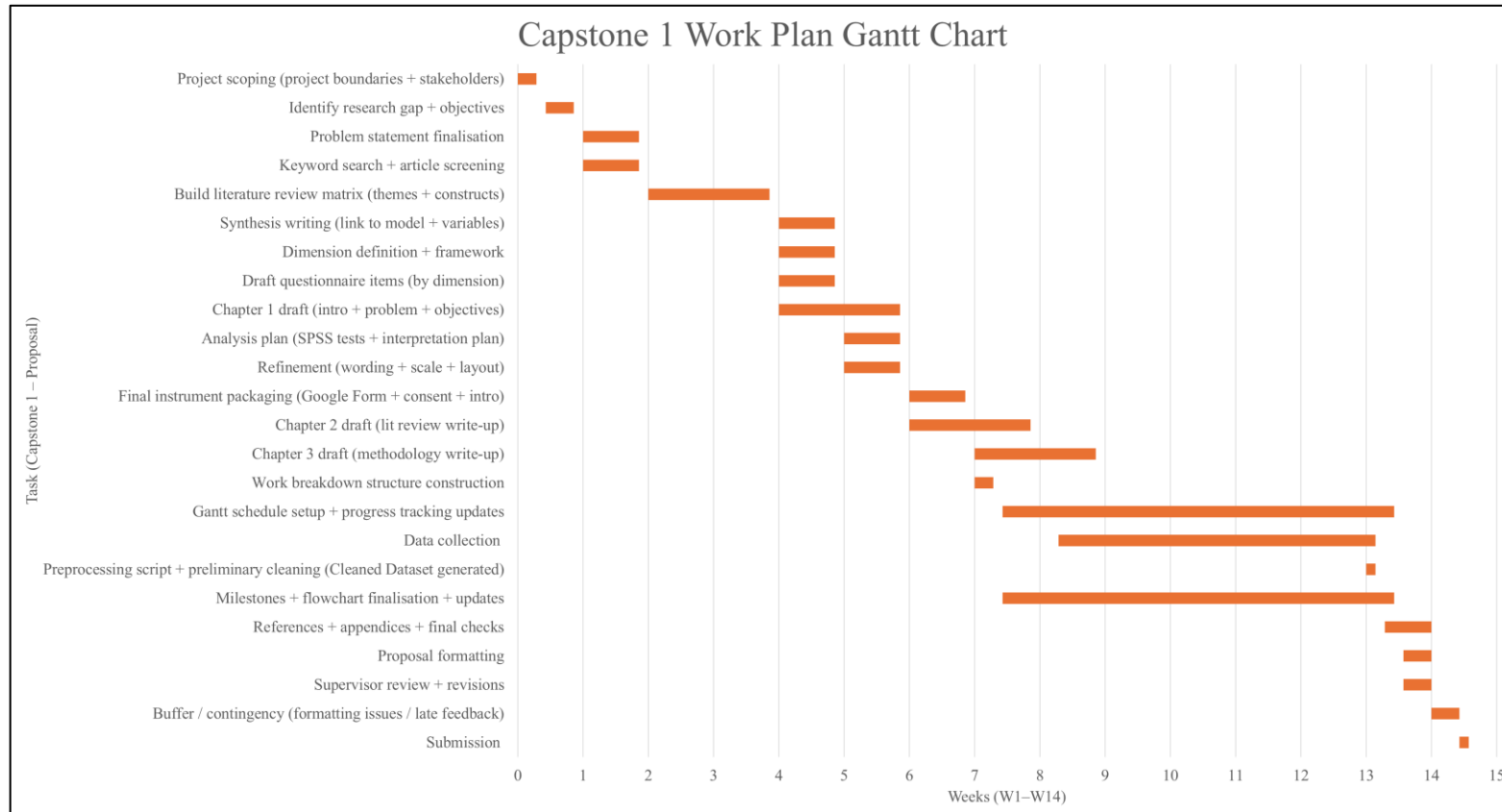


Figure 6.1: Capstone 1 Work Plan Gantt Chart

6.1.2 Capstone 2

Figure 6.2 presents the Capstone 2 work plan (Gantt chart), showing the sequencing of analysis, dashboard work, and reporting tasks.

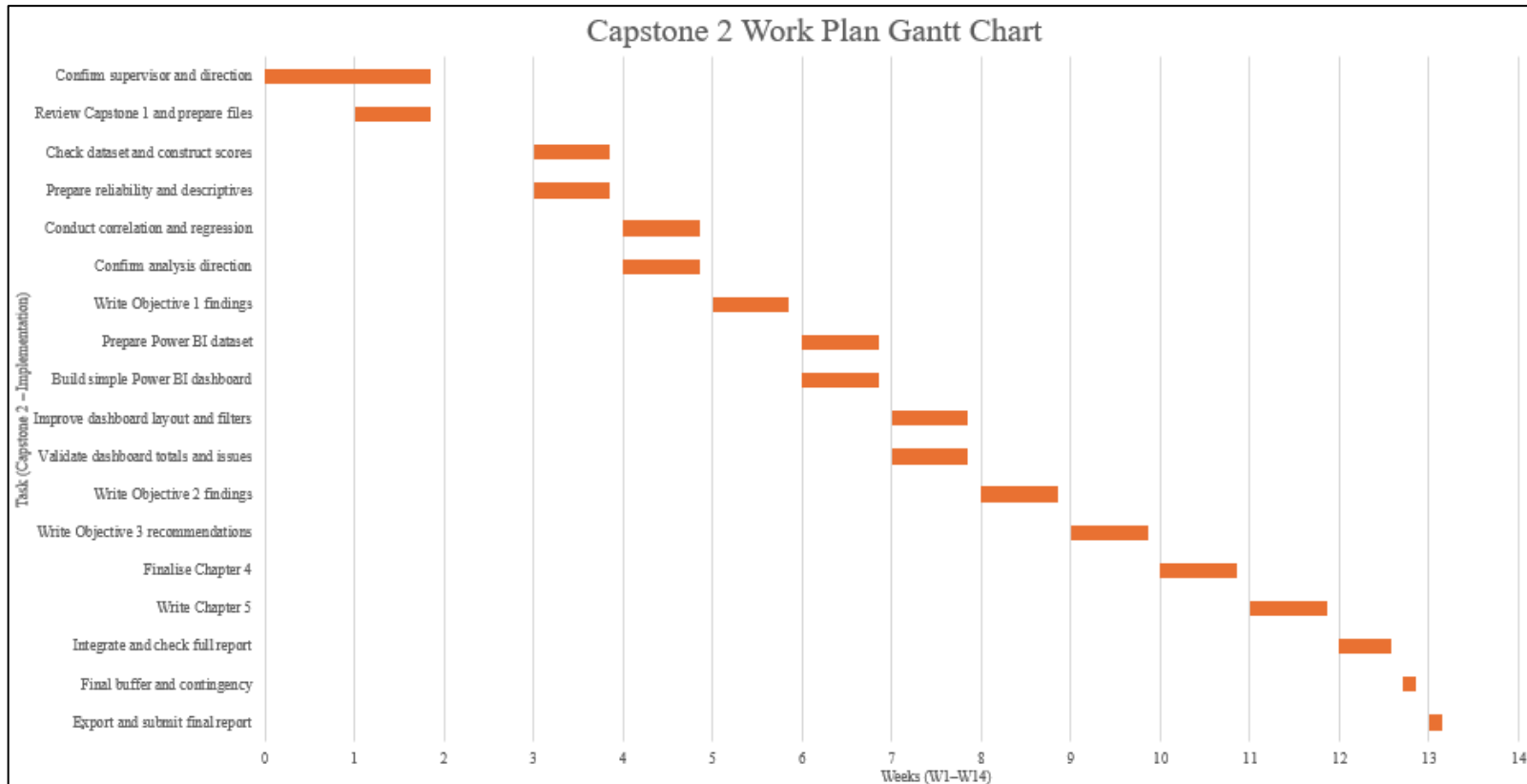


Figure 6.2: Capstone 2 Work Plan Gantt Chart

6.2 Milestones

The milestone timeline chart and tables render the Gantt chart schedule into tangible “checkpoints” that can be more readily followed and talked about. Each milestone includes what needs to be produced, the target date, the criteria for when it can be called finalized, and tasks that must be completed in advance. The time required to produce each deliverable is modeled with three practical points. First, the schedule includes the effort involved in meeting the required academic standard. Second, the schedule takes into account dependencies. Among the dependencies between the tasks, some tasks cannot be performed simultaneously. Third, the schedule incorporates delay points that are common, including low survey response, slow supervisor feedback, formatting, and submission issues. A small buffer is also placed near the final submission period to reduce the chance of missing key deadlines.

6.2.1 Capstone 1

Figure 6.3 presents the scheduled Capstone 1 proposal milestones on a single timeline. Overall, it highlights the sequencing and clustering of late-stage milestones near submission, emphasizing the need for tight coordination and buffer protection at the end of the schedule.

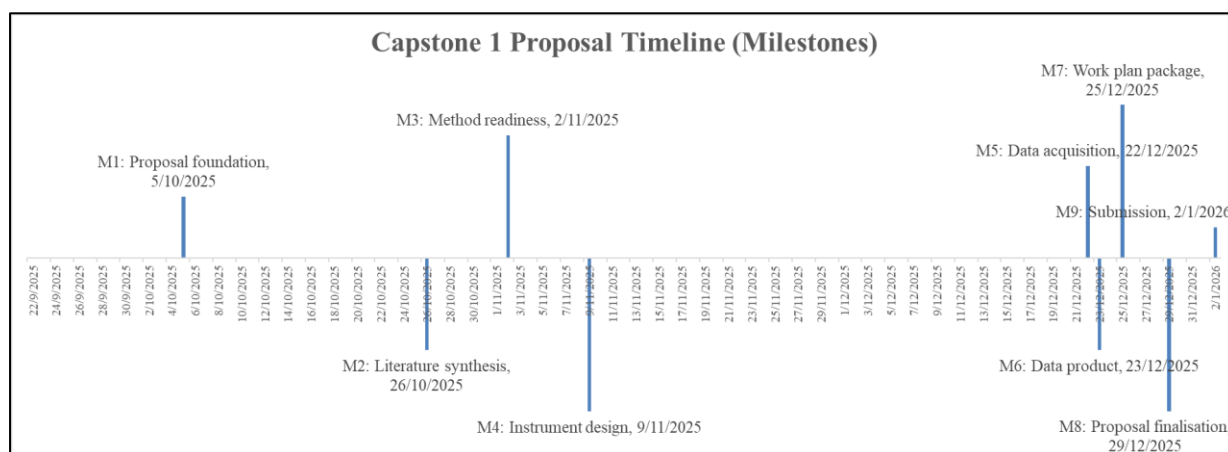


Figure 6.3: Capstone 1 Proposal Milestone Timeline

Table 6.1 summarises the key milestones for Capstone 1, showing the planned deliverables, target dates, and clear completion criteria for each checkpoint. It also states the task dependencies for every milestone, so the project sequence, handoffs, and critical prerequisites are transparent from proposal foundation through final submission.

Table 6.1: Milestones for Capstone 1

Milestone	Deliverable	Date	Completion criteria	Depends on
M1: Proposal foundation	Project scope and stakeholders confirmed	5/10/2025	• Scope boundaries and stakeholders documented. • Research gap and objectives are stated clearly. • Problem statement is finalised.	• Project scoping. • Identify research gap and objectives. • Problem statement finalisation.
M2: Literature synthesis	Literature review matrix and synthesis completed	26/10/2025	• Literature matrix completed with key themes and constructs. • Synthesis links constructs to framework and variables.	• Keyword search and article screening. • Build literature review matrix. • Synthesis writing. • Dimension definition and framework.
M3: Method readiness	Analysis plan finalised (SPSS tests and interpretation plan)	2/11/2025	• SPSS tests selected and justified. • Scoring and interpretation steps documented. • Plan aligns with constructs and framework.	• Analysis plan.
M4: Instrument design	Questionnaire finalised and packaged (Google Form ready)	9/11/2025	• Items finalised by dimension. • Wording, scale, and layout refined. • Consent and introduction included.	• Draft questionnaire items. • Refinement.

			• Final Google Form is ready to deploy.	Final instrument packaging.
M5: Data acquisition	Data collection closed	22/12/2025	• Data collection ended. Responses exported. • Minimum usable responses achieved after removing incomplete or invalid entries.	Data collection.
M6: Data product	Cleaned analysis dataset frozen	23/12/2025	• Missing or invalid values handled. • Coding and labels standardised. • Dataset is ready for SPSS analysis. • Dataset version is frozen and archived.	Preprocessing script and preliminary cleaning.
M7: Work plan package	Work Plan completed (WBS, Gantt chart, milestones, flowchart)	25/12/2025	• WBS structured into phases and work packages. • Gantt shows dates, dependencies, and buffer. • Milestone table and process flowchart are completed and consistent.	• Work breakdown structure construction. • Gantt schedule setup and progress tracking updates. • Milestones and flowchart finalisation and updates.
M8: Proposal finalisation	Proposal ready for submission	1/1/2026	• Proposal formatting completed. • References and appendices included.	• References and appendices and final checks.

			• Final checks completed. Supervisor revisions applied.	• Proposal formatting. • Supervisor review and revisions. • Buffer and contingency.
M9: Submission	Capstone 1 submitted	2/1/2026	• Final proposal submitted successfully. • Submission confirmation recorded.	Submission.

Table 6.2 shows the planned time length for each main Capstone 1 work block. It also gives a short reason for each time allocation. Each block is linked to its related milestone(s) and to the risks that the schedule aims to control.

Table 6.2: Duration Justification for Major Work Blocks (Capstone 1)

Work block	Planned duration (days)	Justification	Related milestone(s)	Key risks addressed
A) Chapter drafting blocks (Ch.1, Ch.2, Ch.3)	13 days each	Each chapter draft needs time for planning, checking the rubric, adding and matching citations, improving clarity, and checking that the scope, constructs, and method stay consistent across sections. This time also supports repeated revision before the final version is locked.	Supports readiness for M8	Risk 3, Risk 5
B) Data collection	33 days	The data collection window includes survey distribution, reminder rounds, reaching a minimum number of usable responses, removing incomplete or invalid entries, and exporting the dataset for cleaning. A longer duration is needed because the response rate is uncertain and may change week by week.	M5: Data acquisition	Risk 1, Risk 2
C) Planning and monitoring (work plan	42 days	Planning continues throughout the project. Time is set aside for progress tracking, handling dependency changes, and keeping	M7: Work plan package	Risk 4 (primary),

build & maintenance)		the WBS, Gantt chart, milestone table, and workflow diagram consistent until the full work plan package is finalised.		supports Risk 3
D) Preprocessing script & preliminary cleaning	1 day	Data cleaning is treated as a focused step after collection ends. The task includes generating a clean dataset output, handling obvious missing or invalid values, standardising labels and codes, and saving one stable analysis-ready version. This stage covers preliminary cleaning only, not full statistical analysis.	M6: Data product	Risk 2
E) References, appendices & final checks	5 days	Time is also reserved for final documentation checks. This includes making sure the reference list matches the in-text citations, packaging the appendices (questionnaire, code, and work breakdown), checking figure and table numbering, and reducing avoidable submission errors.	M8: Proposal finalisation	Risk 5
F) Proposal formatting	3 days	Layout and formatting take more time than anticipated. Alignment of headings, tables, figures, spacing and page layout with faculty needs. Readability and consistent style across chapters.	M8: Proposal finalisation	Risk 5

G) Supervisor review & revisions	3 days	The process includes a review phase to integrate supervisor inputs, address areas where the rubric has gaps, and refine the overall flow before submitting it. This phase ensures that the report aligns well with the evaluation criteria and enhances the overall coherence across sections, leading to a clearer, more coherent, and better-explained final report that can be easily assessed.	M8: Proposal finalisation	Risk 3
H) Buffer / contingency	3 days	The last phase of the schedule includes a short buffer to account for late supervisor's feedback, unforeseen formatting problems, or last-minute revisions. This buffer does not affect the submission date but protects the critical path of the project and ensures it is completed gracefully even in the face of modest delays.	Protects M9 readiness	Risk 3, Risk 5
I) Submission	1 day	Submission is one of the briefest steps but still treated as a separate stage in the workflow. It involves the final export to check that files are complete and correctly formatted, time set for uploading, and preparations for the submission recording. Keeping this record serves as a buffer in case of potential technical issues,	M9: Submission	Risk 5

		ensuring that proof of submission can be retrieved and the process can be counted on.		
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Table 6.3 outlines the key delivery risks for Capstone 1. It explains how each risk may affect the work plan and the timeline. It also lists practical actions to reduce the risk and links these actions to the relevant milestones (M5–M9). This helps keep the proposal on track and ready for submission.

Table 6.3: Key risks and mitigation actions for Capstone 1

Risk	Impact	Mitigation	Related milestone(s)
Risk 1: Low response rate / slow participation during data collection	The survey may not reach the minimum number of usable responses by the planned closing date.	Begin survey distribution early and run reminder rounds during the collection period. Use more channels to reach different groups, such as student communities and working adults, and use mixed platforms where appropriate. If the response rate stays slow, activate a backup contact list and increase outreach while keeping the closing date unchanged.	M5: Data acquisition
Risk 2: Data quality issues (incomplete, inconsistent, or invalid responses)	More responses may be removed than expected because of incomplete or invalid entries. This can reduce the final sample size and push back the preliminary cleaning stage.	Apply the screening rules in a consistent way and record each cleaning decision. Once the minimum usable response target is met, save one cleaned dataset version and keep that version fixed for the next steps.	M6: Data product

Risk 3: Late supervisor feedback close to submission	Late revisions can squeeze the time for formatting and final checks near submission. This increases the chance of mistakes in tables, figures, and references.	Place the supervisor review inside the planned finalisation window and keep the contingency buffer for late changes. If feedback arrives close to the deadline, focus first on rubric-critical fixes, such as scope clarity, method alignment, and work plan completeness.	M8: Proposal finalisation, M9: Submission
Risk 4: Work plan artefacts become inconsistent (WBS, Gantt, milestone table, flowchart)	If the work plan artefacts do not match each other, markers may see the plan as unclear or unprofessional. This can weaken the overall impression of project control.	Treat tracking and updates as continuous work until M7. Only lock the final package after all artefacts match the same task names, dates, and dependencies across the WBS, Gantt chart, milestones, and workflow diagram.	M7: Work plan package
Risk 5: Formatting/submission errors (export issues, broken references, last-minute edits)	Submission may be delayed or the final document may contain avoidable technical issues.	Use the final checks period and the buffer to reduce avoidable errors. Complete a final review for reference matching, figure and table numbering, and file export checks before submission.	M8: Proposal finalisation, M9: Submission

6.2.2 Capstone 2

Figure 6.4 presents the Capstone 2 milestone timeline from 4 May 2026 to 3 August 2026. It shows the progression from supervisor confirmation and statistical analysis to Power BI dashboard development, recommendation preparation, final report integration and submission. The timeline also includes a short contingency period before the final submission date.

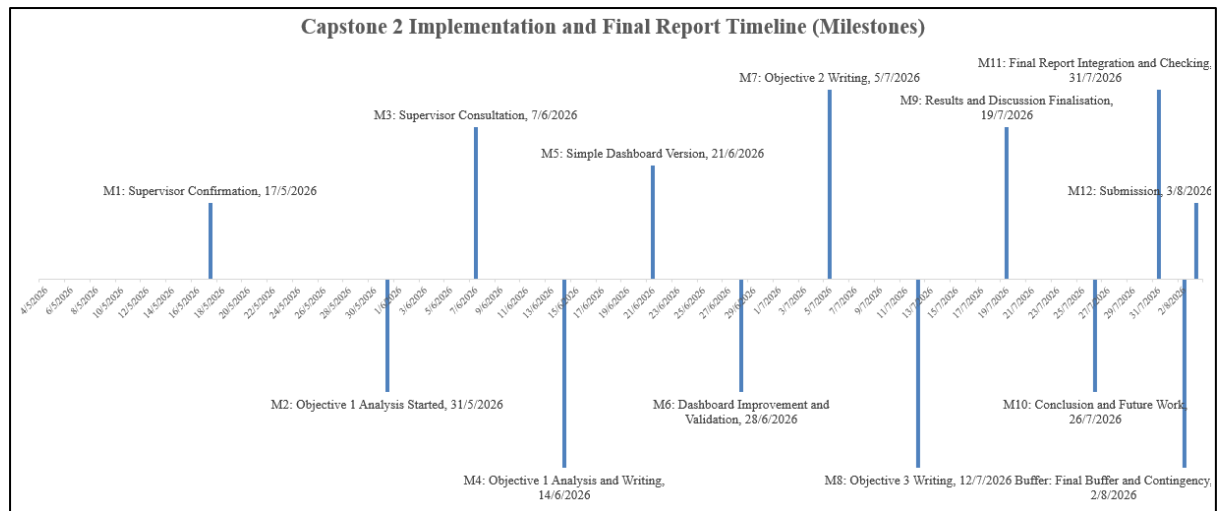


Figure 6.4: Capstone 2 Proposal Milestone Timeline

Table 6.4 lists the main Capstone 2 milestones and shows what must be delivered at each stage. It includes target dates, clear completion criteria, and key dependencies, starting from supervisor confirmation and statistical analysis preparation, followed by dashboard development and validation, recommendations, final writing, report integration, a contingency buffer, and submission. The table also shows how each stage depends on earlier outputs, so progress can be tracked step by step until the final submission.

Table 6.4: Milestones for Capstone 2

Milestone	Deliverable	Date	Completion criteria	Depends on
M1: Supervisor Confirmation	Supervisor and Capstone 2 direction confirmed	17/05/2026	- Supervisor list checked - Supervisor confirmed	Capstone 1 proposal and supervisor list

			• Capstone 2 direction reviewed based on Capstone 1 proposal.	
M2: Objective 1 Analysis Started	Statistical analysis preparation started	31/05/2026	• Cleaned dataset checked • Construct scores reviewed; • Reliability and descriptive analysis outputs prepared.	Cleaned dataset and measurement items
M3: Supervisor Consultation	Analysis direction confirmed with supervisor	07/06/2026	• Supervisor contacted; • Analysis direction, Chapter 4 structure, and SPSS output interpretation checked.	Initial Objective 1 analysis
M4: Objective 1 Analysis and Writing	Statistical findings written for Chapter 4	14/06/2026	Reliability, descriptive statistics, Pearson correlation, and multiple linear regression results completed and written.	SPSS outputs and supervisor feedback
M5: Simple Dashboard Version	Initial Power BI dashboard created	21/06/2026	Basic KPI cards, customer satisfaction score, service-quality dimension scores, and main filters created.	Objective 1 findings and dashboard-ready dataset
M6: Dashboard Improvement and Validation	Improved Power BI dashboard completed and checked	28/06/2026	• Dashboard pages improved • Filter functions checked • Dashboard values compared with analysis	Simple dashboard version

			outputs to reduce inconsistencies.	
M7: Objective 2 Writing	Dashboard findings written in Chapter 4	05/07/2026	Dashboard pages, KPIs, issue monitoring, segment comparison, and visual insights explained and linked to Objective 2.	Completed and checked Power BI dashboard
M8: Objective 3 Writing	Data-driven recommendations completed	12/07/2026	• Recommendations written based on statistical findings and dashboard insights • Priorities, responsible areas, and monitoring KPIs included.	Objective 1 findings and Objective 2 dashboard insights
M9: Results and Discussion Finalisation	Chapter 4 completed	19/07/2026	• Objective 1, Objective 2, and Objective 3 discussion completed • Tables and figures referenced correctly • Results linked to research objectives.	Objective 1, Objective 2, and Objective 3 writing
M10: Conclusion and Future Work	Chapter 5 completed	26/07/2026	• Overall conclusion, objective summary, practical contribution, limitations, and future work completed.	Finalised Chapter 4
M11: Final Report Integration and Checking	Full report checked and ready for submission	31/07/2026	• Chapters 1 to 6 checked • Abstract updated	Completed full report draft and

			• References, appendices, formatting, table numbering, and figure numbering reviewed • Supervisor corrections applied.	supervisor feedback
Buffer: Final Buffer and Contingency	Final formatting, export, and technical checking completed	01/08/2026	• Final supervisor corrections applied • References, tables, figures, and appendices checked • Dashboard and website links tested • Correct file version prepared for submission.	M11: Final Report Integration and Checking
M12: Submission	Final Capstone 2 report submitted	03/08/2026	Final report exported, uploaded, and submission confirmation recorded.	Final buffer and contingency period

Table 6.5 shows the planned time for each main Capstone 2 work block and explains why the time is needed. Each block is linked to its related milestone(s), and the main risks are stated so the reason for the timing is clear. Overall, the schedule is designed to support a stable measurement model, dependable analysis results, clear KPI definitions, a dashboard that is checked and validated, and enough time for recommendations, final report integration, and a buffer before submission.

Table 6.5: Duration Justification for Major Work Blocks (Capstone 2)

Work block	Planned duration (days)	Justification	Related milestone(s)	Key risks addressed
A) Supervisor confirmation and project direction checking	14 days	The first two weeks were used to confirm the supervisor, review Capstone 1 feedback, and clarify the direction for Capstone 2. This was necessary because the project moved from proposal planning into implementation and final reporting.	M1	Risk 7
B) Objective 1 statistical analysis	14 days	Objective 1 required checking the cleaned dataset, reviewing construct scores, preparing reliability outputs, descriptive statistics, correlation analysis, and multiple linear regression. This stage was important because the later dashboard and recommendations depended on accurate statistical findings.	M2, M3	Risk 1, Risk 2
C) Objective 1 writing	7 days	The statistical results had to be explained clearly in Chapter 4. The writing included reliability interpretation, respondent profile, service-quality dimension results,	M4	Risk 2, Risk 6

		correlation findings, regression model interpretation, and discussion linked to the research objective.		
D) Dashboard development	14 days	A simple Power BI dashboard version was first created to present key measures such as customer satisfaction, service-quality scores, issue prevalence, and main filters. The dashboard was then improved into a fuller version with clearer pages, filters, and issue analysis over a two-week development period.	M5, M6	Risk 3, Risk 4
E) Dashboard validation and Objective 2 writing	14 days	Dashboard values were checked against the analysis outputs to reduce errors in totals, filters, and issue counts. The Objective 2 section was then written to explain the dashboard pages, KPI cards, issue monitoring, segment comparisons, and insights.	M6, M7	Risk 3, Risk 4, Risk 6
F) Objective 3 recommendations writing	7 days	The recommendations were developed after the analysis and dashboard findings were completed. Each recommendation was linked to evidence, service-quality dimensions, priority level, responsible area, and monitoring KPI so that the recommendations were not too general.	M8	Risk 5
G) Chapter 4 finalisation	7 days	Chapter 4 was finalised after the Objective 1, Objective 2, and Objective 3 sections were written.	M9	Risk 2, Risk 6

		This stage ensured that the results, dashboard insights, and recommendations were connected clearly to the three project objectives.		
H) Chapter 5 conclusion and future work	7 days	Chapter 5 was written after Chapter 4 was finalised. This allowed the conclusion to summarise the actual findings, discuss limitations, state practical contributions, and propose future work based on the completed project.	M10	Risk 6
I) Final report integration and checking	5 days	The full report was checked for consistency across Chapters 1 to 6. This included updating the abstract, applying supervisor corrections, checking references, table and figure numbering, formatting, appendices, and final export quality.	M11	Risk 6
J) Final buffer and contingency	2 days	A short buffer was reserved for final supervisor corrections, reference and citation checks, table and figure checks, Appendix Confirmation, dashboard and website link testing, PDF review, and preparation of the correct submission file.	Protects M12 readiness	Risk 6
K) Submission	1 day	Submission was treated as a separate stage to allow time for exporting the final document, uploading the correct file, and recording proof of submission.	M12	Risk 6

Table 6.6 lists the main delivery risks for Capstone 2 and explains how each risk may affect the project results. It also provides practical actions to reduce each risk and links the risk to the milestone(s) that could be delayed or affected. This structure makes it easier to monitor risks in line with the timeline and the planned deliverables.

Table 6.6: Key risks and mitigation actions for Capstone 2

Risk	Impact	Mitigation actions	Related milestone(s)
Risk 1: Statistical output inconsistency	Reliability, correlation, or regression results may need to be rerun if the wrong dataset or scoring method is used.	Use one fixed cleaned dataset version, document construct scoring rules, and save SPSS outputs clearly.	M2, M3, M4
Risk 2: Weak or unclear interpretation of Objective 1	The Chapter 4 discussion may become too descriptive and not sufficiently linked to the research objectives.	Interpret results using effect size, significance, and relationship to customer satisfaction. Link findings to the objective and prior literature where appropriate.	M4, M9
Risk 3: Dashboard calculation or filter errors	Dashboard values may not match analysis results, causing inconsistency between the report and dashboard.	Check dashboard totals, filters, and issue counts against Excel/SPSS outputs before writing the dashboard discussion.	M5, M6, M7
Risk 4: Dashboard design not supporting decision-making	The dashboard may look visual but fail to explain the key	Use clear KPI cards, dimension comparison, issue monitoring, and insight summary pages so that the	M5, M6, M7

	service-quality issues clearly.	dashboard supports Objective 2.	
Risk 5: Recommendations too general	Recommendations may appear weak if they are not clearly connected to actual findings.	Link each recommendation to statistical findings, dashboard insights, issue frequency, responsible area, priority level, and monitoring KPI.	M8
Risk 6: Final formatting or submission error	Incorrect file version, missing references, wrong numbering, or upload issues may affect final submission quality.	Reserve final checking time for formatting, references, appendices, figure/table numbering, file export, and submission proof.	M11, M12
Risk 7: Supervisor availability or late direction changes	Delayed confirmation or a late change in project direction may postpone the start of analysis or require completed work to be revised.	Contact the supervisor early, prepare focused questions and Capstone 1 materials before consultation, document the agreed direction, and follow up promptly when clarification is needed.	M1, M3

6.3 Flowchart

In this section, the flowchart describes the work plan as a sequence of processes instead of a timeline. The Gantt chart still shows when tasks happen, and the milestone tables still list what needs to be delivered, but the flowchart focuses on how work moves from one phase to the next and how outputs feed forward as inputs. For this chapter, it is organised into three phases, which makes it easier to explain the overall workflow during supervision meetings and progress updates.

6.3.1 Phase 1

This workflow explains how the study finds which e-banking service quality dimensions have the strongest link with customer satisfaction, using a clear survey-to-statistics path. It first states the analytical goal of identifying key drivers, then turns this goal into specific survey questions and a repeatable analysis sequence that ranks the dimensions based on the evidence. Figure 6.5 illustrates the workflow for Objective 1, from survey data preparation to model-ready datasets.

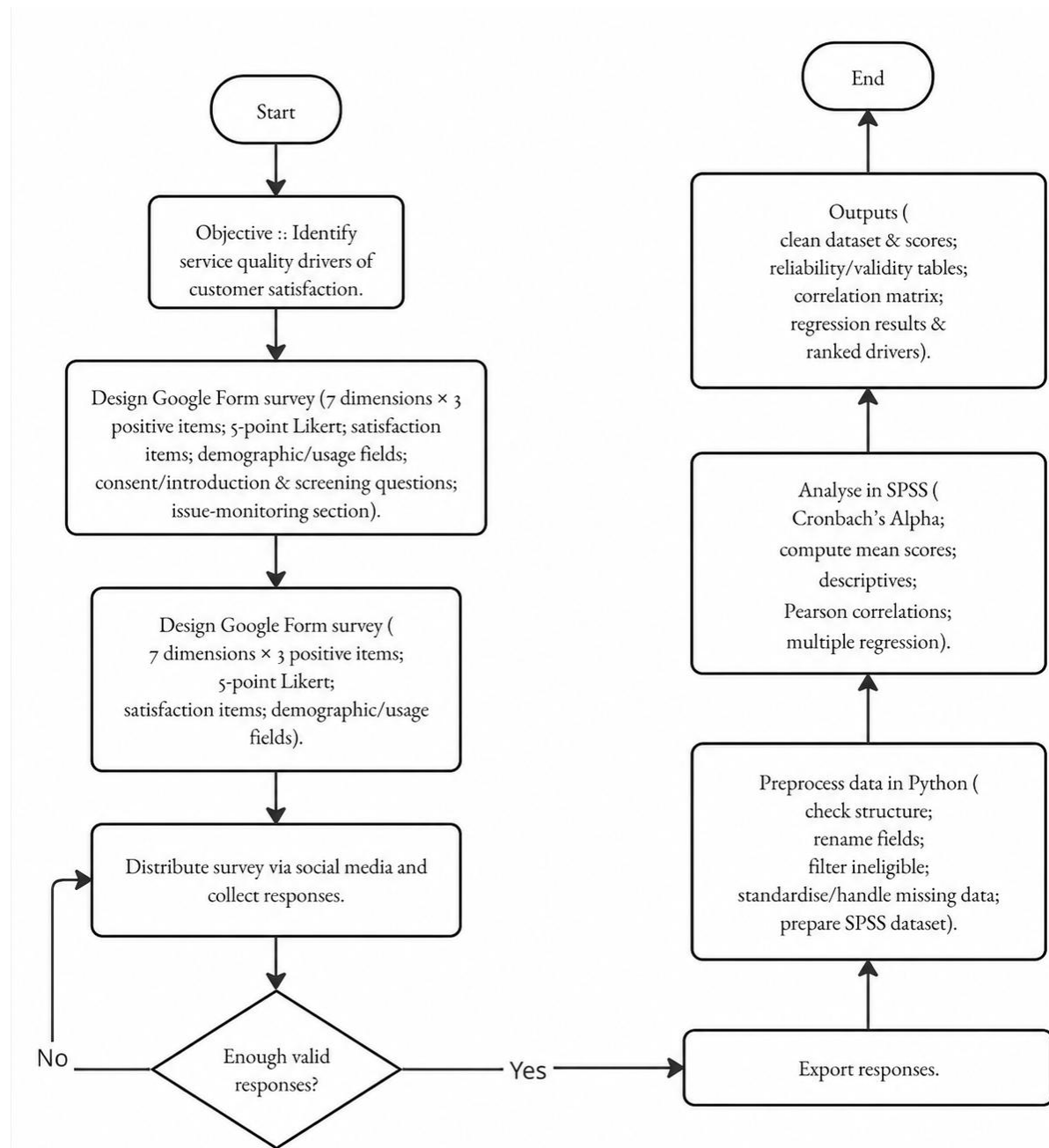


Figure 6.5: Objective 1 workflow

6.3.2 Phase 2

This workflow shows how the project turns the survey results into a dashboard that managers can read like a story and use to make choices. The aim is not only to show charts. The aim is to help users answer clear questions, such as what should be fixed first, which customer group is most affected, and where the biggest service gaps are. Figure 6.6 presents the Objective 2 workflow and shows the steps for dashboard preparation and visual analysis in Power BI.

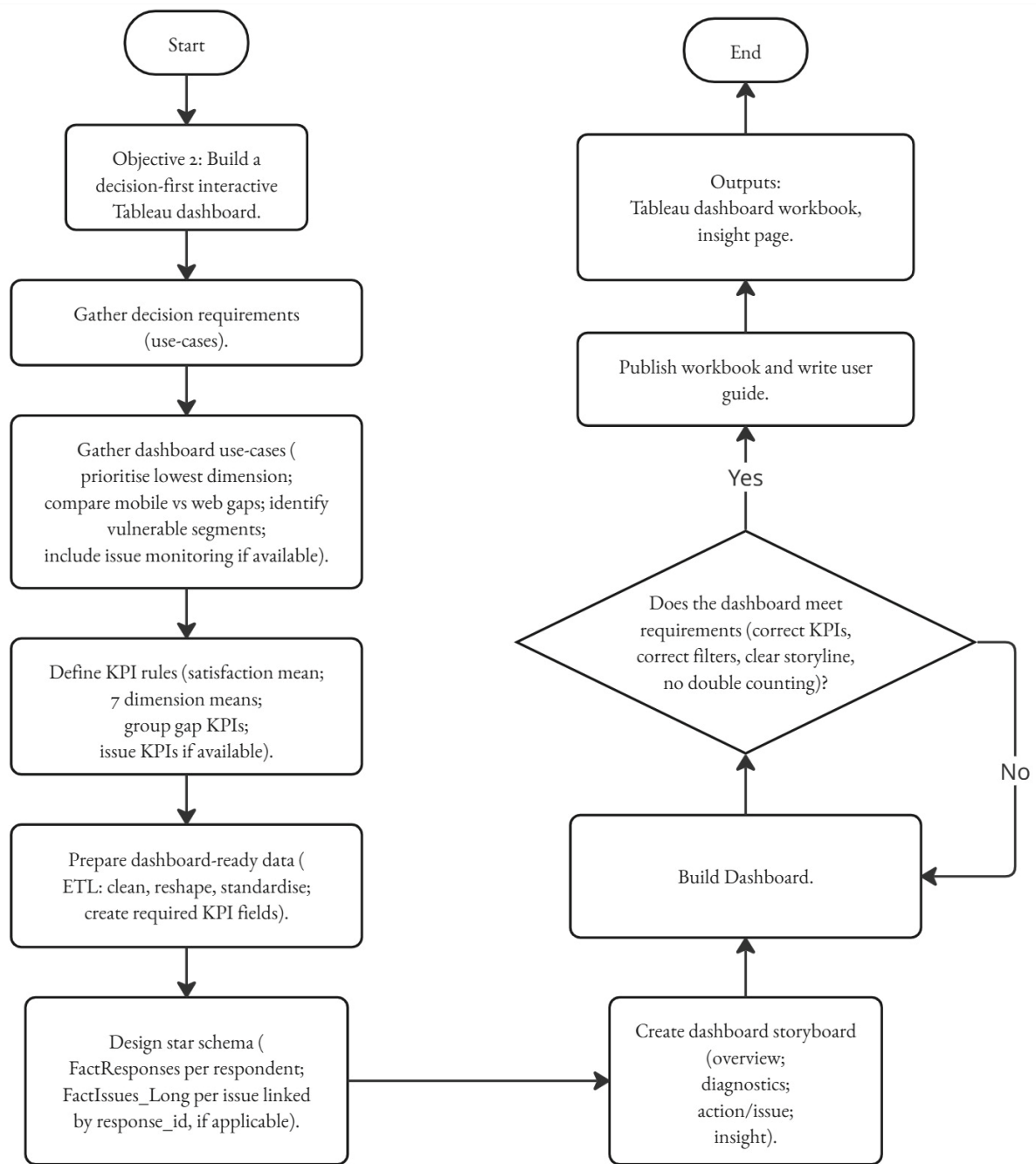


Figure 6.6: Objective 2 workflow

6.3.3 Phase 3

This workflow uses statistical results and dashboard patterns to turn evidence into practical recommendations. Each recommendation is linked to a responsible party and to simple indicators that can be tracked over time. This way, the project supports real service improvements and does not stop at reporting findings. Figure 6.7 shows the Objective 3 process and explains how the Evidence-to-Action approach is used to map findings into specific actions.

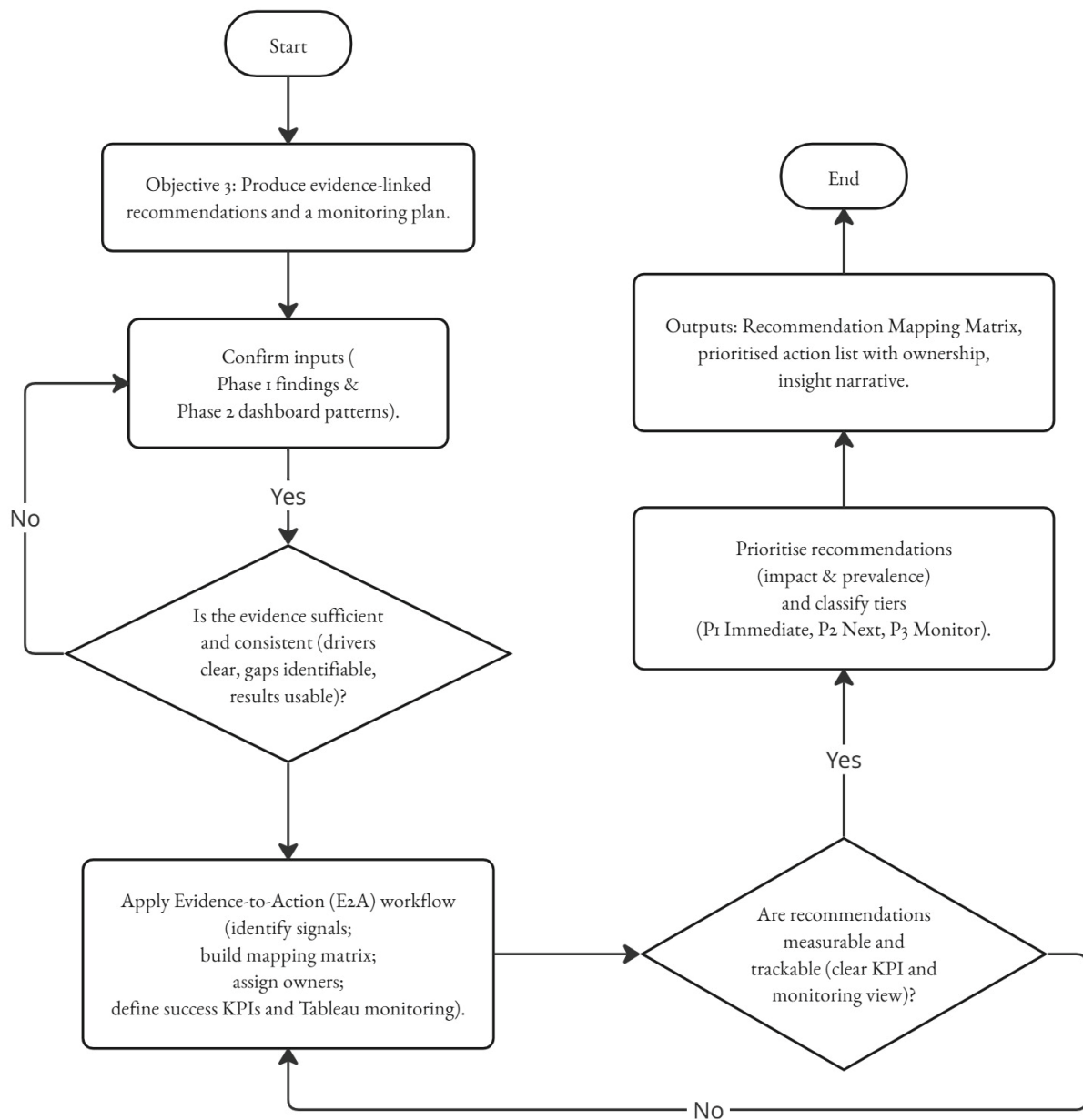


Figure 6.7: Objective 3 workflow

6.4 Summary

This chapter brings the project schedule and delivery controls for both capstones into one place by linking three views: time planning through Gantt charts, measurable checkpoints through milestones with clear completion criteria and dependencies, and process flow through a phase-based flowchart. The milestone plan uses time estimates that match real academic workload, task dependencies, and common project risks. A small buffer is also placed close to submission to protect the critical path if delays happen. Appendix D adds a more detailed task breakdown with start and end dates and estimated durations. This allows progress to be tracked at the milestone level and also through day-to-day tasks.

Capstone 2 focused on implementation, dashboard development, validation and final reporting. The work period ran from early May to mid-August 2026, beginning with reliability assessment and construct-score preparation. Descriptive analysis, correlation analysis and multiple linear regression were then conducted to identify the service-quality dimensions associated with Customer Satisfaction. A KPI dictionary and Power BI-ready datasets were subsequently prepared through ETL and data modelling. The dashboard was developed and checked for total consistency, filter behaviour and drill-down accuracy. Finally, the findings were translated into an Evidence-to-Action recommendation set, followed by report integration and preparation for submission.

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Appendix

Appendix A

Digital Supporting Materials

The supporting materials for this final-year project are available in the GitHub repository through the link below:

GitHub Repository:

<https://github.com/wenssy-gif/fyp-appendix-files>



Figure A: QR Code for the Final Year Project GitHub Repository

The repository contains the anonymised dataset, processed datasets, SPSS output results, SPSS syntax files and HTML dashboard files used in this project. All personally identifiable respondent information has been removed to maintain data privacy and confidentiality.

Appendix B

Interactive Web-Based Dashboard

The Malaysia E-Banking Service Quality Dashboard can be accessed through the following link:

<https://wenssy-gif.github.io/fyp-appendix-files/malaysia-ebanking-service-quality-analysis/dashboard/>



Figure B: QR Code for the Interactive E-Banking Dashboard

Appendix C

Google Form Survey Questionnaire

12/06/25, 6:31 AM	Malaysian E-Banking Service Quality & Customer Satisfaction Survey	12/06/25, 6:31 AM	Malaysian E-Banking Service Quality & Customer Satisfaction Survey
Malaysian E-Banking Service Quality & Customer Satisfaction Survey This survey is conducted as part of my Final Year Project for the Bachelor of Information Systems (Honours) (Data Analytics) at Sunway University, titled "A Data-Driven Web-Based Service Quality Analysis for Enhancing Customer Satisfaction in Malaysian E-Banking". The purpose of this study is to examine how different aspects of e-banking service quality influence customer satisfaction. The insights collected will be analysed and visualised through a web-based dashboard developed for this study. Your responses are anonymous and will be used solely for academic research. No sensitive identifiers (e.g., name, IC, bank account number) will be collected. Participation is voluntary, and you may exit the survey at any time without penalty. * Indicates required question 1. I have read the information above and agree to participate. * Mark only one oval. ☐ Yes, I agree ☐ No, I do not agree SECTION A — Demographic Information This section collects basic background information to help understand how different groups of users experience e-banking services.		2. Age Group * Mark only one oval. ☐ Below 18 years ☐ 18–20 years ☐ 21–30 years ☐ 31–40 years ☐ 41–50 years ☐ Above 50 years 3. Nationality * Mark only one oval. ☐ Malaysian ☐ Non-Malaysian 4. Ethnicity * Mark only one oval. ☐ Malay ☐ Chinese ☐ Indian ☐ Other: _____ 5. Gender * Mark only one oval. ☐ Male ☐ Female	
https://docs.google.com/forms/d/1g2JHFD3USXBCGQ2b7gYkLUQDjU0U2N-zZW1d9edR 1/16		https://docs.google.com/forms/d/1g2JHFD3USXBCGQ2b7gYkLUQDjU0U2N-zZW1d9edR 2/16	
12/06/25, 6:31 AM Malaysian E-Banking Service Quality & Customer Satisfaction Survey 6. Employment Status * Mark only one oval. ☐ Employed ☐ Student ☐ Retired ☐ Unemployed 7. Monthly Income (RM) * Mark only one oval. ☐ Below 2,000 ☐ 2,001 – 4,000 ☐ 4,001 – 6,000 ☐ 6,001 – 8,000 ☐ Above 8,000 8. Number of Bank Accounts * Mark only one oval. ☐ One ☐ Two ☐ More than two 9. Experience Using E-Banking * Mark only one oval. ☐ Less than 1 year ☐ 1 – 3 years ☐ 4 – 6 years ☐ More than 6 years		10. Frequency of E-Banking Use * Mark only one oval. ☐ Daily ☐ Weekly ☐ Several times a month ☐ About once a month ☐ Less than once a month 11. Main Device Used for E-Banking * Mark only one oval. ☐ Smartphone / Mobile App ☐ Computer / Laptop (Web/Online Banking) ☐ Tablet ☐ Both Mobile and Web 12. Region / Residence * Mark only one oval. ☐ Urban ☐ Rural	
https://docs.google.com/forms/d/1g2JHFD3USXBCGQ2b7gYkLUQDjU0U2N-zZW1d9edR 3/16		https://docs.google.com/forms/d/1g2JHFD3USXBCGQ2b7gYkLUQDjU0U2N-zZW1d9edR 4/16	

12/26/25, 6:31 AM
Malaysian E-Banking Service Quality & Customer Satisfaction Survey

13. Which bank do you use **most often** for your e-banking services? *

Mark only one oval.

☐ Maybank
☐ CIMB Bank
☐ Public Bank
☐ RHB Bank
☐ Hong Leong Bank
☐ Bank Islam Malaysia
☐ Bank Rakyat
☐ Ambank
☐ UOB Malaysia
☐ OCBC Bank
☐ Alliance Bank
☐ Affin Bank
☐ HSBC Bank Malaysia
☐ Standard Chartered Bank Malaysia
☐ Agribank
☐ BSN
☐ MBSB Bank
☐ Bank of China Malaysia
☐ ICBC Malaysia
☐ Citibank Malaysia
☐ GXBank
☐ AEON Bank
☐ Boost Bank
☐ SeaBank Malaysia
☐ Al Rajhi Bank Malaysia
☐ Other: _____

12/26/25, 6:31 AM
Malaysian E-Banking Service Quality & Customer Satisfaction Survey

SECTION B — E-Banking Service Quality

This section evaluates your experience with various aspects of e-banking service quality and measures your overall satisfaction with your bank's e-banking services and whether the platform meets your expectations.

Please rate each statement based on your experience using your primary bank's e-banking platform.

Scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree

Reliability

Rate the accuracy and consistency of your bank's e-banking services.

14. My bank's digital banking services are delivered accurately and without errors. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

15. My bank consistently provides the digital services it promises. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

16. Transactions are completed correctly the first time. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

12/26/25, 6:31 AM
Malaysian E-Banking Service Quality & Customer Satisfaction Survey

RESPONSIVENESS

Rate how promptly your bank responds during e-banking interactions.

17. My bank responds promptly when I need assistance with digital services. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

18. Digital services are processed quickly when requested. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

19. My bank clearly informs me when online services will be completed. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

Efficiency

Evaluate the speed and smoothness of the digital platform.

20. It is easy and quick to complete transactions using my bank's online platform. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

12/26/25, 6:31 AM
Malaysian E-Banking Service Quality & Customer Satisfaction Survey

21. I can easily find the information or service I need on the platform. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

22. The digital platform loads and operates smoothly without delay. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

Privacy and Security

Rate your level of trust in your bank's security measures.

23. I feel safe providing my personal or financial information online. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

24. My bank protects my transaction details from unauthorized access. *

Mark only one oval.

1 2 3 4 5

Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree

12/26/25, 6:31 AM Malaysian E-Banking Service Quality & Customer Satisfaction Survey 25. I trust my bank's digital services to handle data securely. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree Ease of Use Rate how easy it is to use your bank's digital interface. 26. The e-banking interface is easy to understand and use. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree 27. I can navigate the e-banking platform confidently without guidance. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree 28. Performing any banking task on the website or app requires little effort. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree System Availability Evaluate how accessible and stable the platform is. https://docs.google.com/forms/d/11g5JHFD3USXBcCqQb7gVcLkUGDlU02N-cZW1d9gdlr 9/16	12/26/25, 6:31 AM Malaysian E-Banking Service Quality & Customer Satisfaction Survey 29. I can access the e-banking system whenever I need it. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree 30. The pages and functions load quickly and rarely crash. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree 31. The e-banking platform works smoothly across all my devices. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree Website Design Rate the visual appeal and design of the platform. 32. The e-banking website has an attractive and professional design. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree https://docs.google.com/forms/d/11g5JHFD3USXBcCqQb7gVcLkUGDlU02N-cZW1d9gdlr 10/16
12/26/25, 6:31 AM Malaysian E-Banking Service Quality & Customer Satisfaction Survey 33. The layout and menus are well organized and easy to follow. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree 34. Visual elements such as colors, icons, and fonts make the interface appealing. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree SECTION C — Customer Satisfaction This section measures your overall satisfaction with your bank's e-banking services. Please rate how well the platform meets your expectations and whether you would recommend it to others. 35. I am satisfied with the overall quality of my bank's e-banking services. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree 36. My experience using the e-banking system meets my expectations. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree https://docs.google.com/forms/d/11g5JHFD3USXBcCqQb7gVcLkUGDlU02N-cZW1d9gdlr 11/16	12/26/25, 6:31 AM Malaysian E-Banking Service Quality & Customer Satisfaction Survey 37. I would recommend my bank's e-banking services to others. * Mark only one oval. 1 2 3 4 5 Stro ☐ ☐ ☐ ☐ ☐ Strongly Agree SECTION D — Issue Monitoring Please answer the following questions about any issues you may have encountered with your bank's e-banking service. 38. Have you faced any issues when using your bank's e-banking services? * Mark only one oval. ☐ Yes Skip to question 39 ☐ No SECTION D — Issue Monitoring Please answer the following questions about any issues you may have encountered with your bank's e-banking service. 39. When did the issue occur (within the last 3 months)? * Mark only one oval. ☐ Within the last 7 days ☐ 1-4 weeks ago ☐ 1-3 months ago https://docs.google.com/forms/d/11g5JHFD3USXBcCqQb7gVcLkUGDlU02N-cZW1d9gdlr 12/16

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Malaysian E-Banking Service Quality & Customer Satisfaction Survey

40. Which e-banking platform did you experience the issue on? *

Mark only one oval.

- ☐ Mobile App
☐ Web / Online Banking
☐ Both

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Malaysian E-Banking Service Quality & Customer Satisfaction Survey

41. Which bank did you encounter the problem with? *

Mark only one oval.

- ☐ Maybank
☐ CIMB Bank
☐ Public Bank
☐ RHB Bank
☐ Hong Leong Bank
☐ Bank Islam Malaysia
☐ Bank Rakyat
☐ Ambank
☐ UOB Malaysia
☐ OCBC Bank
☐ Alliance Bank
☐ Affin Bank
☐ HSBC Bank Malaysia
☐ Standard Chartered Bank Malaysia
☐ Agrobank
☐ BSN
☐ MBSB Bank
☐ Bank of China Malaysia
☐ ICBC Malaysia
☐ Citibank Malaysia
☐ GXBank
☐ AEON Bank
☐ Boost Bank
☐ SeaBank Malaysia
☐ Al Rajhi Bank Malaysia
☐ Other: _____

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Malaysian E-Banking Service Quality & Customer Satisfaction Survey

42. What issue(s) did you experience? (Select all that apply) *

Tick all that apply.

- ☐ [REL] Transaction failure
☐ [REL] Delayed confirmation
☐ [REL] Service outage
☐ [REL] Transfer credited very late
☐ [REL] Scheduled payment not processed as planned
☐ [RES] Slow customer support
☐ [RES] Unhelpful chatbot / automated replies
☐ [EFF] Slow performance
☐ [EFF] OTP takes too long / expires
☐ [PS] Suspicious activity / unusual login
☐ [PS] Concern about data security
☐ [PS] Not sure if messages/links are genuine (scam concern)
☐ [EOU] Confusing interface
☐ [EOU] Hard-to-find functions
☐ [EOU] Difficulty logging in
☐ [EOU] Too many steps for simple tasks
☐ [SA] App crashes
☐ [SA] Site/app unresponsive
☐ [WBD] Poor layout
☐ [WBD] Unattractive/confusing colour scheme

THANK YOU!!!

Thank you for completing this survey. Your responses contribute to the development of a web-based dashboard aimed at enhancing customer satisfaction in e-banking systems.

This content is neither created nor endorsed by Google.

Google Forms

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Appendix D

A full Python code for data preprocessing in this project.

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In [47]:

```
#Setup
import pandas as pd
import numpy as np
import re
from IPython.display import display

# FILE PATHS
INPUT_XLSX = r"C:/Users/Admin/Downloads/Malaysian E-Banking Service Quality & Customer Satisfaction Survey (Responses).xlsx"

df_raw = pd.read_excel(INPUT_XLSX)
```

In [48]:

```
# Step 1: Data Exploration
# Preview
print("\nPreview (first 5 rows)")
display(df_raw.head())

# Dtypes
print("\nInfo:")
df_raw.info()

# Missing values count
print("\nMissing values per column:")
display(df_raw.isnull().sum().sort_values(ascending=False).rename("missing").to_frame())

# Duplicate responses (common check: full-row duplicate)
print("\nDuplicate rows:", df_raw.duplicated().sum())
```

Preview (first 5 rows)

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	Timestamp	I have read the information above and agree to participate.	Age Group	Nationality	Ethnicity	Gender	Employment Status	Monthly Income (RM)	Number of Bank Accounts	Experience Using E-Banking	...	The layout and menus are well organized and easy to follow.	Visual elements such as colors, icons, and fonts make the interface appealing
0	2025-11-26 19:54:01.767	Yes, I agree	18-20 years	Malaysian	Chinese	Female	Student	Below 2,000	One	1 - 3 years	...	3.0	2.0
1	2025-11-26 19:54:46.725	Yes, I agree	31-40 years	Malaysian	Chinese	Female	Employed	4,001 - 6,000	One	More than 6 years	...	4.0	5.0
2	2025-11-26 19:58:50.045	Yes, I agree	31-40 years	Malaysian	Malay	Female	Employed	Above 8,000	More than two	1 - 3 years	...	5.0	4.0
3	2025-11-26 20:01:06.017	Yes, I agree	31-40 years	Malaysian	Malay	Male	Employed	6,001 - 8,000	More than two	More than 6 years	...	5.0	4.0
4	2025-11-26 20:08:17.798	Yes, I agree	41-50 years	Malaysian	Indian	Female	Employed	2,001 - 4,000	One	1 - 3 years	...	4.0	5.0

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5 rows × 43 columns

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```
Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 491 entries, 0 to 490
Data columns (total 43 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Timestamp                                491 non-null    datetime64[ns]
1   I have read the information above and agree to participate.  491 non-null    object
2   Age Group                                486 non-null    object
3   Nationality                              486 non-null    object
4   Ethnicity                                486 non-null    object
5   Gender                                    486 non-null    object
6   Employment Status                        486 non-null    object
7   Monthly Income (RM)                      486 non-null    object
8   Number of Bank Accounts                  486 non-null    object
9   Experience Using E-Banking                486 non-null    object
10  Frequency of E-Banking Use                486 non-null    object
11  Main Device Used for E-Banking            486 non-null    object
12  Region / Residence                        486 non-null    object
13  Which bank do you use most often for your e-banking services?  486 non-null    object
14  My bank's digital banking services are delivered accurately and without errors.  486 non-null    float64
15  My bank consistently provides the digital services it promises.  486 non-null    float64
16  Transactions are completed correctly the first time.              486 non-null    float64
17  My bank responds promptly when I need assistance with digital services.  486 non-null    float64
18  Digital services are processed quickly when requested.            486 non-null    float64
19  My bank clearly informs me when online services will be completed.  486 non-null    float64
20  It is easy and quick to complete transactions using my bank's online platform.  486 non-null    float64
21  I can easily find the information or service I need on the platform.  486 non-null    float64
22  The digital platform loads and operates smoothly without delay.    486 non-null    float64
23  I feel safe providing my personal or financial information online.  486 non-null    float64
24  My bank protects my transaction details from unauthorized access.    486 non-null    float64
25  I trust my bank's digital services to handle data securely.         486 non-null    float64
26  The e-banking interface is easy to understand and use.             486 non-null    float64
27  I can navigate the e-banking platform confidently without guidance.  486 non-null    float64
28  Performing any banking task on the website or app requires little effort.  486 non-null    float64
29  I can access the e-banking system whenever I need it.              486 non-null    float64
30  The pages and functions load quickly and rarely crash.             486 non-null    float64
31  The e-banking platform works smoothly across all my devices.        486 non-null    float64
32  The e-banking website has an attractive and professional design.    486 non-null    float64
33  The layout and menus are well organized and easy to follow.         486 non-null    float64
34  Visual elements such as colors, icons, and fonts make the interface appealing.  486 non-null    float64
```

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```
35 I am satisfied with the overall quality of my bank's e-banking services. 486 non-null float64
36 My experience using the e-banking system meets my expectations. 486 non-null float64
37 I would recommend my bank's e-banking services to others. 486 non-null float64
38 Have you faced any issues when using your bank's e-banking services? 486 non-null object
39 When did the issue occur (within the last 3 months)? 338 non-null object
40 Which e-banking platform did you experience the issue on? 338 non-null object
41 Which bank did you encounter the problem with? 338 non-null object
42 What issue(s) did you experience? (Select all that apply) 338 non-null object
dtypes: datetime64[ns](1), float64(24), object(18)
memory usage: 165.1+ KB
```

Missing values per column:

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missing

What issue(s) did you experience? (Select all that apply)	153
Which bank did you encounter the problem with?	153
Which e-banking platform did you experience the issue on?	153
When did the issue occur (within the last 3 months)?	153
The e-banking platform works smoothly across all my devices.	5
My bank protects my transaction details from unauthorized access.	5
I trust my bank's digital services to handle data securely.	5
The e-banking interface is easy to understand and use.	5
I can navigate the e-banking platform confidently without guidance.	5
Performing any banking task on the website or app requires little effort.	5
I can access the e-banking system whenever I need it.	5
The pages and functions load quickly and rarely crash.	5
The layout and menus are well organized and easy to follow.	5
The e-banking website has an attractive and professional design.	5
The digital platform loads and operates smoothly without delay.	5
Visual elements such as colors, icons, and fonts make the interface appealing.	5
I am satisfied with the overall quality of my bank's e-banking services.	5
My experience using the e-banking system meets my expectations.	5
I would recommend my bank's e-banking services to others.	5
Have you faced any issues when using your bank's e-banking services?	5
I feel safe providing my personal or financial information online.	5
I can easily find the information or service I need on the platform.	5

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	missing
Main Device Used for E-Banking	5
It is easy and quick to complete transactions using my bank's online platform.	5
Ethnicity	5
Gender	5
Employment Status	5
Monthly Income (RM)	5
Number of Bank Accounts	5
Experience Using E-Banking	5
Frequency of E-Banking Use	5
Region / Residence	5
Age Group	5
Which bank do you use most often for your e-banking services?	5
My bank's digital banking services are delivered accurately and without errors.	5
My bank consistently provides the digital services it promises.	5
Transactions are completed correctly the first time.	5
My bank responds promptly when I need assistance with digital services.	5
Digital services are processed quickly when requested.	5
My bank clearly informs me when online services will be completed.	5
Nationality	5
I have read the information above and agree to participate.	0
Timestamp	0

Duplicate rows: 0

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```

In [49]: #Step 2: Attribute Renaming
# Normalize headers
def normalize_header(s):
    s = str(s)
    s = s.replace("\n", " ").replace("\r", " ").replace("\t", " ")
    s = re.sub(r"^\s+", " ", s)
    s = s.strip()
    s = s.replace(" ", "").replace("'", '').replace('"', '')
    return s

df_raw2 = df_raw.copy()
df_raw2.columns = [normalize_header(c) for c in df_raw2.columns]

#Build rename mapping
rename_map = {
    "Timestamp": "response_time",
    "I have read the information above and agree to participate.": "consent_agreed",
    "Age Group": "age_group",
    "Nationality": "nationality",
    "Ethnicity": "ethnicity",
    "Gender": "gender",
    "Employment Status": "employment_status",
    "Monthly Income (RM)": "monthly_income_rm",
    "Number of Bank Accounts": "num_bank_accounts",
    "Experience Using E-Banking": "ebanking_experience",
    "Frequency of E-Banking Use": "usage_frequency",
    "Main Device Used for E-Banking": "main_device",
    "Region / Residence": "residence_region",
    "Which bank do you use most often for your e-banking services?": "main_bank",
    "Have you faced any issues when using your bank's e-banking services?": "issue_experience",
    "When did the issue occur (within the last 3 months)?": "issue_timing",
    "Which e-banking platform did you experience the issue on?": "issue_platform",
    "Which bank did you encounter the problem with?": "issue_bank",
    "What issue(s) did you experience? (Select all that apply)": "issue_types_raw",
}

#Rename Likert items → REL/RES/EFF/PS/EQU/SA/WBD/CS
likert_map = {
    # Reliability
    "My bank's digital banking services are delivered accurately and without errors.": "REL1",

```

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```
"My bank consistently provides the digital services it promises.": "REL2",
"Transactions are completed correctly the first time.": "REL3",

# Responsiveness
"My bank responds promptly when I need assistance with digital services.": "RES1",
"Digital services are processed quickly when requested.": "RES2",
"My bank clearly informs me when online services will be completed.": "RES3",

# Efficiency
"It is easy and quick to complete transactions using my bank's online platform.": "EFF1",
"I can easily find the information or service I need on the platform.": "EFF2",
"The digital platform loads and operates smoothly without delay.": "EFF3",

# Privacy & Security
"I feel safe providing my personal or financial information online.": "PS1",
"My bank protects my transaction details from unauthorized access.": "PS2",
"I trust my bank's digital services to handle data securely.": "PS3",

# Ease of Use
"The e-banking interface is easy to understand and use.": "EQU1",
"I can navigate the e-banking platform confidently without guidance.": "EQU2",
"Performing any banking task on the website or app requires little effort.": "EQU3",

# System Availability
"I can access the e-banking system whenever I need it.": "SA1",
"The pages and functions load quickly and rarely crash.": "SA2",
"The e-banking platform works smoothly across all my devices.": "SA3",

# Website Design
"The e-banking website has an attractive and professional design.": "WBD1",
"The layout and menus are well organized and easy to follow.": "WBD2",
"Visual elements such as colors, icons, and fonts make the interface appealing.": "WBD3",

# Customer Satisfaction
"I am satisfied with the overall quality of my bank's e-banking services.": "CS1",
"My experience using the e-banking system meets my expectations.": "CS2",
"I would recommend my bank's e-banking services to others.": "CS3",
}

# 3) Normalize mapping keys
rename_map_norm = {normalize_header(k): v for k, v in rename_map.items()}
```

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```
likert_map_norm = {normalize_header(k): v for k, v in likert_map.items()}
full_map_norm = {**rename_map_norm, **likert_map_norm}

# 4) Apply renaming
df = df_raw2.rename(columns=rename_map_norm).rename(columns=likert_map_norm).copy()

# 5) Rename table (no Found_in_File)
full_map_norm = {**rename_map_norm, **likert_map_norm}
rename_table = pd.DataFrame(
    [(old, new) for old, new in full_map_norm.items()],
    columns=["Original", "Renamed"]
)

display(rename_table)
```

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	Original	Renamed
0	Timestamp	response_time
1	I have read the information above and agree to...	consent_agreed
2	Age Group	age_group
3	Nationality	nationality
4	Ethnicity	ethnicity
5	Gender	gender
6	Employment Status	employment_status
7	Monthly Income (RM)	monthly_income_rm
8	Number of Bank Accounts	num_bank_accounts
9	Experience Using E-Banking	ebanking_experience
10	Frequency of E-Banking Use	usage_frequency
11	Main Device Used for E-Banking	main_device
12	Region / Residence	residence_region
13	Which bank do you use most often for your e-ba...	main_bank
14	Have you faced any issues when using your bank...	issue_experience
15	When did the issue occur (within the last 3 mo...	issue_timing
16	Which e-banking platform did you experience th...	issue_platform
17	Which bank did you encounter the problem with?	issue_bank
18	What issue(s) did you experience? (Select all ...	issue_types_raw
19	My bank's digital banking services are deliver...	REL1
20	My bank consistently provides the digital serv...	REL2
21	Transactions are completed correctly the first...	REL3

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	Original	Renamed
22	My bank responds promptly when I need assistan...	RES1
23	Digital services are processed quickly when re...	RES2
24	My bank clearly informs me when online service...	RES3
25	It is easy and quick to complete transactions ...	EFF1
26	I can easily find the information or service I...	EFF2
27	The digital platform loads and operates smooth...	EFF3
28	I feel safe providing my personal or financial...	PS1
29	My bank protects my transaction details from u...	PS2
30	I trust my bank's digital services to handle d...	PS3
31	The e-banking interface is easy to understand ...	EOU1
32	I can navigate the e-banking platform confiden...	EOU2
33	Performing any banking task on the website or ...	EOU3
34	I can access the e-banking system whenever I n...	SA1
35	The pages and functions load quickly and rarel...	SA2
36	The e-banking platform works smoothly across a...	SA3
37	The e-banking website has an attractive and pr...	WBD1
38	The layout and menus are well organized and ea...	WBD2
39	Visual elements such as colors, icons, and fon...	WBD3
40	I am satisfied with the overall quality of my ...	CS1
41	My experience using the e-banking system meets...	CS2
42	I would recommend my bank's e-banking services...	CS3

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```
In [50]: #Step 3: Inclusion Filtering

df_before = len(df)

# Count how many match each exclusion rule
consent_no = (df["consent_agreed"].astype(str).str.strip() == "No, I do not agree").sum()
below_18 = (df["age_group"].astype(str).str.strip() == "Below 18 years").sum()
less_month = (df["usage_frequency"].astype(str).str.strip() == "Less than once a month").sum()

# Apply the filters
consent_no = (df["consent_agreed"] == "No, I do not agree").sum()
below_18 = (df["age_group"] == "Below 18 years").sum()
less_month = (df["usage_frequency"] == "Less than once a month").sum()

df = df[
    (df["consent_agreed"] != "No, I do not agree") &
    (df["age_group"] != "Below 18 years") &
    (df["usage_frequency"] != "Less than once a month")
].copy()

df_after = len(df)

print("Excluded: No consent =", consent_no)
print("Excluded: Below 18 years =", below_18)
print("Excluded: Less than once a month =", less_month)
print("Rows before filter:", df_before)
print("Rows after filter:", df_after)
print("Total removed:", df_before - df_after)

Excluded: No consent = 5
Excluded: Below 18 years = 3
Excluded: Less than once a month = 57
Rows before filter: 491
Rows after filter: 426
Total removed: 65
```

```
In [51]: # Step 4: Value Standardisation

# Ethnicity: keep Malay/Chinese/Indian else "Other"
allowed_ethnicity = {"Malay", "Chinese", "Indian"}
```

```
df["ethnicity"] = df["ethnicity"].astype(str).str.strip()
df["ethnicity"] = df["ethnicity"].where(df["ethnicity"].isin(allowed_ethnicity), "Other")

# Bank standardisation for main_bank and issue_bank
approved_banks = {
    "Maybank",
    "CIMB Bank",
    "Public Bank",
    "RHB Bank",
    "Hong Leong Bank",
    "Bank Islam Malaysia",
    "Bank Rakyat",
    "AmBank",
    "UOB Malaysia",
    "OCBC Bank",
    "Alliance Bank",
    "Affin Bank",
    "HSBC Bank Malaysia",
    "Standard Chartered Bank Malaysia",
    "Agrobank",
    "BSN",
    "MBSB Bank",
    "Bank of China Malaysia",
    "ICBC Malaysia",
    "Citibank Malaysia",
    "GXBANK",
    "AEON Bank",
    "Boost Bank",
    "SeaBank Malaysia",
    "Al Rajhi Bank Malaysia"
}

def standardize_bank(x):
    if pd.isna(x):
        return np.nan
    s = str(x).strip()
    return s if s in approved_banks else "Other"

df["main_bank"] = df["main_bank"].apply(standardize_bank)
df["issue_bank"] = df["issue_bank"].apply(standardize_bank)
```

```

# Ethnicity counts (fixed order)
eth_counts = (
    df["ethnicity"]
    .value_counts(dropna=False)
    .reindex(["Malay", "Chinese", "Indian", "Other"], fill_value=0)
    .reset_index()
)
eth_counts.columns = ["ethnicity", "count"]
display(eth_counts)

# Helper: force "Other" to appear (and optionally put it last)
def counts_with_other(series, label_col):
    vc = series.value_counts(dropna=False)

    # Ensure "Other" shows up even if 0
    if "Other" not in vc.index:
        vc.loc["Other"] = 0

    # Optional: move "Other" to the bottom
    vc = vc.drop("Other").sort_values(ascending=False)
    vc.loc["Other"] = series.eq("Other").sum()

    out = vc.reset_index()
    out.columns = [label_col, "count"]
    return out

# main_bank and issue_bank counts (include Other even if 0)
display(counts_with_other(df["main_bank"], "main_bank"))
display(counts_with_other(df["issue_bank"], "issue_bank"))

print("Ethnicity total:", df["ethnicity"].value_counts(dropna=False).sum())
print("Main bank total:", df["main_bank"].value_counts(dropna=False).sum())
print("Issue bank total:", df["issue_bank"].value_counts(dropna=False).sum())

```

	ethnicity	count
0	Malay	143
1	Chinese	191
2	Indian	92
3	Other	0

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	main_bank	count
0	Agrobank	30
1	Standard Chartered Bank Malaysia	22
2	OCBC Bank	22
3	GXBank	21
4	RHB Bank	20
5	Affin Bank	20
6	Citibank Malaysia	20
7	HSBC Bank Malaysia	19
8	ICBC Malaysia	19
9	SeaBank Malaysia	18
10	Bank Rakyat	18
11	Bank Islam Malaysia	18
12	Bank of China Malaysia	17
13	AEON Bank	17
14	CIMB Bank	16
15	Alliance Bank	16
16	MBSB Bank	16
17	BSN	14
18	UOB Malaysia	13
19	Boost Bank	13
20	Public Bank	12
21	Hong Leong Bank	11

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	main_bank	count
22	Al Rajhi Bank Malaysia	10
23	Maybank	8
24	Other	16

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	issue_bank	count
0	NaN	126
1	Bank of China Malaysia	20
2	HSBC Bank Malaysia	17
3	Affin Bank	16
4	Alliance Bank	15
5	Standard Chartered Bank Malaysia	15
6	CIMB Bank	14
7	SeaBank Malaysia	14
8	UOB Malaysia	14
9	Agrobank	13
10	Bank Rakyat	13
11	GXBank	13
12	MBSB Bank	12
13	AEON Bank	12
14	Public Bank	11
15	BSN	11
16	ICBC Malaysia	11
17	Hong Leong Bank	10
18	Bank Islam Malaysia	10
19	OCBC Bank	10
20	Al Rajhi Bank Malaysia	9
21	RHB Bank	8

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	issue_bank	count
22	Maybank	7
23	Citibank Malaysia	7
24	Boost Bank	6
25	Other	12

Ethnicity total: 426
Main bank total: 426
Issue bank total: 426

```
In [52]: # Step 5: Data Cleaning
# Structural Missing Handling (+ flag inconsistencies + blank out inconsistent follow-ups)

followup_fields = ["issue_timing", "issue_platform", "issue_bank", "issue_types_raw"]

def compute_followup_status(row):
    exp = row.get("issue_experience")
    exp = "" if pd.isna(exp) else str(exp).strip()

    followup_has_any = any(
        pd.notna(row.get(f)) and str(row.get(f)).strip() != ""
        for f in followup_fields
    )

    if exp == "No":
        return "Inconsistent" if followup_has_any else "Not Applicable"
    if exp == "Yes":
        return "Applicable" if followup_has_any else "Missing"
    return "Missing" # unexpected Label

# 1) Create followup_status
df["followup_status"] = df.apply(compute_followup_status, axis=1)

# 2) If "No" but follow-up has values -> flag is already "Inconsistent"
# Now blank out the follow-up entries to align with Google Forms Logic
incon_mask = df["followup_status"].eq("Inconsistent")
df.loc[incon_mask, followup_fields] = pd.NA
```

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```
# become "Not Applicable" after cleaning
df["followup_status"] = df.apply(compute_followup_status, axis=1)

# QA summaries
display(df["issue_experience"].value_counts(dropna=False).to_frame("count"))
display(df["followup_status"].value_counts(dropna=False).to_frame("count"))

# Inspect inconsistent rows
display(df.loc[incon_mask, ["issue_experience"] + followup_fields].head(30))
print("Total inconsistent rows:", incon_mask.sum())
```

count	
issue_experience	
Yes	303
No	123

count	
followup_status	
Applicable	279
Not Applicable	123
Missing	24

	issue_experience	issue_timing	issue_platform	issue_bank	issue_types_raw
102	No	<NA>	<NA>	<NA>	<NA>
105	No	<NA>	<NA>	<NA>	<NA>
108	No	<NA>	<NA>	<NA>	<NA>
112	No	<NA>	<NA>	<NA>	<NA>
114	No	<NA>	<NA>	<NA>	<NA>
119	No	<NA>	<NA>	<NA>	<NA>
131	No	<NA>	<NA>	<NA>	<NA>
132	No	<NA>	<NA>	<NA>	<NA>
145	No	<NA>	<NA>	<NA>	<NA>
146	No	<NA>	<NA>	<NA>	<NA>
147	No	<NA>	<NA>	<NA>	<NA>
148	No	<NA>	<NA>	<NA>	<NA>
153	No	<NA>	<NA>	<NA>	<NA>
156	No	<NA>	<NA>	<NA>	<NA>
157	No	<NA>	<NA>	<NA>	<NA>
158	No	<NA>	<NA>	<NA>	<NA>
163	No	<NA>	<NA>	<NA>	<NA>
165	No	<NA>	<NA>	<NA>	<NA>
166	No	<NA>	<NA>	<NA>	<NA>
168	No	<NA>	<NA>	<NA>	<NA>
169	No	<NA>	<NA>	<NA>	<NA>

Total inconsistent rows: 21

```

In [53]: # Handle missing values
objective1_cols = (
    [f"REL{i}" for i in range(1, 4)] +
    [f"RES{i}" for i in range(1, 4)] +
    [f"EFF{i}" for i in range(1, 4)] +
    [f"PS{i}" for i in range(1, 4)] +
    [f"EOU{i}" for i in range(1, 4)] +
    [f"SA{i}" for i in range(1, 4)] +
    [f"WBD{i}" for i in range(1, 4)] +
    [f"CS{i}" for i in range(1, 4)]
)

# BEFORE
before_rows = len(df)
before_missing_rows = df[objective1_cols].isna().any(axis=1).sum()

print("Before deletion:")
print("Total rows:", before_rows)
print("Rows that have missing in Objective 1 columns:", before_missing_rows)

# DELETE (Listwise deletion)
df_obj1 = df.dropna(subset=objective1_cols).copy()

# AFTER
after_rows = len(df_obj1)

print("\nAfter deletion:")
print("Total rows:", after_rows)
print("Rows removed:", before_rows - after_rows)
print("\nObjective 1 rows after listwise deletion:", len(df_obj1))

Before deletion:
Total rows: 426
Rows that have missing in Objective 1 columns: 0

After deletion:
Total rows: 426
Rows removed: 0

Objective 1 rows after listwise deletion: 426

```

```

In [54]: import re
import numpy as np
import pandas as pd

# 1) Copy + make response_id
df_dash = df.reset_index(drop=True).copy()
df_dash["response_id"] = df_dash.index + 1

# 2) Keep only the columns you need for Tableau issues table
issues = df_dash[[
    "response_id", "issue_experience", "followup_status",
    "issue_timing", "issue_platform", "issue_bank", "issue_types_raw",
    "main_bank", "ethnicity", "age_group", "gender"
]].copy()

# 3) Keep only people who should have issue rows (Yes cases)
issues = issues[(issues["issue_experience"] == "Yes") & (issues["followup_status"] == "Applicable")]

# 4) Split checkbox string into rows (explode)
issues_long = (
    issues.assign(issue_option_raw=issues["issue_types_raw"].fillna("").str.split(","))
    .explode("issue_option_raw")
)

# 5) Clean spaces and drop blanks
issues_long["issue_option_raw"] = issues_long["issue_option_raw"].astype(str).str.strip()
issues_long = issues_long[issues_long["issue_option_raw"] != ""]

# 6) Extract tag like [REL] and remove it from Label
# - tag is letters inside [ ]
issues_long["issue_dimension_tag"] = issues_long["issue_option_raw"].str.extract(r"^\s*([A-Z]+)\s*$")
issues_long["issue_label"] = issues_long["issue_option_raw"].str.replace(r"^\s*([A-Z]+)\s*$", "", regex=True)

print("Issue long table rows:", len(issues_long))
display(issues_long.head(10))

Issue long table rows: 2956

```

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response_id	issue_experience	followup_status	issue_timing	issue_platform	issue_bank	issue_types_raw	main_bank	ethnicity	age_grou
0	1	Yes	Applicable	1-4 weeks ago	Both	AEON Bank [REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-4 year
0	1	Yes	Applicable	1-4 weeks ago	Both	AEON Bank [REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-4 year
0	1	Yes	Applicable	1-4 weeks ago	Both	AEON Bank [REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-4 year
0	1	Yes	Applicable	1-4 weeks ago	Both	AEON Bank [REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-4 year
0	1	Yes	Applicable	1-4 weeks ago	Both	AEON Bank [REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-4 year
0	1	Yes	Applicable	1-4 weeks ago	Both	AEON Bank [REL] Scheduled payment not processed as plann...	OCBC Bank	Chinese	31-4 year
1	2	Yes	Applicable	1-4 weeks ago	Web / Online Banking	Bank Rakyat [REL] Transaction failure. [REL] Delayed confi...	Citibank Malaysia	Malay	31-4 year
1	2	Yes	Applicable	1-4 weeks ago	Web / Online Banking	Bank Rakyat [REL] Transaction failure. [REL] Delayed confi...	Citibank Malaysia	Malay	31-4 year

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response_id	issue_experience	followup_status	issue_timing	issue_platform	issue_bank	issue_types_raw	main_bank	ethnicity	age_grou
1	2	Yes	Applicable	1-4 weeks ago	Web / Online Banking	Bank Rakyat [REL] Transaction failure. [REL] Delayed confi...	Citibank Malaysia	Malay	31-4 year
1	2	Yes	Applicable	1-4 weeks ago	Web / Online Banking	Bank Rakyat [REL] Transaction failure. [REL] Delayed confi...	Citibank Malaysia	Malay	31-4 year

```
In [55]: import os

OUTPUT_DIR = r"C:/Users/Admin/Downloads/"
os.makedirs(OUTPUT_DIR, exist_ok=True)

issues_long.to_excel(os.path.join(OUTPUT_DIR, "issues_long.xlsx"), index=False)
df_dash.to_excel(os.path.join(OUTPUT_DIR, "respondents_dash.xlsx"), index=False)
df_obj1.to_excel(os.path.join(OUTPUT_DIR, "objective1_cleaned.xlsx"), index=False)

print("Saved to:", OUTPUT_DIR)
```

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Appendix E

Table for Work breakdown

Table E.1: Work breakdown for Capstone 1

Task (Capstone 1 – Proposal)	Start	End	Milestone	Task Duration (Days)
Project scoping (project boundaries + stakeholders)	22/9/2025	24/9/2025		2
Identify research gap + objectives	25/9/2025	28/9/2025		3
Problem statement finalisation	29/9/2025	5/10/2025	M1: Proposal foundation	6
Keyword search + article screening	29/9/2025	5/10/2025		6
Build literature review matrix (themes + constructs)	6/10/2025	19/10/2025		13
Synthesis writing (link to model + variables)	20/10/2025	26/10/2025		6
Dimension definition + framework	20/10/2025	26/10/2025	M2: Literature synthesis	6
Draft questionnaire items (by dimension)	20/10/2025	26/10/2025		6
Chapter 1 draft (intro + problem + objectives)	20/10/2025	2/11/2025		13
Analysis plan (SPSS tests + interpretation plan)	27/10/2025	2/11/2025	M3: Method readiness	6
Refinement (wording + scale + layout)	27/10/2025	2/11/2025		6
Final instrument packaging (Google Form + consent + intro)	3/11/2025	9/11/2025	M4: Instrument design	6

Chapter 2 draft (lit review write-up)	3/11/2025	16/11/2025		13
Chapter 3 draft (methodology write-up)	10/11/2025	23/11/2025		13
Work breakdown structure construction	10/11/2025	12/11/2025		2
Gantt schedule setup + progress tracking updates	13/11/2025	25/12/2025		42
Data collection	19/11/2025	22/12/2025	M5: Data acquisition	34
Preprocessing script + preliminary cleaning (Cleaned Dataset generated)	22/12/2025	23/12/2025	M6: Data product	1
Milestones + flowchart finalisation + updates	13/11/2025	25/12/2025	M7: Work plan package	42
References + appendices + final checks	24/12/2025	29/12/2025	M8: Proposal finalisation	5
Proposal formatting	26/12/2025	29/12/2025		3
Supervisor review + revisions	26/12/2025	29/12/2025		3
Buffer / contingency (formatting issues / late feedback)	29/12/2025	1/1/2026		3
Submission	1/1/2026	2/1/2026	M9: Submission	1

Capstone 2

Table E.2: Work breakdown for Capstone 2

Task (Capstone 2 – Implementation)	Start	End	Milestone	Task Duration (Days)
Confirm supervisor list and Capstone 2 direction	04/05/2026	17/05/2026	M1: Supervisor Confirmation	14 days
Review Capstone 1 proposal and prepare Capstone 2 files	11/05/2026	17/05/2026	M1: Supervisor Confirmation	7 days
Check cleaned dataset and construct scores	25/05/2026	31/05/2026	M2: Objective 1 Analysis Started	7 days
Prepare reliability and descriptive outputs	25/05/2026	31/05/2026	M2: Objective 1 Analysis Started	7 days
Conduct correlation and regression analysis	01/06/2026	07/06/2026	M3: Supervisor Consultation	7 days
Contact supervisor and confirm analysis direction	01/06/2026	07/06/2026	M3: Supervisor Consultation	7 days
Write Objective 1 results and discussion	08/06/2026	14/06/2026	M4: Objective 1 Analysis and Writing	7 days
Prepare Power BI dashboard dataset	15/06/2026	21/06/2026	M5: Simple Dashboard Version	7 days
Build simple Power BI dashboard version	15/06/2026	21/06/2026	M5: Simple Dashboard Version	7 days
Improve dashboard layout, visuals, and filters	22/06/2026	28/06/2026	M6: Dashboard Improvement and Validation	7 days
Validate dashboard totals and issue counts	22/06/2026	28/06/2026	M6: Dashboard Improvement and Validation	7 days
Write Objective 2 dashboard findings	29/06/2026	05/07/2026	M7: Objective 2 Writing	7 days

Write Objective 3 recommendations	06/07/2026	12/07/2026	M8: Objective 3 Writing	7 days
Finalise Chapter 4 Results and Discussion	13/07/2026	19/07/2026	M9: Results and Discussion Finalisation	7 days
Write Chapter 5 conclusion, limitations, and future work	20/07/2026	26/07/2026	M10: Conclusion and Future Work	7 days
Integrate and check full report, apply supervisor corrections	27/07/2026	31/07/2026	M11: Final Report Integration and Checking	5 days
Final buffer and contingency	01/08/2026	02/08/2026	Protects M12 readiness	2 days
Export final report and submit	03/08/2026	03/08/2026	M12: Submission	1 day